

THE ROSEN FIBRATION

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ABSTRACT. Equipped with little mathematical technology but a great deal of inventive spirit, the biologist Robert Rosen carried out foundational work at the interface of mathematical biology and category theory, having fallen in love with the latter under the influence of Eilenberg and Mac Lane. Today the time is ripe to introduce more powerful mathematical tools and to seek a modern reformulation of Rosen’s ideas using the expressive resources of contemporary category theory.

We thus reformulate Rosen’s programme inside an arbitrary monoidal closed category where we cast the theory of (M,R) -systems, diagrams of shape $A \rightarrow B \rightarrow [A, B]$ that can be interpreted as a process-repair couple.

This yields what we call an *incoherent* (M,R) -system. These assemble into a total category carrying a canonical span over C , fibred in the input variable though only *displayed* in the output one, and they have interesting models in vector spaces, quantales, Chu spaces, and combinatorial species under both convolution and substitution.

Repair becomes *polymorphic* once the repair map is required to be natural. We compare a hierarchy of naturality conditions, preferring to focus on those where the dependence on A, B is functorial; thus, we organize *coherent* (M,R) -systems into a profunctor, whose tabulator is the *Rosen fibration* of the title. Naturality is a severe constraint: in **Set**, and more generally in every localic topos, coherent (M,R) -systems are trivial. Nontrivial examples require the ambient category to carry isotropy, in a sense measured by a certain object of C , the end $\int_X [X, X]$, which classifies repair maps and which we call the *Bernstein centre*.

We then formalize Rosen’s *metabolic closure*, classify the universal closure points, and extend the theory (both the incoherent and the coherent one) to diagrams $A \rightarrow B \rightarrow [A, B] \rightarrow [B, [A, B]] \rightarrow \dots$ of arbitrary, even infinite, length. Finally, we analyse Arbib’s correspondence between (M,R) -systems and Mealy automata; we upgrade it to a fibred functor, and observe that metabolic closure translates into a fact of classical representation theory: a system is closed exactly when the initial state of the associated automaton has nontrivial isotropy, and a universal closure point is an element of its stabilizer. A substantial part of the mathematics below has been formalized and machine-checked in the Agda proof assistant.

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1. INTRODUCTION

Robert Rosen’s programme of *relational biology* sought to characterize living systems not by the physical and chemical substrate that realizes them, but by the abstract pattern of relations among their constituent processes [Ros58a, Ros59]. Working at Chicago’s Committee on Mathematical Biology, Rosen sharpened this into the idea that a biological system should be modelled as a category of components together with functorial relations among them, adopting the then-recent theory of Eilenberg and Mac Lane [EML45] as its native formal language [Ros58b]; the programme culminated around 1964–1966 in the definition of what Rosen dubbed an (M,R) -system:

a diagram of shape

$$A \rightarrow B \rightarrow H(A, B) \tag{1.1}$$

where A, B are sets, and $H(A, B)$ is an appropriately determined subset of the set B^A of functions from A to B ; this is meant to encode a ‘metabolic’ process $A \rightarrow B$, together with a ‘repair’ map $B \rightarrow H(A, B)$ that reconstructs the very component responsible for carrying it out ($H(A, B)$ is notation for the ‘hom’ set of functions $A \rightarrow B$, of which the process is a term).

Subsequent work recast (M,R)-systems as *sequential machines*, establishing their basic structural theory [Ros64a]: the state space of the associated machine is $S = H(A, B)$, of cardinality $|B|^{|A|}$, while the input alphabet is A itself, of size $|A|$, so that the alphabet is always negligible compared to the state space as soon as $|B| \geq 2$. Since reversibility of environmentally induced structural change was shown to require strong connectedness, one is led to restrict the ambient category of realizable maps, discarding enough set-theoretic functions for strong connectedness to become likely again [Ros64b]. The third paper in this line, in a terse remark whose consequences occupy an entire section of the present paper, observes that the totality of (M,R)-systems formed over a given category assembles, in a natural way, into a category [Ros66a]; it is precisely this remark that we unpack in full generality, in §3–4 below.

Demetrius promptly extended this identification, characterizing when an (M,R)-system is *behaviorally reversible* as a condition on the associated sequential machine [Dem66]. The automata-theoretic reading was soon taken up by Arbib, who showed that (M,R)-systems with a fixed ‘genome’ (his term for the repair map held constant) correspond to a subcategory of automata whose states depend on the output produced [Arb66]; Warner dropped the fixed-genome restriction and represented general (M,R)-systems by categories of automata, proving two comparison theorems between the resulting categories [War82]; and Băianu, first alone and then together with Marinescu, studied further algebraic and functorial aspects of this correspondence, in particular proving that categories of general (M,R)-systems admit projective limits [Băi73, BM74]. This is a completeness result we generalize considerably in Remark 3.30 and Remark 3.33.

Several decades later, Palmer, Williams and Gatherer revisited the correspondence in the language of *X-machines*: automata equipped with a memory set, whose transitions are driven by a chosen type of partial functions on that memory rather than by raw input symbols. They show that ordinary finite-state machines, and even memory-equipped *stream X-machines*, are too weak to model the self-referential structure of an (M,R)-system, and that only *communicating X-machines* (networks of stream X-machines exchanging messages) can evade the obstruction [PWG16]. §5 below takes a different route back to Arbib’s own, older correspondence between (M,R)-systems and Mealy automata [Arb66]: Arbib relates the two only at the level of objects, and we show this assignment does not extend to a functor landing in the category of Mealy automata ordinarily considered in the literature; replacing the target by a category of ‘twisted’ Mealy automata, whose morphisms reindex input and output with opposite variance, repairs exactly this failure and turns Arbib’s correspondence into a genuine (indeed fibred) functor.

With the passing of time, Rosen’s emphasis gradually shifted from this automata-theoretic reading towards the notion of *closure to efficient causation*: informally, the requirement that every component responsible for carrying out one of its processes be itself produced by another process internal to the system, so that no efficient cause is left unentailed from outside [Ros12]. Rosen argued this closure to be incompatible with simulation by a Turing machine, a claim revisited and given precise categorical and λ -calculus formulations by Letelier and colleagues, who work it out explicitly in the compact closed category **Rel** and in complete lattices [LSAGA⁺04, LSAGA⁺06]; by Mossio, Longo and Stewart, who realize closure via a fixed-point combinator and argue that closure alone does not entail non-computability [MLS09]; and by Cárdenas and colleagues, who respond by insisting on Rosen’s own distinction between *simulating* and *modelling* a system [CLG⁺10].

These works seem aware of, but generally fail to make explicit use of, the fact that closure to efficient causation is essentially the property, of certain models of the λ -calculus (i.e. Cartesian closed categories), of admitting a so-called *reflexive object*: an object X equipped with an isomorphism $X \cong X^X$, exhibiting X itself as a space of (X -valued) functions on X . Nontrivial such models of λ -calculus were first constructed by Dana Scott [Sco75], and accordingly the categories

that they form go under the name ‘Scott domains’. In Lemma 4.2 we show that a rather general notion of ‘coherent’ (M, R) -system (one whose repair map is natural, rather than merely a bare arrow) is forced to be trivial in the category **Set** of sets and functions; in Proposition 4.23, by contrast, we exhibit a category that is still, in the relevant sense, a model of the λ -calculus, but admits genuinely nontrivial coherent (M, R) -systems. The obstruction responsible for triviality in **Set**, isolated in Remark 4.6 below, is exactly that the terminal object of **Set** happens to be a *generator* (or *separator*) of the whole category [AR94]. This property fails, and so no longer obstructs nontriviality, once one leaves **Set** for a suitable non-well-pointed topos.

Connected with the notion of reflexivity is the idea that the defining diagram of an (M, R) -system is an instance of self-application, ‘ $f(f) = f$ ’ in the loose notation of [SAJGL11], an idea those authors trace back to Lawvere’s fixed-point theorem for Cartesian closed categories, itself a vast categorical generalization unifying the diagonal arguments of Cantor, Russell, Gödel and Tarski [Law69], and which Soto-Andrade and Varela extended to a broader self-referential setting, relating it explicitly to Rosen’s programme and to the notion of biological autonomy [SAV84]. The uncurrying-recurrying operation of Definition 2.19, by which we pass between a repair map $\Phi : B \rightarrow [A, B]$ and an endomap of $[B, B]$, is the categorical-algebra incarnation of exactly this self-referential move, stripped of any commitment to a specific ambient category of biological relevance.

The notion of closure was, in the following decades, generalized well beyond (M, R) -systems proper into an account of *biological organization* as closure of constraints, by Montévil and Mossio, who reformulate closure in terms of constraints (structures that canalize a process without themselves being altered by it) mutually enabling one another’s maintenance [MM15]. Louie’s two monographs remain the most extensive syntheses of the whole programme available to date, and we lean on their terminology and conventions throughout: the earlier [Lou09] a direct continuation of Rosen’s own *Life Itself*, the later [Lou17] a still more systematic functorial treatment. Rosen’s own, older notion of a *modelling relation*—the bidirectional encoding/decoding correspondence between a natural system and a formal system representing it—has been recast in explicitly adjunction-theoretic terms by Zafiris, who exhibits encoding and decoding as a pair of adjoint functors between categories of observables [Zaf12]; and a Cartesian fragment of Maturana and Varela’s autopoiesis has been cast categorically by Nomura [Nom01].

Another chapter of the story involves Rosen’s disciple, A. Louie, who in the early 2000s strengthened Rosen’s non-simulability claim into an explicit noncomputability result for any material realization of an (M, R) -system: his [Lou05] gives the general impredicativity argument and rebuts several published attempts at explicit computable counterexamples, while a sharper 2007 note targets one specific critique [CH06] directly, arguing it rests on an incorrect definition of ‘mechanism’ and on a mishandling of categorical products and coproducts [Lou07]. The claim has not gone unchallenged: Chu and Ho have argued at length, reconstructing Rosen’s argument from *Life Itself* in careful categorical detail, that several of his central notions are ill-defined and that his central proof does not, in fact, establish non-simulability [CH06].¹

With essentially no exception, the literature surveyed above, foundational or contemporary, casts (M, R) -systems in the category **Set** of sets and functions, or at best in a Cartesian category, improperly treated in all respects as if it were **Set**; Remark 3.2 and Remark 3.4 below record a suspicion, seemingly not articulated before, that this is a rather degenerate special case of a much richer phenomenon.

Structure of the paper. The present paper undertakes a systematic reformulation of Rosen’s (M, R) -system (3.1) in an arbitrary monoidal closed category C , i.e. a category equipped with a tensor product $-\otimes-$ and an internal hom $[-, -]$ satisfying $C(X\otimes A, B) \cong C(X, [A, B])$, so that $[A, B]$ plays the role of ‘the object of maps from A to B ’ even when C is not **Set**. The reformulation is carried out with a deliberately minimal set of standing hypotheses: we do not assume C symmetric (i.e. we assume no identification $A\otimes B \cong B\otimes A$), and throughout the paper we track exactly where symmetry, braiding, *biclosure* (the existence of a second, independent internal hom allowing one to

¹At the moment we take no side in this debate, which concerns the physical and computational instantiation of (M, R) -systems rather than their mathematical structure, the latter being the sole object of the present paper.

‘curry in the other direction’, needed precisely because symmetry is not assumed), or the existence of limits and colimits becomes necessary, rather than assuming them wholesale from the outset (Remark 3.3).

In §2 we assemble the categorical toolkit that our task requires: we recall the statement and consequences of the Yoneda lemma, the construction and basic properties of comma categories and inserters, coends, closed and monoidal closed categories, and isolate, as Definition 2.19, the *uncurrying-recurrying* operation already discussed above, by which a repair map $\Phi : B \rightarrow [A, B]$ is uncurried into $B \otimes A \rightarrow B$ and, granted biclosure or symmetry, recurried along the other tensor slot into an endomap of $[B, B]$. This simple construction recurs several times throughout the paper, from the definition of composition of (M,R)-systems to the dinatural end construction of Remark 4.8.

In §3 we study *incoherent* (M,R)-systems based at (A, B) , i.e. diagrams of shape $A \rightarrow B \rightarrow [A, B]$: an ‘incoherent’ (M,R)-system, in our terminology, is simply Rosen’s original diagram taken at face value, with Φ a bare arrow rather than a coherently-varying family, and naturality is only demanded starting in section 4. We assemble incoherent (M,R)-systems into a total category **iMRS** that fits in a canonical span over C, C^{op} , providing a setting in which one can cast at once Rosen’s claim [Ros66a] and Băianu’s completeness result in a sufficiently (co)complete monoidal closed category (Proposition 3.29 onward). Along the way we identify (Proposition 3.38) incoherent (M,R)-systems with *lenses* $[A/B] \rightarrow [B/B]$, i.e. datatypes equipped with a ‘get’ map $A \rightarrow B$ paired with a ‘put’ map $B \otimes A \rightarrow B$; such structures are familiar from functional programming [FGM⁺07, JRW12, PGW17, Ril18, CEG⁺24]; we then show that the mismatch between the general lens-composition formula and the sequential composition of (M,R)-systems explains, from the optics perspective, why the latter is associative but only left-unital.

A gallery of examples, in Equation 3—3.19, grounds the theory outside **Set**, each illustrating a different way in which the ambient category interprets the type of an incoherent (M,R)-system. In vector spaces, an (M,R)-system based at the ground field recovers a bilinear pairing; something similar happens in compact closed categories such as the category **Rel** of relations and the category $\vee$ **Lat** of join-semilattices, and in *quantales*, complete join semilattices equipped with a monoid multiplication possessing both a left and a right residual (the order-theoretic shadow of a (bi)closed monoidal category) [Yet90, Ros90]. We also study the cases of pointed sets, or pointed spaces; of endofunctors of a complete category C under composition, where the internal hom is given by right Kan extension; and of categories $(\mathbf{Cat}, \boxtimes)$, equipped with the so-called ‘funny’ tensor product $A \boxtimes B$ (which freely combines the morphisms of A and B *without* imposing the interchange law that holds automatically for the ordinary product $A \times B$). At greater length, we study (M,R)-systems in the category **Spc** of combinatorial species under Day convolution [Joy86, BLL98], where an (M,R)-system based at the representable species $y(1)$ is shown to be exactly a pointed module for the species of linear orders, its repair map a coalgebra for the species-derivative comonad (the categorified analogue of differentiating a generating function), and its iterated repair a trajectory in a total category of elements; and (M,R)-systems in Chu spaces [Bar91, Pra99, Abr10], pairs of a set of ‘points’ and a set of ‘tests’ related by a pairing valued in a fixed truth-value object k , where taking k to be a three-element chain models Łukasiewicz–Kleene three-valued logic, and taking the Chu space built from a Hilbert space recovers, via Abramsky’s construction, Wigner’s theorem on physical symmetries as a statement about which Chu-transforms exist. We close the section by exhibiting incoherent (M,R)-systems as dynamical systems in a coKleisli category, whose morphisms $X \rightarrow Y$ are maps out of a comonadic ‘context’ around X , here the Cartesian product with A , when C is Cartesian, connecting the theory back to the classical, Rosen-native setting.

In §4 we reach the core of the paper, considering the problem of upgrading the repair map Φ of an incoherent (M,R)-system to a natural transformation, so that repair becomes *polymorphic* and can vary together with the process it is supposed to repair. We survey a hierarchy of increasingly abstract naturality conditions, distinguished by exactly what is required to vary naturally: an *id-natural* (M,R)-system (§4.1) asks $\Phi_X : X \rightarrow [A, X]$ to vary naturally over *every* object X of C , with no reference to any process at all; a *cod-natural* (§4.3.1) one instead asks Φ to vary naturally over every *process* $h : X \rightarrow Y$ (formally, over the arrow category of C), so that repair is tied to how

a system was actually produced, which is arguably the more biologically faithful reading; an *A/C-natural* (§4.3.2) one weakens this further, demanding naturality only among processes sharing a fixed domain A ; and a fully *profunctorial* (M, R) -system (§4.3.5) drops the reference to a fixed A altogether.

These conditions reveal a tension between two opposing desiderata of the theory, tractability and nontriviality. More naturality can force a great many almost-trivial models:

- First, as a consequence of the Yoneda lemma, in the category **Set** no nontrivial *id*-natural (M, R) -system exists (Lemma 4.2); this mirrors, in a sense, the Yoneda-flavoured argument Rosen himself sketches informally in [Ros12, §3.5]; we observe that outside well-pointed toposes, by contrast, nontrivial Cartesian coherent (M, R) -systems genuinely exist, as we exhibit concretely in the topos of C_2 -sets (Proposition 4.23). Toposes of monoid actions, toposes of graphs, sheaves over nontrivial topological spaces. . . are all non well-pointed.
- Second, more naturality does not automatically yield a *profunctor* of (M, R) -systems, which is the structure needed to reindex an (M, R) -system along a change of A and B (reindexing along B was not possible for incoherent (M, R) -systems): we prove that *id*-natural and *cod*-natural (M, R) -systems are in fact in bijection, because the codomain functor of the arrow category is a *lax epimorphism* (Lemma 4.33), and that this shared notion of coherent (M, R) -system does assemble into a profunctor; but *A/C*-natural (M, R) -systems, despite being arguably the most semantically natural of the four, provably do not give a profunctor (subsubsection 4.3.2).

When this has been made precise, we assemble the profunctor of *cod/id*-natural (M, R) -systems into a fibration taking its tabulator, the *Rosen fibration* of the title (Definition 4.10): the category whose objects are (M, R) -systems and whose fibres over $C^{\text{op}} \times C$ record, at each pair (A, B) , exactly the coherent (M, R) -systems $\langle f, \Phi \rangle$ based there.

We then formalize Rosen’s own notion of *metabolic closure* (informally, that the process f is itself recoverable by applying repair to some input, $f = \Phi_a$ for some a) and of a *universal closure point*, an a_0 achieving this for every possible process simultaneously; we show the latter notion is compatible with reindexing along the Rosen fibration, and extend the whole theory to (M, R) -systems of arbitrary finite length $A \rightarrow B \rightarrow [A, B] \rightarrow [B, [A, B]] \rightarrow \cdots$, the categorical avatar of the very construction Rosen sketches informally, as equations (3.80)–(3.81) of *Anticipatory Systems*, under the heading ‘Another Relational Metaphor’ [Ros12]. That extension is not merely a matter of drawing longer diagrams. A system of length n carries $n - 1$ natural transformations rather than one, all of whose components are available at once (Construction 4.50), and the obstructions to closure multiply accordingly: one may fail to find a transformation of the required type at some level of the tower, or find one that no lower level can see. Taking the limit of the tower gives the category $\mathbf{MRS}_\infty$ of *infinitely extended* (M, R) -systems (Definition 4.51), for which the literature we have surveyed offers evocative pictures but no definition we could use. Metabolic closure does not extend to these systems by itself: granting each level an independently chosen closure point is too weak to say that the hierarchy sustains itself, since nothing then relates one level to the next. What is needed is an extra compatibility across levels, which we isolate as *tower-coherence* (Definition 4.56), and the point of the notion is that it collapses: a tower is coherent exactly when its whole countable family of repair transformations is induced by a single one out of the colimit of the tower (Proposition 4.57), and, when the monoidal unit is a generator, exactly when a single global *label* governs the entire hierarchy. The condition has real content precisely where the theory is not degenerate — it is vacuous in **Set**, where there is nothing to choose at any level, and genuinely restrictive in $\mathbf{Vect}_k$ (Remark 4.60).

Finally, §5 relates incoherent (M, R) -systems to *Mealy automata*, refurbishing an old paper by Arbib [Arb66]: a total category of Mealy machines has been the subject of several papers by the first author, and it has been proved in [Lor25b, Lor25a] to yield a two-sided fibration; this fits nicely with the perspective of the present paper, which builds on Arbib and Warner’s correspondence (specified by them only at the level of objects, whereas we treat it as a—fibred—functor); to do so, however, we have to consider a category of ‘twisted’ Mealy automata, whose

morphisms reindex the input and output variables with opposite variance rather than covariantly in lockstep.

We close by translating metabolic closure into the vocabulary of classical representation theory [Ste16], reading $[A, B]$ as the carrier of a representation of the monoid A^* of iterated inputs: closure turns out to be exactly the condition that the initial state of the associated Mealy automaton has nontrivial *isotropy* (a nonidentity element fixing it), and a universal closure point is exactly an element of the *stabilizer* of that state. In this way a diagram-chasing biological notion is translated into a single, classical fact of algebra.

A remark on method. We would rather make a closing point plainly than leave it to be inferred. The literature we have just surveyed spans nearly seventy years and, even restricted to the sources we were able to assemble for the present project, runs to close to three thousand pages; a considerable fraction of it consists of appeals to intuition never checked against any shared formal standard; claims that are, in the sense Wolfgang Pauli gave the phrase, *not even wrong*: too loosely stated for their truth or falsity to be settled by anything beyond further appeal to intuition. Chu and Ho’s contention, recounted above, that several of Rosen’s own central notions are ill-defined is not an isolated skirmish but a symptom of exactly this state of affairs: a tradition that has, after decades, still not agreed on whether its own foundational proof is even correct, because no version of that proof was ever pinned down precisely enough to admit a definitive answer.

We read this charitably rather than as an indictment. Much of this literature was written by scientists who arrived at Rosen’s intuition from biology, cybernetics, and philosophy rather than from mathematics, and who worked without the categorical machinery, developed only in the decades since, that could have pinned that intuition down; that so many of them nonetheless sensed there was something worth pursuing is itself a testament to how good the intuition was. But insight, of which Rosen had no shortage, needs precision to harden into knowledge, and precision is what twenty-first-century mathematics can now supply.

This is why a substantial part of the mathematics below has been formalized and machine-checked in the Agda proof assistant (v2.8.0); the codebase is maintained as a subfolder (Rosen ) in a fork of the `agda-categories`  library.

We state precisely what that claim covers, since a formalization is worth exactly as much as the accuracy of the claims made on its behalf. Each statement that has been formalized carries a marker linking to the corresponding line of the codebase, and the marker comes in two colours:

- a black ‘’ means the statement is proved outright: its proof term is complete and appeals to no postulate. See for example Definition 3.5;
- a red ‘’ means the opposite, and is a warning: the formalization is complete in *shape* but rests on an explicitly postulated step, so *nothing* about that statement has been verified. See Proposition 3.51.

A statement carrying no marker at all has not been formalized: this is the case for the whole of section 2, which recalls standard material, and for a number of statements elsewhere. As of this writing 34 statements are black, none is red, and no statement carries an unresolved placeholder — the red marker is kept because it is the honest thing to have available, not because we currently need it.

Two qualifications, which we prefer to state than to have discovered. First, of the 37 modules in the codebase 29 pass Agda’s `--safe` check, which mechanically refuses every postulate; most of the 34 black statements live in one of those. The rest are proved outright as well, but sit in modules that are not `--safe` — either because they are instantiated over `Set` and so assume function extensionality, or because they share a file with a postulate they do not themselves use. Second, function extensionality is an ordinary axiom rather than a gap, and it is the *only* postulate in the whole development, isolated in a single file, `Axioms.agda`, so that its use can be audited. Where we have found a claim of ours to be false we have removed it and documented the obstruction in its place, rather than assuming it: a postulate asserting a falsehood is worse than an absence, because it reads as a result. Both lists are recoverable from the codebase in one `grep`, and we would rather the reader run it than take our word.

With that scope fixed, the point stands: a claim checked by a proof assistant is not merely argued for. Within the black markers it is true in a sense no one can reasonably contest, since a machine, applying rules of deduction fixed in advance and answerable to no one's authority or intuition, has verified it — and, just as importantly, the red markers record exactly where we have *not* earned that right.

We believe *verified truth* is an appropriate tribute to a mind as brilliant and imaginative as Rosen's, decades ahead of the category theory that could substantiate his ideas.

Remark 1.1. The task of building this codebase has been heavily helped by various generative AI models; primarily ChatGPT 5.4, ChatGPT 5.6 Luna and Sol, Claude Opus 5, and Claude Sonnet 5. The same models were also used to assemble, retrieve and organize the literature survey above, and to help with proofreading and streamlining the text. No human-readable mathematical argument has been AI-generated.

2. PRELIMINARIES

The purpose of the present section is to fix notation and help readers who are not fluent in category theory to grasp the main ideas of the subsequent sections; we invite even the experienced reader to glance at it, as it establishes the notation we will employ in section 3; the style of exposition in the present section is quite terse. After each definition, we include a brief remark pointing to a few references and explaining where the notion will enter our discussion.

Definition 2.1 (Yoneda embedding). Every small category C admits a fully faithful functor into the category $[C^{\text{op}}, \mathbf{Set}]$ (denoted $\widehat{C}$ for short) of *presheaves* on C , i.e. contravariant functors from C into sets. This functor is called the *Yoneda embedding*

$$\mathbb{Y} : C \longrightarrow \widehat{C} \quad (2.1)$$

of C and it sends each object C to the *representable* hom-functor $\mathbb{Y}C = C(-, C)$.

Theorem 2.2 (Yoneda lemma). For every $F \in \widehat{C}$ and every object $C \in C$, there is an isomorphism

$$\varepsilon_{F,C} : \widehat{C}(\mathbb{Y}C, F) \xrightarrow{\cong} FC \quad (2.2)$$

induced by evaluation at id_C , namely the function

$$\varepsilon_{F,C}(\alpha) := \alpha_C(\text{id}_C) \in FC \quad (2.3)$$

is bijective and natural in C (and in F).

To put it more tersely, following the fully natural formulation given in [Rie16, §2.2], there is an isomorphism of functors between the *evaluation* of F at C ,

$$\text{ev}_C : C^{\text{op}} \times \widehat{C} \rightarrow \mathbf{Set}, \quad (C, F) \mapsto FC, \quad (2.4)$$

and the composite functor $\widehat{C}(\mathbb{Y}, \text{id}_{\widehat{C}})$, filling the triangle

$$\begin{array}{ccc} & (\widehat{C})^{\text{op}} \times \widehat{C} & \\ \mathbb{Y}^{\text{op}} \times \text{id}_{\widehat{C}} \nearrow & \Downarrow \varepsilon & \searrow \widehat{C}(-, -) \\ C^{\text{op}} \times \widehat{C} & \xrightarrow{\text{ev}_C} & \mathbf{Set} \end{array} \quad (2.5)$$

Remark 2.3. The Yoneda lemma originates in [Yon60], and the reader can consult [ML98] for the classical textbook statement; the ‘fully-natural’ formulation used above is contained in [Rie16]; the Yoneda lemma is the main obstruction in the passage from *incoherent* (M,R)-systems (Definition 3.5) to *coherent* ones (Lemma 4.2).

Definition 2.4 (Comma category). Given categories $\mathcal{A}, \mathcal{B}, \mathcal{C}$ and functors $F : \mathcal{A} \rightarrow \mathcal{C}$ and $G : \mathcal{B} \rightarrow \mathcal{C}$ with the same codomain, the *comma category* (F/G) is the category whose

- objects are triples (A, B, p) , where $p \in C(FA, GB)$ is an arrow of $\mathcal{C}$;

- arrows $(a, b) : (A, B, p) \rightarrow (A', B', q)$ are pairs of arrows $a : A \rightarrow A'$ and $b : B \rightarrow B'$ such that the following square commutes:

$$\begin{array}{ccc} FA & \xrightarrow{p} & GB \\ Fa \downarrow & & \downarrow Gb \\ FA' & \xrightarrow{q} & GB'. \end{array} \quad (2.6)$$

Some useful properties of comma categories are collected in the following proposition.

Proposition 2.5 (Some properties of (F/G)).

- The comma category comes equipped with two *projection* functors:

$$\mathcal{A} \xleftarrow{\pi_F} (F/G) \xrightarrow{\pi_G} \mathcal{B}. \quad (2.7)$$

The left projection $\pi_F : (F/G) \rightarrow \mathcal{A}$ is a fibration, while the right projection $\pi_G : (F/G) \rightarrow \mathcal{B}$ is an opfibration (these are in a suitable sense the free fibration on F and the free opfibration on G ; however, this remark is inessential for our purposes).

- The comma category (F/G) enjoys the following universal property. The square

$$\begin{array}{ccc} (F/G) & \xrightarrow{\pi_F} & \mathcal{A} \\ \pi_G \downarrow & \not\parallel_{\theta_{FG}} & \downarrow F \\ \mathcal{B} & \xrightarrow{G} & \mathcal{C} \end{array} \quad (2.8)$$

together with the 2-cell filling it, whose component at (A, B, p) is $p : FA \rightarrow GB$, is terminal. Concretely, every other span $\mathcal{B} \leftarrow \mathcal{X} \rightarrow \mathcal{A}$, equipped with a 2-cell

$$\chi : F \circ l \Rightarrow G \circ r, \quad (2.9)$$

factors uniquely through θ_{FG} ; i.e. there is a unique functor $\langle l, r \rangle : \mathcal{X} \rightarrow (F/G)$ such that $\chi = \theta_{FG} * \langle l, r \rangle$ (the ‘whiskering’ of the natural transformation θ_{FG} and the functor $\langle l, r \rangle$):

$$\begin{array}{ccccc} \mathcal{X} & & & & \\ & \searrow \langle l, r \rangle & & \searrow l & \\ & (F/G) & \xrightarrow{p_F} & \mathcal{A} & \\ & \downarrow p_G & \not\parallel_{\theta_{FG}} & \downarrow F & \\ & \mathcal{B} & \xrightarrow{G} & \mathcal{C} & \end{array} \quad (2.10)$$

- The comma category can be computed as a limit in **Cat**. More precisely, there is a pullback square of functors

$$\begin{array}{ccc} (F/G) & \longrightarrow & C^{\rightarrow} \\ \langle p_F, p_G \rangle \downarrow & \lrcorner & \downarrow \langle d_0, d_1 \rangle \\ \mathcal{A} \times \mathcal{B} & \xrightarrow{F \times G} & C \times C. \end{array} \quad (2.11)$$

Remark 2.6. Comma categories have been a standard tool of category theory since Lawvere’s thesis [Law63]; the reader can consult [ML98, Jac99] for a systematic treatment; comma constructions are ubiquitous in describing categories of ‘machines’ and will appear in §5 relating (M, R) -systems and ‘Mealy automata’.

The comma category (F/G) is an important example of a 2-categorical *limit*. Another important example is the *serter* of two functors.

Definition 2.7 (serter of two functors). Given two parallel functors $F, G : \mathcal{B} \rightarrow \mathcal{C}$, the *serter* $(F \multimap G)$ is the category whose

- objects are pairs (B, h) , where $h : FB \rightarrow GB$;

- arrows $(B, h) \rightarrow (B', k)$ are arrows $u : B \rightarrow B'$ such that

$$Gu \circ h = k \circ Fu. \quad (2.12)$$

The inserter $(F \multimap G)$ enjoys a universal property analogous to that of (F/G) . The square

$$\begin{array}{ccc} (F \multimap G) & \xrightarrow{U} & \mathcal{B} \\ U \downarrow & \not\cong_{\theta_{FG}} & \downarrow F \\ \mathcal{B} & \xrightarrow{G} & C \end{array} \quad (2.13)$$

where $U : (F \multimap G) \rightarrow \mathcal{B}$ is the evident forgetful functor, comes equipped with a 2-cell

$$\iota : F \circ U \Rightarrow G \circ U \quad (2.14)$$

and is terminal among squares of this form. Thus, although there need not be a natural transformation $F \Rightarrow G$ on the whole of $\mathcal{B}$, there is a canonical one after restriction along U .

The difference between a comma category and an inserter is worth making explicit. In an inserter, the two functors are parallel and there is a single projection $U : (F \multimap G) \rightarrow \mathcal{B}$. By contrast, a comma category has two generally different projections. In particular, when F and G are parallel, both (F/G) and $(F \multimap G)$ are defined, but they are different categories.

An important example of an inserter is the category of algebras, or coalgebras, for an endofunctor.

Definition 2.8 (Algebras and coalgebras as inserters). Let $F : C \rightarrow C$ be an endofunctor. The category $\mathbf{Alg} F$ of F -algebras is defined as the inserter

$$\begin{array}{ccc} \mathbf{Alg} F & \xrightarrow{U} & C \\ U \downarrow & \not\cong & \downarrow F \\ C & \xrightarrow{=} & C. \end{array} \quad (2.15)$$

Dually, the category $\mathbf{coAlg} F$ of F -coalgebras is defined as the inserter

$$\begin{array}{ccc} \mathbf{coAlg} F & \xrightarrow{U} & C \\ U \downarrow & \not\cong & \parallel \\ C & \xrightarrow{F} & C. \end{array} \quad (2.16)$$

Remark 2.9. The presentation of algebras and coalgebras for an endofunctor as inserters is standard; the reader can consult [Jac99, ML98]; we will use the notion of endofunctor algebra for the characterization in Proposition 3.29.

Definition 2.10 (Module for an endofunctor). Let $F : C \rightarrow C$ be an endofunctor; a *module* for F (see [Dub70]) consists of a functor $X : \mathcal{X} \rightarrow C$ and a natural transformation $\xi : F \circ X \Rightarrow X$.

Thus, the category of algebras for F can be characterized as the universal module for F , via the structure given by the forgetful functor $U : \mathbf{Alg}(F) \rightarrow C$.

Remark 2.11. The notion of module for an endofunctor, in the enriched setting, is contained in [Dub70]; we will observe in Remark 4.13 that the repair map of a coherent (M, R) -system can be seen as a module in this sense.

The next technical tool we need is the *dinatural equivalence relation* used in the concrete construction of a coend, a particular kind of colimit associated with functors of type $C^{\text{op}} \times C \rightarrow \mathcal{D}$. We shall only need the case in which $\mathcal{D} = \mathbf{Set}$.

Definition 2.12 (Dinatural equivalence relation). Let $T : C^{\text{op}} \times C \rightarrow \mathbf{Set}$ be a functor. The relation of dinatural equivalence is the equivalence relation $\sim$ on the disjoint union $\sum_X T(X, X)$ generated by declaring

$$(X, \xi) \sim (Y, \theta) \quad (2.17)$$

whenever $\xi \in T(X, X)$, $\theta \in T(Y, Y)$, and there exists an arrow $f : X \rightarrow Y$ such that

$$T(X, f)(\xi) = T(f, Y)(\theta). \quad (2.18)$$

Definition 2.13. The *coend* of T is, when it exists, the quotient of the coproduct $\sum_X T(X, X)$ by the dinatural equivalence relation,

$$\int^X T(X, X) := \left(\sum_X T(X, X) \right) / \sim, \quad (2.19)$$

denoted simply by $\int T$ when there is no risk of confusion. It comes equipped with a canonical family of functions

$$\iota_A : T(A, A) \longrightarrow \int^X T(X, X) \quad (2.20)$$

satisfying the following *cowedge condition*:

For every arrow $u : A \rightarrow B$ in C , the square

$$\begin{array}{ccc} T(B, A) & \xrightarrow{T(u, A)} & T(A, A) \\ T(B, u) \downarrow & & \downarrow \iota_A \\ T(B, B) & \xrightarrow{\iota_B} & \int^X T(X, X) \end{array} \quad (2.21)$$

commutes.

The family $(\iota_A \mid A \in C)$ is the ‘initial cowedge’, i.e. it is initial among families satisfying this condition.

Remark 2.14. Ends, coends, and dinatural transformations are treated systematically in [Lor21], with the classical account given in [ML98]; we will use this notion in Remark 4.8, where we reduce an *id*-coherent (M, R) -system to a single map into an appropriate *end*, and in subsection 4.6, where we characterize a ‘procomonad’ structure on the profunctor of *id*-coherent (M, R) -systems.

Definition 2.15 (Closed category). A category is *closed* if it is equipped with the following data:

- a functor $[-, -] : C^{\text{op}} \times C \rightarrow C$, called the *internal hom* or simply the *hom* of C ;
- an object $I \in C$, called the *unit object*;
- a natural isomorphism $i : \text{Id}_C \xrightarrow{\cong} [I, -]$, exhibiting each object as its internal hom from the unit;
- a family of morphisms $j_X : I \rightarrow [X, X]$ forming an extranatural family in the variable X , meaning that for every $h : X \rightarrow Y$ in C , the square

$$\begin{array}{ccc} I & \xrightarrow{j_X} & [X, X] \\ j_Y \downarrow & & \downarrow [X, h] \\ [Y, Y] & \xrightarrow{[h, Y]} & [X, Y] \end{array} \quad (2.22)$$

commutes. Note that this is a *wedge* condition, dual to the one in (2.21) above. It models how one chooses an identity for each object X , compatibly with the morphisms of C .

These data are subject to certain coherence conditions, expressing the expected interaction among the *hom*, I , i , and j , and generalizing the usual Yoneda-type identities. However, we do not dwell on the precise definition, as the notion of closed category is a relatively impoverished one.

By contrast, the categories one encounters most often are *monoidal* closed, among which the *Cartesian closed* categories form a particularly important subclass; see [Lam80].

Definition 2.16. A monoidal category is equipped with a bifunctor

$$- \otimes - : C \times C \longrightarrow C \quad (2.23)$$

satisfying the axioms of associativity, left and right unitality (with respect to a *monoidal unit* object $I \in C_0$), as well as the coherence conditions of [ML98].

Again, we do not give a self-contained account, since the theory of monoidal categories is classical and easily accessible references are available; see [ML98, EGNO15, FS19]. Instead, we focus on monoidal closure:

Definition 2.17 (Monoidal closed category). A closed category $(\mathcal{V}, [-, -])$ is *monoidal closed* if it is equipped with a monoidal product $- \otimes -$, with unit I , and a natural isomorphism

$$\mathcal{V}(X, [Y, Z]) \cong \mathcal{V}(X \otimes Y, Z). \quad (2.24)$$

Remark 2.18. The original definition of closed category is due to [EK66]; its monoidal refinement, treated above, is discussed in [Day70b, Koc70]; it is a ubiquitous assumption of the present paper to work in an ambient (monoidal) closed category.

Definition 2.19 (Uncurrying-recurrying). In a monoidal closed category C as in Definition 2.17, every object $Y \in C$ induces a natural isomorphism

$$C(X \otimes Y, Z) \cong C(X, [Y, Z]) \quad (2.25)$$

exhibiting $- \otimes Y$ as left adjoint to $[Y, -]$; we call the passage from left to right *currying*, and its inverse *uncurrying*.

Given a map $\Phi : B \rightarrow [A, B]$, *uncurrying* Φ (i.e. applying the isomorphism above right-to-left, with $X = B, Y = A, Z = B$) yields

$$\varphi := \text{ev}_{A,B} \circ (\Phi \otimes \text{id}_A) : B \otimes A \rightarrow B, \quad (2.26)$$

where $\text{ev}_{A,B} : [A, B] \otimes A \rightarrow B$ is the counit of $- \otimes A \dashv [A, -]$.

We now describe the converse operation of *recurrying*, which exchanges the roles played by A and B , turning $\varphi : B \otimes A \rightarrow B$ into a map $A \rightarrow [B, B]$. Unlike uncurrying, this requires additional structure on C , in one of the following two forms.

- C is *biclosed*: besides the internal hom $[-, -]$ above, right adjoint to $- \otimes Y$ for every Y , there is a second internal hom $\{-, -\} : C^{\text{op}} \times C \rightarrow C$, right adjoint instead to $Y \otimes -$, i.e. a natural isomorphism

$$C(Y \otimes X, Z) \cong C(X, \{Y, Z\}). \quad (2.27)$$

Taking $Y = B, X = A, Z = B$ in this second adjunction, $\varphi : B \otimes A \rightarrow B$ curries directly into a map $A \rightarrow \{B, B\}$, with no need for a symmetry.

- $- \otimes -$ is *symmetric*, with symmetry isomorphism $\sigma_{B,A} : B \otimes A \xrightarrow{\cong} A \otimes B$. Precomposing φ with $\sigma_{B,A}^{-1}$ gives a map $\varphi \circ \sigma_{B,A}^{-1} : A \otimes B \rightarrow B$, which currying along the original adjunction (with $X = A, Y = Z = B$) turns into a map $A \rightarrow [B, B]$.

When C is symmetric and biclosed, the two homs $[-, -]$ and $\{-, -\}$ are canonically isomorphic, and the two recipes above agree up to this isomorphism; we do not distinguish between them notationally, writing $\Phi^\vee : A \rightarrow [B, B]$ for the map obtained by either route, and calling it the *uncurrying-recurrying* of Φ .

In short: starting from $\Phi : B \rightarrow [A, B]$, one uncurries into $\varphi : B \otimes A \rightarrow B$; assuming either that C is biclosed, or that $- \otimes -$ is symmetric, one then recurries φ , along the other tensor slot, into $\Phi^\vee : A \rightarrow [B, B]$.

Remark 2.20. Mere closedness provides the minimal structure needed for the curried formulation of an (M,R)-system, whereas a monoidal closed category allows one to pass freely between the curried map $\Phi : B \rightarrow [A, B]$ and its uncurried counterpart $\Phi' : B \otimes A \rightarrow B$. Neither formulation is indispensable in complete generality. In a monoidal category C that is not closed, one may take the uncurried map Φ' as primitive; conversely, in a closed category whose closed structure does not arise from a monoidal product, one may work directly with the curried map Φ . The most convenient formulation therefore depends on the ambient category.

A full review of the theory of fibrations would take us too far afield. Standard references are [Jac99, Chapter 1] and [LR20]. We shall nevertheless make repeated use of one of that theory's fundamental results: the *Grothendieck–Bénabou equivalence*.

Theorem 2.21. There is an equivalence of categories

$$\mathbf{Fib}/C \cong \mathbf{Cat}_{ps}(C^{\text{op}}, \mathbf{Cat}) \quad (2.28)$$

between the category of *fibrations* over C and the category of *pseudofunctors* $C^{\text{op}} \rightarrow \mathbf{Cat}$ ([Jac99, Definition 1.4.4]), induced by the correspondence of [Jac99, Proposition 1.4.5]. This equivalence restricts to one between the category of *presheaves* $C^{\text{op}} \rightarrow \mathbf{Set}$ and the category of *discrete fibrations* over C .

We will also make somewhat implicit use of the most general version of this correspondence, which in recent years has generally been dubbed a ‘straightening’ construction (because it allows one to straighten the fibres of a fibration into a presheaf).

Theorem 2.22. There is an equivalence of categories between $\mathbf{Cat}/\mathcal{B}$, the slice category of functors $F : \mathcal{X} \rightarrow \mathcal{B}$, whose arrows $H : (\mathcal{X}, F) \rightarrow (\mathcal{Y}, G)$ are commutative triangles $G \circ H = F$, and the category of *normal lax functors*

$$\mathcal{B} \longrightarrow \mathbf{Prof} \quad (2.29)$$

sending an object B of $\mathcal{B}$ to a category $F_{\natural}B$ (the fibre of F over B) and a morphism $u : B \rightarrow B'$ to a *profunctor* (see below for the definition) $F_{\natural}u : (F_{\natural}B)^{\text{op}} \times (F_{\natural}B') \rightarrow \mathbf{Set}$.

Remark 2.23. Fibrations and the Grothendieck construction are covered extensively in [Bén85, Str23, Gra66].

In fact, the above statement made use of the notion of profunctor, which we now have a good motivation to introduce.

A *profunctor* P between categories $\mathcal{A}, \mathcal{B}$ consists of a functor of type

$$P : \mathcal{A} \times \mathcal{B}^{\text{op}} \longrightarrow \mathbf{Set}. \quad (2.30)$$

Definition 2.24 (Profunctors and tabulators). The *tabulator* of P (also called the *category of elements* or simply the *total category* of P ; but we reserve such names for different constructions here) is defined as follows:

- objects are triples $(A, B; \xi)$ where $\xi \in P(A, B)$ is an element;
- arrows $(A, B; \xi) \rightarrow (A', B'; \theta)$ are pairs of arrows $u \in \mathcal{A}(A, A'), v \in \mathcal{B}(B, B')$ such that

$$P(A, v)(\xi) = P(u, B')(\theta) \quad \text{as elements of } P(A, B'). \quad (2.31)$$

The verification that this defines a category $\oplus P$ is easy; note that the category $\oplus P$ comes equipped with two *projection* functors

$$\mathcal{A} \xleftarrow{\pi^l} \oplus P \xrightarrow{\pi^r} \mathcal{B} \quad (2.32)$$

and a natural transformation of type

$$\oplus P(_, _) \quad P(\pi^l _, \pi^r _) \quad (2.33)$$

with components given by sending (u, v) as above to the common value $t = P(A, v)(\xi) = P(u, B')(\theta)$, and satisfying the following universal property: every cell of shape

factors uniquely as

(2.34)

Remark 2.25. The tabulator of a distributor is discussed at length in [Béner], and its relationship with (co)ends is treated in the literature on ‘double categories’ [GP99b, Gra19]; tabulators will allow us to construct, in Definition 4.10, a two-sided discrete fibration associated to the profunctor of coherent (M,R)-systems.

A similar construction, which however reverses the direction of $u : A \rightarrow A'$ into $u : A' \rightarrow A$, is the following.

Definition 2.26 (Category of twisted elements of a profunctor). Let P be a profunctor, in the notation of (2.30); the *category of twisted elements* of P is defined as follows:

- the objects are triples $(A, B; \xi)$ where $\xi \in P(A, B)$ is an element;
- a morphism $(A, B; \xi) \rightarrow (A', B'; \theta)$ is a pair of arrows $u \in \mathcal{A}(A', A), v \in \mathcal{B}(B, B')$ such that the function $P(u, v) : P(A, B) \rightarrow P(A', B')$ sends ξ to θ .

It is clear that this gives rise to a category $\tau\mathcal{EL}[P]$; the category comes equipped with a forgetful functor $V : \tau\mathcal{EL}[P] \rightarrow C^{\text{op}} \times C$.

Remark 2.27. The twisted arrow category is a standard construction attached to profunctors, essential to the theory of ends and coends, as well as to some applications to categorical logic; see [ML98, Lor21] for more details; this construction is essential to the development of Construction 4.50, where we build categories of ‘long’ (and possibly infinitely extended) (M, R) -systems. (Strictly speaking the category of Proposition 3.42 is not an example of a twisted arrow category, since a profunctor of incoherent (M, R) -systems does not exist; but the notation comes in handy.)

3. INCOHERENT (M, R) -SYSTEMS

Notation 3.1. Informally speaking, an ‘ (M, R) -system’ consists of a diagram of shape

$$A \xrightarrow{f} B \xrightarrow{\Phi} [A, B] \quad (3.1)$$

This follows Rosen’s notation closely, where f is dubbed the ‘process’ of the system, and Φ is dubbed the ‘repair’ map (more on the meaning of this terminology will follow in Remark 3.6). It is diagrams of this shape, or suitable variations thereof, that we want to study.

For the entire paper C denotes a monoidal closed category. We do not insist on assuming that C is symmetric, and we always uncurry a repair map $B \rightarrow [A, B]$ into an arrow $B \otimes A \rightarrow B$.

Whenever needed, we will also assume that C admits certain limits, or colimits (for example, finite products or coproducts, which are necessary to cast results like Proposition 3.29).

We insist on phrasing our basic definition in a generic monoidal category as much as possible, as opposed to working in a *Cartesian* monoidal category; this is a design choice that requires explanation, because, to the best of our knowledge, such a general perspective has been articulated neither in the existing literature on Rosen’s programme [Ros64a, Ros64b, Ros66a] nor in subsequent categorical approaches to theoretical biology [LSAGA⁺06, Nom01, Lou17].

The lack of attention to what we believe is the natural setting for (M, R) -systems can probably be explained by a certain distance between communities, and a lack of awareness that ‘resource-aware logics’ exist; this is, however, taken for granted in mathematical logic and computer science [Gir87, LS86]. Indeed, *resource sensitivity* is the central organizing principle of *linear logic* as envisaged in [Gir87, See89]; the relevance of linear logic to mathematical biology appears to have gone largely unnoticed, a gap that we hope to start filling with the present work.

It should be noted that an approximation of this idea can be found in Rosen’s own writings [Ros64a, Ros12], where he refers to the possibility of working in a category **Rel** of sets and *relations* rather than sets and *functions*: however, Rosen seems to operate on the intuition that **Rel**, equipped with the tensor product defined in [See89], is indeed a model for linear logic.

Remark 3.2 (On Cartesian versus purely monoidal (M, R) -systems). The internal logic of a Cartesian category is a setting in which assumptions may be duplicated and discarded without restriction, via the ‘diagonal’ and ‘terminal’ maps,

$$A \xrightarrow{\Delta} A \times A \quad A \xrightarrow{!} 1 \quad (3.2)$$

In proof theoretic terms, contraction and weakening are built into the logic, so that resources are treated as freely reusable.² From this perspective, Cartesian logic may be regarded as a logic of unrestricted reproducibility [LS86, See89].

By contrast, the internal logic of a *symmetric monoidal* category is, in general, resource-sensitive. *Linear* logic makes this principle explicit by treating assumptions as resources that cannot, in general, be duplicated or discarded (unless what is called an ‘exponential modality’ is present). The logical formalism therefore keeps track of the consumption, preservation, and transformation of resources [Gir87, See89].

Biological systems are fundamentally constrained by the scarcity of resources, both in nature and in the realizability of their processes. Categorical frameworks with a resource-sensitive internal logic therefore appear more natural than Cartesian ones for describing such systems. Notwithstanding the above, we dedicate a brief subsection (subsection 3.2) to the study of (M, R) -systems in Cartesian categories, partly for the sake of completeness, partly because virtually all references on (M, R) -systems in the literature work in that setting, and partly because some results (see Proposition 3.53) are only valid in a Cartesian setting.

Remark 3.3. A further point concerning the philosophy behind the design of the theory: it is still a matter of debate what the *truly* minimal assumptions are under which a reasonably complete theory of (M, R) -systems can be cast; we believe the present work provides a relatively definite answer:

It is sufficient that C is either a monoidal category (where in (3.1) the repair map Φ is assumed to be uncurried), or a closed category (where Φ is curried) in the sense of Eilenberg-Kelly [EK66].

In particular, no symmetry, completeness or cocompleteness is strictly necessary for the theory of ‘bare’ (M, R) -systems. However, the theory of ‘purely closed’ categories is a bit unwieldy, due to the lack of symmetry for the tensor operation; thus, a ‘comfortable’ set of hypotheses may include symmetry, or at least a braiding (without which certain characterizations are still possible, but may become more convoluted), as well as a sufficient supply of limits and colimits in C (without which, certain equivalent representations of (M, R) -systems are not available).

One of the strongest reasons to see the Cartesian setting as a highly degenerate framework for the theory of (M, R) -systems is the following.

Remark 3.4. In §4 we will examine a stronger notion of *coherent* (M, R) -system, where we strengthen the structure of what here we dub the ‘repair’ map of the (M, R) -system into a natural transformation. We will promptly observe that there is a tension between two opposing desiderata: on one hand, one would like to have ‘natural repair maps’, producing a workable theory; on the other hand, one would certainly like to have access to compelling nontrivial examples. In a sense, there is an obstruction to having both: in the category of sets and functions, the only setting in which the notion of (M, R) -system has been cast so far [Ros64a, Ros64b, Ros66a, LSAGA⁺04, LSAGA⁺06, Lou17], an elementary use of the Yoneda lemma shows that coherent (M, R) -systems of shape

$$\Phi^\natural : C \begin{array}{c} \xrightarrow{id} \\ \Downarrow \\ \xrightarrow{[A, -]} \end{array} C \quad (3.3)$$

(so that Φ is simply recovered as $\Phi_B^\natural$) are almost completely uninteresting. A natural environment in which to study Cartesian coherent (M, R) -systems is, however, a nontrivial Grothendieck topos.

²From a more algebraic perspective, we refer the reader to the so-called *Fox theorem* [Fox76]: a monoidal category $(C, \otimes, I)$ is Cartesian if and only if every object $A \in C$ carries a *comonoid* structure

$$\Delta_A : A \rightarrow A \otimes A, \quad e_A : A \rightarrow I,$$

such that, moreover, every arrow $f : A \rightarrow B$ is a comonoid homomorphism (i.e. $\Delta_B \circ f = (f \otimes f) \circ \Delta_A$ and $e_B \circ f = e_A$); Δ and e are then respectively the diagonal and the unique map to the monoidal unit (which can then be proved to be a terminal object of C).

Once again, we stress that this remark has never been made before the present work; all the references investigating (M, R) -systems that we could consult work in a category of sets, interpreting $B \otimes A$ as a Cartesian product. In such a situation, there is just *one* natural transformation $\Phi^\natural$ of the above type, the components of which must all uncurry to projections $\pi_X : X \times A \rightarrow X$; see Lemma 4.2.

After this long series of preliminaries, we are ready to give the first definition of ‘incoherent’ (M, R) -system, in which we do not require the repair map to be natural in any sense. An extensive study of this notion will occupy the entire section.

Definition 3.5 (Incoherent (M, R) -system, $\mathcal{U}$). Let A, B be objects of C . An *incoherent (M, R) -system* (of type A, B) in C consists of a diagram of shape

$$A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]. \quad (3.4)$$

We denote an (M, R) -system as $\mathcal{M}$ and we destructure $\mathcal{M}$ as the pair $\langle f, \Phi \rangle$; a *morphism* $\mathcal{M} \rightarrow \mathcal{N}$ of incoherent (M, R) -systems, destructured as $(\ell, r) : \langle f, \Phi \rangle \rightarrow \langle g, \Psi \rangle$ consists of a pair of arrows in C ,

$$\ell : A \rightarrow A' \quad r : B \rightarrow B', \quad (3.5)$$

such that the equations

$$r \circ f = g \circ \ell \quad [\ell, B'] \circ \Psi \circ r = [A, r] \circ \Phi \quad (3.6)$$

hold; in diagrammatic terms, both diagrams in

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\ \ell \downarrow & & \downarrow r & & \downarrow [A, r] \\ & (i) & & (ii) & [A, B'] \\ & & & & \uparrow [\ell, B'] \\ A' & \xrightarrow{g} & B' & \xrightarrow{\Psi} & [A', B'] \end{array} \quad (3.7)$$

are commutative.

We denote by $\mathbf{iMRS}(A, B)$ (or $\mathbf{iMRS}_C(A, B)$ when there is risk of ambiguity) the collection of incoherent (M, R) -systems of type (A, B) .

Remark 3.6 ((M, R) -systems as process-repair pairs). An incoherent (M, R) -system can be thought of as consisting, first, of a process (or ‘metabolism function’) $f : A \rightarrow B$, carrying an input of type A to an output of type B , and, second, of a repair map $\Phi : B \rightarrow [A, B]$. The latter assigns to each state or product b a possible process $A \rightarrow B$: in the terminology developed from Rosen’s original (M, R) systems, Φ is a ‘selector’ or ‘replacement map’, rather than necessarily a material repair operation [Ros64a, Ros64b, Ros66a, LSAGA⁺04, LSAGA⁺06]. In this sense, in a more mathematical dialect, Φ provides a term of type f , internalized in C via the closed structure.

In the Cartesian setting used in Rosen’s early treatments, this is the familiar state-transition reading; the uncurried representation makes it evident that one can think of f as a generalized element of B , and Φ as an action on B , from which we can access the ‘iterates’

$$A \xrightarrow{f} B, \quad A \rightarrow A \times A \xrightarrow{f \times A} B \times A \xrightarrow{\Phi} B, \quad A \xrightarrow{(f \times A \times A) \circ \Delta} B \times A \times A \xrightarrow{\Phi \times A} B \times A \xrightarrow{\Phi} B, \dots \quad (3.8)$$

In a general monoidal category the tensor instead records that state and input are resources whose combination need not admit copying or deletion [Ros64a, Ros12, Gir87, See89].

The analogy was subsequently developed in automata-theoretic directions. Rosen recast (M, R) systems as sequential machines, making the output-dependent state transition explicit [Ros64a, Ros64b, Ros66a]; Arbib related systems with a fixed repair component to Mealy automata [Arb66], and Warner represented more general (M, R) systems through categories of automata [War82]. Later work revisited the same correspondence using communicating X -machines, arguing that ordinary finite-state machines are too weak to capture the self-referential structure [PWG16]. The

present term “process-repair datatype” keeps the common typed interface in view while postponing stronger requirements: in the next section, we ask when the repair assignment can itself vary coherently, rather than remaining a single, unconstrained map attached to one system.

Definition 3.7 (Total category of incoherent (M, R) -systems, $\mathcal{C}^{\text{inco}}$). The incoherent (M, R) -systems are respectively the objects and the arrows of the *total category* of incoherent (M, R) -systems, that we denote **iMRS** (or **iMRS**_C when there is risk of ambiguity).³ Evidently,

- the composition of arrows is performed componentwise and uses the bifactoriality property of $[-, -]$: given a composable pair of arrows $\mathcal{M} \rightarrow \mathcal{N} \rightarrow \mathcal{P}$, destructured as

$$\langle\langle f, \Phi \rangle\rangle \xrightarrow{(\ell, r)} \langle\langle g, \Psi \rangle\rangle \xrightarrow{(\ell', r')} \langle\langle h, \Theta \rangle\rangle, \quad (3.9)$$

the composite $(\ell', r') \circ (\ell, r)$ is $(\ell' \circ \ell, r' \circ r)$ and it is such that

$$\begin{aligned} [\ell' \circ \ell, id] \circ \Theta \circ r' \circ r &= [\ell, id] \circ [\ell', id] \circ \Theta \circ r' \circ r && \text{(hom-functor)} \\ &= [\ell, id] \circ [id, r'] \circ \Psi \circ r && \text{((ii)-comm)} \\ &= ([id, r'] \circ [\ell, id]) \circ \Psi \circ r && \text{(hom-bifunctor)} \\ &= [id, r'] \circ [id, r] \circ \Phi && \text{((ii)-comm)} \\ &= [id, r' \circ r] \circ \Phi. && \text{(hom-functor)} \end{aligned}$$

- The identity $\mathcal{M} \rightarrow \mathcal{M}$ for this composition rule is the pair $(id_A, id_B) : \langle\langle f, \Phi \rangle\rangle \rightarrow \langle\langle f, \Phi \rangle\rangle$.

Remark 3.8. There is an equivalent presentation of the total category **iMRS** of incoherent (M, R) -systems, allowed by the freedom to give the repair map Φ in curried form ($\Phi : B \rightarrow [A, B]$) or in uncurried form ($\Phi : B \otimes A \rightarrow B$); either characterization is more convenient for certain purposes, and less convenient for others. We will not be particularly careful in distinguishing these two presentations of **iMRS** (see for example Proposition 3.36 below, where it is more convenient to argue in uncurried form).

For the sake of clarity, let’s record how (3.6).ii changes for incoherent (M, R) -systems in uncurried form: the second equation in (3.6) is replaced by the simpler

$$\begin{array}{ccc} B \otimes A & \xrightarrow{\Phi'} & B \\ r \otimes A \downarrow & \searrow r \otimes \ell & \downarrow r \\ B' \otimes A & \xrightarrow{B' \otimes \ell} B' \otimes A' \xrightarrow{\Psi'} & B' \end{array} \quad (3.10)$$

(or in equational form, $r \circ \Phi' = \Psi' \circ r \otimes \ell$, where clearly $r \otimes \ell$ is the common value $(B' \otimes \ell) \circ (r \otimes A) = (r \otimes A') \circ (B \otimes \ell)$ given by bifactoriality of the tensor operation).

We collect several examples of incoherent (M, R) -systems in various categories, to illustrate the generality of the notion.

Example 3.9. In the category of vector spaces, one can consider an (M, R) -system where $B = k$ is the ground field, and find that a diagram

$$A \xrightarrow{\alpha} k \xrightarrow{\beta} [A, k] \quad (3.11)$$

consists of a pair of k -linear maps $\alpha : A \rightarrow k, \beta : k \rightarrow [A, k]$; clearly, the second map amounts to another $\beta : A \rightarrow k$, due to the isomorphism $\mathbf{Vect}(k, [A, k]) \cong \mathbf{Vect}(A, [k, k]) \cong \mathbf{Vect}(A, k)$, thus giving a pair $\alpha, \beta : A \rightarrow k$ of covectors of A ; the tensor product $\alpha \otimes \beta : A \otimes A \rightarrow k \otimes k \cong k$, which coincides with the uncurrying of $\beta \circ \alpha$, now amounts to a bilinear pairing on A , and one can recover all (M, R) -systems of type (3.11) as all and only the pairings $A \otimes A \rightarrow k$ that are pure tensors in that they arise as Kronecker products of the form $\alpha \otimes \beta$.

³At the moment of writing, we have found only one circumstance in which considering more than one **iMRS**_C at once is necessary: it is the straightforward observation, for which however we do not have any practical use at the moment, that the assignment $C \mapsto \mathbf{iMRS}_C$ is, in the notation of the above definition, a pseudofunctor $\mathbf{CIMonCat}_{\text{lax}} \rightarrow \mathbf{Cat}$, where we call ‘**CIMonCat**_{lax}’ the 2-category of closed monoidal categories, and lax monoidal functors, see [ML98, §XI.2].

Example 3.10. A similar computation can be performed in the categories **Rel** of relations and $\vee \mathbf{Lat}$ of join semilattices; take $B = 1$, the singleton set that is the monoidal unit of **Rel**; then an (M, R) -system $\langle\langle \alpha, \beta \rangle\rangle$ consists of a pair of relations of type $\alpha : A \rightarrow 1$, $\beta : 1 \rightarrow [A, 1]$ i.e., since **Rel** is compact closed with $A^* \cong A$ and hence $[A, 1] \cong A$, a pair of subsets $\alpha, \beta \subseteq A$; the tensor $\alpha \otimes \beta \subseteq A \times A$, which coincides with the uncurrying of $\beta \circ \alpha$, is exactly the product relation $\alpha \times \beta$, so, as in 3.3.9, every such (M, R) -system is a ‘pure tensor’, now in the sense of a Cartesian product of two subsets of A . The identical computation, with 1 replaced by the two-element lattice $\mathbf{2} = \{\perp < \top\}$ (the monoidal unit of $\vee \mathbf{Lat}$), applies verbatim in $\vee \mathbf{Lat}$.

More generally, we can reproduce this reasoning in every compact closed category; we refrain from doing so in full detail, but the reader will be able to fill in the argument by abstracting from the examples of (finite-dimensional) vector spaces, **Rel** and $\vee \mathbf{Lat}$.

Example 3.11. Recall that a *quantale* [Yet90, Ros90] is a suplattice that, *qua* category, is monoidal (bi)closed. It is then equipped with a monotone monoid operation $*$: $Q \times Q \rightarrow Q$, so that there is a Galois connection

$$x * y \leq z \iff x \leq y \multimap z \iff y \leq x \multimap z, \quad (3.12)$$

where the existence of arbitrary suprema ensures that the adjoint functor theorem holds in the form

$$y \multimap z := \bigvee \{t \mid t * y \leq z\} \quad x \multimap z := \bigvee \{t \mid x * t \leq z\} \quad (3.13)$$

In a quantale Q , an (M, R) -system consists of a chain of inequalities $a \leq b \leq a \multimap b$, which uncurries into the simpler ‘right action’ condition $b * a \leq b$. In all integral quantales (where the top element and the monoid unit for $*$ coincide, cf. [Ros90]) like the Booleans, more generally all Heyting algebras, the suplattice of ideals of a commutative ring R , **Cost** := $([0, \infty], \geq, +, 0)$, or **Ult** := $([0, \infty], \geq, \max, 0)$, the inclusion $b \leq a \multimap b$ is always satisfied, so an (M, R) -system consists simply of an inequality $a \leq b$. In non-integral quantales like the free quantale 2^{S^*} (the powerset of the free monoid on a set S of ‘letters’), the condition $b \leq a \multimap b$ is not vacuous: the inequality $b \leq a \multimap b$ says precisely that b is closed under right concatenation by any word of a , i.e. $b \cdot a \subseteq b$, recovering the right-action condition $b * a \leq b$ directly rather than as an automatic consequence of integrality. For instance, fixing a single letter $s \in S$ and taking $a = \{s\}$, any language b closed under appending s , such as $b = S^*s$ (the words ending in s) or $b = \{s\}^*$ (the words consisting of repeated copies of s), gives a genuine (M, R) -system $a \leq b \leq a \multimap b$ that is not forced by the ambient order alone, in contrast with the automatic inclusion holding for every pair in the integral case.

Example 3.12. In any monoidal closed category C admitting countably infinite coproducts, one can uncurry the repair arrow Φ of an (M, R) -system $\langle\langle f, \Phi \rangle\rangle = (A \rightarrow B, B \rightarrow [A, B])$ into an arrow $B \otimes A \rightarrow B$, which extends into a unique right action $B \otimes A^* \rightarrow B$ of the free monoid $A^* = \sum_{n \in \mathbb{N}} A^n$ on B . In particular, in the category of vector spaces (or more generally in R -modules over a ring R , or more generally in any abelian category), it is a reasonable question to ask what an (M, R) -system that is in addition a *complex* of maps looks like, i.e. such that the composite $\Phi \circ f : A \rightarrow B \rightarrow [A, B]$ is the zero map. In such a situation, Φ reduces to a right action $B \otimes A^* \rightarrow B : (b, \underline{a}) \mapsto b \cdot \underline{a}$, with the property that the image $f(A) \subset B$ is contained in the annihilator

$$\text{Ann}(\Phi) := \{b \in B \mid \forall \underline{a} \in A^*, b \cdot \underline{a} = 0\}. \quad (3.14)$$

We are now ready to collect more lengthy examples.

Remark 3.13 ((M, R)-systems in species, with convolution). A particularly rich example of (M, R) -system in a monoidally cocomplete category is the following; consider the category **Spc** of Joyal’s *combinatorial species* [Joy86, BLL98], i.e. the functor category **Bij** $\rightarrow$ **Set** where **Bij** is the groupoid core of finite sets; among all species, the *representable* ones, denoted $y(n)$, are ‘concentrated on n -sets’ in the sense that $y(n)[S] \neq \emptyset$ if and only if S has n elements, and in that case it is a singleton.

Equipped with *Day convolution*, see [Day70a, Day70b],

$$(F \otimes G)[S] = \bigsqcup_{S=S_1 \sqcup S_2} F[S_1] \times G[S_2], \quad (3.15)$$

the category **Spc** is symmetric monoidal closed, with unit the species $y(0)$ concentrated on the empty set; the internal hom $\{F, G\}$ is defined by $\{F, G\}[S] = \mathbf{Spc}(F, G[S \sqcup _])$, in such a way as to represent the functor $\mathbf{Spc}(_ \otimes F, G)$.

For a species A , we write A^* to denote the free monoid on A in $(\mathbf{Spc}, \otimes)$, i.e. $A^* := \sum_{n \geq 0} A^{\otimes n}$ is the countable coproduct of iterated Day convolutions of A with itself.

By the universal property of free monoids, for any species B an action of A^* on B , i.e. a right module structure $B \otimes A^* \rightarrow B$, is equivalent to a single map $\varphi : B \otimes A \rightarrow B$, resulting from uncurrying the repair function Φ of an incoherent (M, R) -system $\langle f, \Phi \rangle$.

For example, if $A = y(1)$ is the representable at 1, an (M, R) -system of the form

$$b_0 : y(1) \rightarrow B, \quad B \otimes y(1)^* \rightarrow B \quad (3.16)$$

carries no more information than the pair

$$b_0 \in B[1], \quad \varphi : B \otimes L \rightarrow B, \quad (3.17)$$

where $L = \sum_{n \geq 0} y(1)^{\otimes n} = \sum_{n \geq 0} y(n)$ is the species of *linear orders*, see [BLL98, p. 10].

Thus, an incoherent (M, R) -system $\langle f, \Phi \rangle$ based at $(y(1), B)$ in the category **Spc** consists of a pointed L -module B , and more generally, an incoherent (M, R) -system $\langle f, \Phi \rangle$ based at (A, B) consists of a pointed A^* -module B .

More can be said for an incoherent (M, R) -system $\langle f, \Phi \rangle$ based at $(y(1), B)$, given that the repair function $\Phi : B \rightarrow \{y(1), B\}$ has a quite direct combinatorial interpretation: recall from [BLL98, §1.4] that $\{y(1), B\}$ is the *species derivative* of B , i.e. the species ∂B acting on a set S as $B[S \sqcup 1]$.⁴

Given all this, Φ is equivalent to the datum of a ∂ -coalgebra having components $\Phi_{[n]} : B[n] \rightarrow B[n \sqcup 1]$ where $[n]$ is a shorthand for an n -element set; as such, taking also into account that B is ‘pointed’ by $b_0 : y(1) \rightarrow B$, we can map b_0 to $b_1 := \Phi_{[1]}(b_0) \in B[2]$, which then can be mapped to $b_2 := \Phi_{[2]}(b_1) = \Phi_{[2]}(\Phi_{[1]}(b_0))$, etc.

This is a ‘dynamical system’ in the ‘space’ B , in the sense that follows. To each species B one can associate its *category of elements* ∇B , having objects the pairs (S, x) , where S is a finite set and $x \in B[S]$, and morphisms $\sigma : (S, x) \rightarrow (T, y)$ the bijections $\sigma : S \rightarrow T$ such that $B[\sigma](x) = y$; there is a canonical forgetful functor $\pi_B : \nabla B \rightarrow \mathbf{Bij}$ which is a discrete opfibration, with fibre $\pi_B^{-1}([n]) = B[n]$, and allows one to recover the species B under this identification; thus, in light of the equivalence between functors $\mathbf{Bij} \rightarrow \mathbf{Set}$ and discrete opfibrations over $\mathbf{Bij}$, the total space ∇B is precisely the ‘bundle’ obtained by gluing the fibres $B[n]$ along the bijections that act on them.

Now, given an incoherent (M, R) -system $\langle f, \Phi \rangle$, iterating Φ from b_0 traces out a sequence

$$\begin{array}{ccccccccccc} B[1] & & B[2] & & B[3] & & B[4] & & \dots & & B[n+1] & & \dots \\ b_0 \longmapsto & b_1 \longmapsto & b_2 \longmapsto & b_3 \longmapsto & \dots \longmapsto & b_n & & \dots \end{array} \quad (3.18)$$

that can be seen as an orbit in the ‘total space’ ∇B : the generator of $L = y(1)^*$ acts on the ‘seed’ b_0 , driving it stepwise through the successive fibres of ∇B .

Clearly, the two descriptions of an incoherent (M, R) -system based at $(y(1), B)$ given above, as a module $\varphi : B \otimes L \rightarrow B$ and as a coalgebra $\Phi : B \rightarrow \partial B$, are Curry-dual to one another, and the orbit $(b_n)_{n \geq 0}$ obtained by iterating Φ coincides with the trajectory of the semiautomaton φ started at b_0 , as well as with the essentially unique factorization of b_0 through the free construction on A .

The same ‘trace semantics’ for an (M, R) -system holds for arbitrary generating species A , and it once again becomes cleanest in the language of fibrations.

The Day convolution of species $B \otimes A$ has a total category $\nabla(B \otimes A)$, such that an object over S is a decomposition $S = S' \sqcup S''$ into disjoint subsets, together with an object of ∇B over S' and one of ∇A over S'' , so φ becomes a fibred extension map, attaching an A -structured piece to a B -structured piece along a disjoint union of supports. Similarly, an object of $\nabla(A^*)$ over a finite set U is an ordered decomposition $U = S_1 \sqcup \dots \sqcup S_k$ together with elements $a_i \in A[S_i]$: a finite word of disjoint A -blocks covering U . The pointed module map $\psi : A^* \otimes y(1) \rightarrow B$ extending b_0 is then, under the Grothendieck construction, a morphism of discrete opfibrations over $\mathbf{Bij}$,

$$\nabla \psi : \nabla(A^* \otimes y(1)) \longrightarrow \nabla B, \quad (3.19)$$

⁴Adjoint representations of this kind, for algebras and coalgebras arising in **Spc**, have been studied to some extent in [Lor25a].

sending an object over $U \sqcup \{1\}$, i.e. a word of A -blocks anchored at the seed 1, to the point of $B[U \sqcup \{1\}]$ obtained by consuming the blocks of the word in order, starting from b_0 .

In this sense every generator a of A^* leaves a *trace* in the total space ∇B , obtained by looking at the orbit of b_0 : the action φ is nothing but a fibrewise map from the total space of words in A anchored at the seed into the total space of B , and the fibre of $\nabla(A^* \otimes y(1))$ over $U \sqcup \{1\}$ indexes precisely the paths through ∇B , starting at the point over $\{1\}$ determined by b_0 , that this map records.

Example 3.14 ((M,R)-systems in species, with substitution). The category **Spc** carries a second monoidal structure, the *substitution* product of [BLL98, §1.4],

$$(F \circ G)[S] = \coprod_{\pi \vdash S} F[\pi] \times \prod_{P \in \pi} G[P] \cong \int^{T \in \mathbf{Bij}} F[T] \times G^{\otimes T}[S], \quad (3.20)$$

where π ranges over the partitions of S and $G^{\otimes T}$ is the T -indexed convolution power of Remark 3.13 (the two descriptions agree as soon as $G[\emptyset] = \emptyset$, which we assume throughout). Its unit is $y(1)$, and not the $y(0)$ of the convolution structure. Substitution is not symmetric, nor braided, and it is closed on one side only: $_ \circ G$ is cocontinuous, being a coend of copies of its argument, whereas $F \circ _$ preserves no colimit in general. The internal hom is defined as

$$[A, B][T] \cong \mathbf{Spc}(A^{\otimes T}, B), \quad (3.21)$$

so that an incoherent (M,R)-system $\langle f, \Phi \rangle$ based at (A, B) uncurries to a pair $f : A \rightarrow B$, $\varphi : B \circ A \rightarrow B$.

In particular, φ exhibits B as a *right module over an operad* in the sense of [Fre09]: an (M,R)-system based at (A, B) is a right module over the free operad $\mathcal{F}(A)$, pointed by f . The free monoid is no longer $\sum_{n \geq 0} A^{\circ n}$, since that formula needs the tensor to preserve countable coproducts in *both* variables and substitution does not; it is instead the free operad, whose value on S is the set of rooted trees with leaves labelled by S and each node of arity k decorated by an element of $A[k]$; words are replaced by trees.

The comparison with Remark 3.13 is sharpest at $A = y(1)$, which was there the interesting case and is here the degenerate one: $y(1)$ has become the monoidal unit, so $[y(1), B] \cong B$, the repair map is a bare endomorphism of the species B , and the free monoid on it is $\sum_{n \geq 0} y(1)$, a single unary operation iterated. Nothing leaves the fibre it starts in, whereas under convolution the same choice produced the derivative ∂B and a seed travelling through $B[1], B[2], B[3], \dots$. The first nondegenerate case is $A = y(2)$, for which $A^{\otimes T}$ is the species of perfect matchings indexed by T , whence

$$[y(2), B][T] \cong B[T \times 2]^{(\mathbb{Z}/2)^T} : \quad (3.22)$$

the B -structures on a doubled support that are invariant under the swap inside each pair. Iterating Φ from the seed by $c_1 := f$ and $c_{n+1} := \varphi \circ (c_n \circ A) : A^{\circ(n+1)} \rightarrow B$ then doubles the support at every step, 2, 4, 8, $\dots$, against the +1 of the derivative; the trace of a seed is an A -decorated tree in ∇B , not a path. In fibred terms, an object of $\nabla(B \circ A)$ over S is a partition of S together with a B -structure on the set of its blocks and an A -structure on each block, and φ is the fibred contraction of such a configuration, where $\nabla(B \otimes A)$ recorded only a two-block decomposition.

Example 3.15 ((M,R)-systems in $\mathbf{Chu}(\mathbf{Set}, k)$). Let k be a set, interpreted as a space of truth values for propositions; a *Chu space* [Bar91, Pra99] over a set k is a triple $A = (|A|, A^\bullet, e_A)$ consisting of a set of *points* $a \in |A|$, a set of *tests* $q \in A^\bullet$, and a pairing $e_A : |A| \times A^\bullet \rightarrow k$ that assigns to each pair (a, q) an ‘answer’ to the question ‘is a such that q ?’.

A morphism $A \rightarrow B$ of Chu spaces is a pair $(r_! : |A| \rightarrow |B|, r^\bullet : B^\bullet \rightarrow A^\bullet)$ such that $e_B(r_!(a), \beta) = e_A(a, r^\bullet(\beta))$ for all $a \in |A|, \beta \in B^\bullet$; these compose in the evident way, yielding the category $\mathbf{Chu}(\mathbf{Set}, k)$ (or more briefly, $\mathbf{Chu}_k$).

The category $\mathbf{Chu}_k$ is symmetric monoidal closed: $|A \otimes B| = |A| \times |B|$, and the internal hom $A \multimap B$ has $|A \multimap B| = \mathbf{Chu}_k(A, B)$, with test set $(A \multimap B)^\bullet = |A| \times B^\bullet$ and pairing $\langle t, (a, \beta) \rangle = e_B(t_!(a), \beta)$, representing $\mathbf{Chu}_k(_ \otimes A, B)$.

Unlike the examples in 3.3.10 above, $\mathbf{Chu}_k$ is not compact closed for $|K| > 2$: the internal hom $A \multimap B$ is in general not isomorphic to $A^\perp \otimes B$, and $\multimap$ is in fact the ‘cofree constructor for linear implication’, see [Bar91, See89, Gir87].

We now embark on the task of studying (M,R) -systems $\langle f, \Phi \rangle$ based at (A, B) in $\mathbf{Chu}_k$, i.e. a diagram of shape

$$A \xrightarrow{(f, f^\bullet)} B \xrightarrow{(\Phi, \Phi^\bullet)} (A \multimap B). \quad (3.23)$$

Unfolding the definition of the repair arrow Φ , one sees that it assigns to every point $b \in |B|$ a Chu-transform $\Phi_!(b) : A \rightarrow B$ (in turn splitting as a pair $(\Phi_!(b)_!, \Phi_!(b)^\bullet)$), together with a backward map $\Phi^\bullet : |A| \times B^\bullet \rightarrow B^\bullet$ satisfying $e_B(b, \Phi^\bullet(a, \beta)) = e_B(\Phi_!(b)_!(a), \beta)$.

This condition says that testing b against the composite test built from a resource $a \in |A|$ and a B -test β is the same as running b 's repair transform on a and testing the result with β : the observable behaviour of b already fixes what $\Phi_!(b)$ does to every input, with no further datum available.

To see this data in a fully worked instance, let $k = \{0 < \frac{1}{2} < 1\}$; this is an algebraic lattice where conjunction gives a pairing $k \times k \rightarrow k$ and thus a Chu space $K = (k, k, \min)$, modelling Łukasiewicz/Kleene three-valued logic, see [CDM00]. Let $A = B = K$. For each $c \in k$, we can define an (M,R) -system based at (K, K) taking as components of the metabolism $\varphi : K \rightarrow K$ the maps $(\varphi_c)_! = \varphi_c^\bullet = \min(_, c)$; evidently

$$\min((\varphi_c)_!(x), y) = \min(x, c, y) = \min(x, (\varphi_c)^\bullet(y)), \quad (3.24)$$

by associativity and commutativity of $\min$, so we have really defined a Chu transformation. Note also that $\varphi_c \circ \varphi_{c'} = \varphi_{\min(c, c')}$.

We thus fix $c \in k$ and take the metabolism map to be $f = \varphi_c$; we define the repair map $\Phi_! : k \rightarrow \mathbf{Chu}_k(K, K)$ by

$$\Phi_!(b) = \varphi_b \circ \varphi_c = \varphi_{\min(b, c)}, \quad \Phi^\bullet(a, \beta) = \min(a, c, \beta). \quad (3.25)$$

This is a valid Chu-transform $K \rightarrow (K \multimap K)$: for every $b, a, \beta \in k$,

$$e_B(b, \Phi^\bullet(a, \beta)) = \min(b, \min(a, c, \beta)) = \min(\varphi_{\min(b, c)}(a), \beta) = e_B(\Phi_!(b)_!(a), \beta), \quad (3.26)$$

so $\langle f, \Phi \rangle$ is a well-defined incoherent (M,R) -system based at (K, K) .

Currying the composite $A \xrightarrow{f} B \xrightarrow{\Phi} (A \multimap B)$ gives, for $a_1, a_2 \in k$,

$$(a_1, a_2) \mapsto \Phi_!(f_!(a_1))_!(a_2) = \varphi_{\min(a_1, c)}(a_2) = \min(a_1, a_2, c). \quad (3.27)$$

Thus the composite computes the ternary meet $\min(a_1, a_2, c)$: the two, non-duplicated, uses of the resource A are combined by the self-application map into a single Łukasiewicz conjunction of both readings together with the ambient threshold c : repair succeeds to exactly the degree that both uses of the resource, and the intrinsic reliability ceiling c , hold jointly. In particular the construction degenerates neither to the metabolism map nor to the identity, and it makes visible, in closed form, the bilinear character that a Cartesian reading of the same diagram would have obscured behind a diagonal $A \rightarrow A \otimes A$. Nothing in this computation is special to the three-valued chain k : for any complete lattice Q , setting $K = (Q, Q, \wedge)$ and $\varphi_c = _ \wedge c$ (using associativity, commutativity, and idempotency of meet in place of those of $\min$) reproduces verbatim the same (M,R) -system and the same currying identity $(a_1, a_2) \mapsto a_1 \wedge a_2 \wedge c$; the interpretation of f and Φ is unchanged: f still tests a resource against the threshold c , and Φ still assigns to each state b the ‘re-thresholded’ transform $\varphi_{b \wedge c}$.

A genuinely compelling instance of the vector-space case is furnished by Hilbert spaces: fixing a (complex, separable) Hilbert space $\mathcal{H}$, Abramsky [Abr10] represents the associated quantum system as the Chu space $K_{\mathcal{H}} = (\mathcal{H}_o, L(\mathcal{H}), e_{\mathcal{H}})$, where $\mathcal{H}_o$ is the set of non-zero vectors, $L(\mathcal{H})$ is the complete ortholattice of closed subspaces of $\mathcal{H}$, and $e_{\mathcal{H}}(\psi, S) = \|P_S \psi\|^2 / \|\psi\|^2$ is the Born-rule probability that a measurement of S succeeds on state ψ ; remarkably, the Chu-transforms $K_{\mathcal{H}} \rightarrow K_{\mathcal{H}}$ are exactly the unitary and antiunitary operators on $\mathcal{H}$, recovering Wigner’s theorem on physical symmetries. Building on this object (though departing from Abramsky’s probabilistic pairing, whose asymmetry makes a state-dependent repair map considerably more delicate to construct), the lattice $L(\mathcal{H})$ itself carries a Chu structure of the kind discussed above: taking $K := (L(\mathcal{H}), L(\mathcal{H}), \wedge)$, with $\wedge$ the (associative, commutative, idempotent) intersection of closed subspaces, an (M,R) -system based at K is an instance of the general complete-lattice construction described above, with $Q = L(\mathcal{H})$: fixing a closed subspace C playing the role of the threshold c ,

the metabolism $f = \varphi_C := (- \wedge C)$ and repair $\Phi_!(B) := \varphi_{B \wedge C}$ define an incoherent (M, R) -system whose currying $(S_1, S_2) \mapsto S_1 \wedge S_2 \wedge C$ computes the joint intersection of two tested subspaces together with a fixed reference subspace: the quantum-logical analogue, for closed subspaces rather than truth values, of the Łukasiewicz example above. Unlike the Boolean or Łukasiewicz cases, $L(\mathcal{H})$ is not distributive, so this instance genuinely lives outside the world of Heyting algebras while still fitting the general lattice-theoretic pattern.

Example 3.16. Let $\mathbf{N}$ denote the set of natural numbers; we regard $\mathbf{N}$ as a discrete category, equipped with a monoidal structure given by addition.

Recall that the category $\mathbf{Set}^{\mathbf{N}}$ of $\mathbf{N}$ -graded sets can be defined as the functor category $\mathbf{Cat}(\mathbf{N}, \mathbf{Set})$, equipped with the monoidal closed structure coming from Day convolution; it is a natural question whether an intuition similar to the one in Remark 3.13 holds in $\mathbf{Set}^{\mathbf{N}}$ as well. Indeed, this is the case, as the category has a similar ‘tensor-by-[1]’ and ‘derivative’ adjunction; one can easily compute that an incoherent (M, R) -system in $\mathbf{Set}^{\mathbf{N}}$ consists of the following data:

- a ‘process’ given by $\{f_n : A_n \rightarrow B_n \mid n \in \mathbf{N}\}$;
- a ‘dynamics’ given by $\{\Phi_n : (B \otimes A)_n \rightarrow B_n\}$.

The latter is in particular equivalent to a point $b \in B_1$ and a family of functions $B_1 \rightarrow B_2 \rightarrow \dots$ when $A = y(1)$ is the family given by $y(1)_n = \{\bullet\}$ if $n = 1$ and empty otherwise.

All in all, a similar line of reasoning works whenever one considers categories of presheaves over bases that are ‘graded’ in a suitable sense; among many, the case of single-sorted PROPs [ML65] (symmetric monoidal categories where the monoid of objects is isomorphic to $(\mathbf{N}, +, 0)$) is emblematic.

Example 3.17. Recall that the category of *pointed sets* consists of the coslice $1/\mathbf{Set}$: objects are pairs (X, x_0) made of a set X and a distinguished point $x_0 \in X$; arrows are basepoint-preserving functions, in the obvious sense.

Much as for the category of pointed sets, we can speak of the category of pointed topological spaces, or more generally of pointed objects in each category $\mathcal{C}$ that admits a terminal object 1 ; as long as $\mathcal{C}$ admits at least the pushouts that are necessary to give this definition, the category of such pointed objects is monoidal closed, with respect to a structure where

- the tensor is called the *smash product* of pointed objects: a pair $(X, x_0), (Y, y_0)$ of pointed objects is tensored into the object appearing in the pushout

$$\begin{array}{ccc} (X \times \{y_0\}) \cup (\{x_0\} \times Y) & \longrightarrow & X \times Y \\ \downarrow & & \downarrow \\ 1 & \longrightarrow & X \wedge Y, \end{array} \quad (3.28)$$

where the left vertical map collapses the *wedge* $X \vee Y := (X \times \{y_0\}) \cup (\{x_0\} \times Y) \subseteq X \times Y$ to a point, and the induced basepoint of $X \wedge Y$ is the image of 1 .

- Each functor $- \wedge Y$ (and thus each $X \wedge -$) admits a right adjoint $[Y, -]$, realizing the isomorphism

$$C_*((X \wedge Y, x_0 \wedge y_0), (Z, z_0)) \cong C_*((X, x_0), ([Y, Z], [y_0, z_0])); \quad (3.29)$$

the set of functions $Y \rightarrow Z$ is indeed pointed, by the function that is constant at the basepoint of Z , and this definition is enough to ensure that there is a bijection with the set of basepoint-preserving functions $(X \wedge Y, x_0 \wedge y_0) \rightarrow (Z, z_0)$.

It is natural to ask what an incoherent (M, R) -system in the category of pointed sets looks like; the category of pointed sets is exactly the Kleisli category of the maybe monad on $\mathbf{Set}$: a basepoint-preserving map $A \rightarrow B$ is the same datum as a partial function $A \rightarrow B$ (which is undefined if $f(t) = b_0$ is the basepoint). Consequently, an incoherent (M, R) -system $f : A \rightarrow B, \Phi : B \rightarrow [A, B]$ in the category of pointed sets is literally metabolism and repair implemented as partial functions allowed to fail: f may send a resource to ‘depleted’ (the basepoint), and Φ assigns to every state $b \in B$ a partial repair recipe that need not be defined on every input. This is a genuinely different flavour of resource-sensitivity from the one exhibited by the Chu-space examples above: there, every process comes with a

dual test that is always defined (an *observational* form of accountability), whereas here a repair process can simply fail to fire, rather than being required to answer every test, and fail if it doesn't pass one.

More generally, by a theorem of Kock [Koc70, Koc72], the Kleisli category of any *commutative* strong monad T on a Cartesian closed category inherits a *symmetric monoidal* structure, with $\mathbf{Set}_T$ recovering the maybe-monad case above as the simplest instance; commutativity of T is exactly the hypothesis needed to make this induced tensor well defined and symmetric. Kock's theorem stops there, however: it does not, by itself, make $\mathbf{Set}_T$ *closed*. Whether the Kleisli category of a commutative strong monad is monoidal closed is a considerably more delicate question in general, addressed only under further hypotheses on T by Keigher [Kei78] and, more recently and in the specific setting of algebraic λ -calculi, by Valiron [Val13]; it does not follow from commutativity alone. The two instances below are closed for an independent reason, namely that each resulting Kleisli category happens to be a well-known compact closed category in its own right, quite apart from any general Kleisli-closure theorem. Taking $T = \mathcal{P}$, the (covariant) powerset monad, $\mathbf{Set}_{\mathcal{P}}$ is exactly the category $\mathbf{Rel}$ of 3.3.10, compact closed with $[A, B] \cong A \times B$: an incoherent (M, R) -system there is precisely the pair of subsets described above, now read as a 'resource is present/absent' pairing. Taking instead $T = \mathcal{M}$, the finite-multiset (free commutative monoid) monad, $\mathbf{Set}_{\mathcal{M}}$ restricted to finite sets is the likewise compact closed category $\mathbf{Mat}_{\mathbb{N}}$ of finite sets and $\mathbb{N}$ -valued matrices under composition by matrix multiplication over the semiring $(\mathbb{N}, +, \times)$, again with $[A, B] \cong A \times B$. Taking $B = 1$ as in 3.3.10, an incoherent (M, R) -system there consists of a pair $\alpha, \beta : A \rightarrow \mathbb{N}$ of finitely-supported multiplicity functions, and the tensor $\alpha \otimes \beta : A \times A \rightarrow \mathbb{N}$, $(a, a') \mapsto \alpha(a)\beta(a')$, records the number of ways of jointly witnessing one copy from α and one from β ; this refines the merely Boolean, existence-only reading of $\mathbf{Rel}$ to an honest count of resource copies, closer in spirit to counting distinct molecules of a metabolite than to recording their mere presence.

Example 3.18 ((M, R)-systems in 'funny' categories). Recall that, for categories A, B , the *funny tensor product* $A \boxtimes B$ [FLK80] has the same objects as the Cartesian product $A \times B$, whereas its arrows are freely generated by pairs $(f, 1_b) : (a, b) \rightarrow (a', b)$ for $f : a \rightarrow a'$ in A , and $(1_a, g) : (a, b) \rightarrow (a, b')$ for $g : b \rightarrow b'$ in B , subject only to the relations that make each 'horizontal' slice $A \rightarrow A \boxtimes B$ (at a fixed object of B) and each 'vertical' slice $B \rightarrow A \boxtimes B$ (at a fixed object of A) into functors, crucially without imposing the interchange relation $(1, g) \circ (f, 1) = (f, 1) \circ (1, g)$ that holds automatically in the Cartesian product $A \times B$. The category $\mathbf{Cat}$, equipped with $\boxtimes$, is monoidal closed, with internal hom $[B, C]_u$ having as objects the functors $B \rightarrow C$, and as morphisms $F \Rightarrow G$ arbitrary object-indexed families of morphisms $F(b) \rightarrow G(b)$ in C : an 'unnatural transformation'.

An incoherent (M, R) -system $f : A \rightarrow B$, $\Phi : B \rightarrow [A, B]_u$ in $(\mathbf{Cat}, \otimes)$ assigns to every object $b \in B$ a whole functor $\Phi_b : A \rightarrow B$ (a full metabolic subsystem indexed by the state b), but as b varies along a morphism $b \rightarrow b'$ of B , the induced comparison between Φ_b and $\Phi_{b'}$ is not required to satisfy any coherence condition. From the point of view of the semantics, such a system corresponds to a $\mathcal{B}_0$ -indexed family of arrows $\Phi_X : X \rightarrow [A, X]$ able to reconstruct a process of any type; in particular, one of type $f : A \rightarrow B$, if given one.

Example 3.19. In this example we take C to be a small-complete category (i.e. a—large—category admitting all limits of diagrams with small domain), and consider the—even larger—category $[C, C]$ of endofunctors of C , (strict) monoidal under composition $F \otimes G := G \circ F$, with (strict) unit the identity functor id_C , and (right) internal hom $[F, G] := \mathrm{Ran}_F G$, the right Kan extension of G along F , which exists because C is complete.

This is precisely the monoidal category in which monads on C are the internal monoids, see [ML98, p. 138]: we now make the point that an incoherent (M, R) -system here is one step short of the classical theory of monads and their modules.

Concretely, an incoherent (M, R) -system consists of a natural transformation $f : A \Rightarrow B$ of endofunctors of C , together with $\Phi : B \Rightarrow \mathrm{Ran}_A B$; uncurrying gives an isomorphism $\mathrm{Nat}(B, \mathrm{Ran}_A B) \cong \mathrm{Nat}(B \circ A, B)$, so Φ transposes to a natural transformation $\mu : B \circ A \Rightarrow B$: this is the structure exhibiting B as a right module (algebra) for the endofunctor A , acting by precomposition. Currying

the self-application composite $A \circ A \xRightarrow{f \circ A} B \circ A \xRightarrow{\mu} B$ recovers a natural transformation $A \circ A \Rightarrow B$, the ‘two-resource’ action one would expect of an iterated repair.

The absence-of-a-diagonal phenomenon already discussed in Remark 3.2 recurs here in a sharp form: for a ‘duplication’ $A \Rightarrow A \circ A$ to exist, A must be a *comonad* on C .

Remark 3.20. The total category **iMRS** comes equipped with two functors

$$C \xleftarrow{[-]_A} \mathbf{iMRS} \xrightarrow{[-]_B} C \quad (3.30)$$

sending, respectively, the typical object $(A, B; \langle f, \Phi \rangle)$ to A and to B , and the typical arrow $(\ell, r) : (A, B; \langle f, \Phi \rangle) \rightarrow (A', B'; \langle g, \Psi \rangle)$ to ℓ and r .

Definition 3.21 (Fibres of **iMRS**, $\mathcal{U}$ and $\mathcal{V}$). Given objects $A, B \in C_0$, we can access the fibres over A and over B of the total category **iMRS**:

- The fibre $\mathbf{iMRS}^A$ has objects the (M,R)-systems with fixed input A and variable output B ; this means that the typical arrow $v : (B; \langle f, \Phi \rangle) \rightarrow (B'; \langle g, \Psi \rangle)$ consists of an arrow $v : B \rightarrow B'$ in C such that $v \circ f = g$ and $[A, v] \circ \Phi = \Psi \circ v$, i.e. such that the diagram

$$\begin{array}{ccccc} & & B & \xrightarrow{\Phi} & [A, B] \\ & f \nearrow & \downarrow v & & \downarrow [A, v] \\ A & & & & \\ & g \searrow & B' & \xrightarrow{\Psi} & [A, B'] \end{array} \quad (3.31)$$

commutes;

- The fibre $\mathbf{iMRS}_B$ has objects the (M,R)-systems with fixed output B and variable input A ; this means that the typical arrow $u : (A; \langle f, \Phi \rangle) \rightarrow (A'; \langle g, \Psi \rangle)$ consists of an arrow $u : A \rightarrow A'$ in C such that $g \circ u = f$ and $[u, B] \circ \Psi = \Phi$, i.e. such that the diagram

$$\begin{array}{ccccc} A & & & & [A, B] \\ & f \searrow & & \nearrow \Phi & \\ & & B & & \\ u \downarrow & & \nearrow g & & \downarrow [u, B] \\ A' & & & & [A', B] \end{array} \quad (3.32)$$

commutes.

Remark 3.22. The verification that these are indeed the fibres of the functors $[-]_A$ and $[-]_B$ is immediate;⁵ note that the discrete category ($=\text{set}$) $\mathbf{iMRS}(A, B)$ of Definition 3.5 arises as the fibre of the functor

$$\langle [-]_A, [-]_B \rangle : \mathbf{iMRS} \longrightarrow C \times C \quad (3.33)$$

induced thanks to the universal property of the product category (in order to avoid confusion with different meanings for the angled brackets, it is probably better to rename this functor to ‘ $[-]_{(A,B)}$ ’).

Proposition 3.23. The functor $[-]_A$ of (3.30) is a Grothendieck fibration.

Proof. It is sufficient to exhibit the reindexing functors

$$u^* : \mathbf{iMRS}^{A'} \longrightarrow \mathbf{iMRS}^A \quad (3.34)$$

induced by an arrow $u : A \rightarrow A'$ in the base C ; u^* acts on an object $\langle f, \Phi \rangle$, based at (A', B) , sending it to the incoherent (M,R)-system $\langle f \circ u, [u, B] \circ \Phi \rangle$ based at (A, B) ; that this assignment

⁵Recall that the *fibre* of a functor $p : \mathcal{E} \rightarrow \mathcal{B}$ consists of the subcategory of $\mathcal{E}$ defined as

$$p^{-1}B := \{p(f) = id_B\},$$

i.e. the subcategory spanned by the arrows that p sends to identities.

is functorial on morphisms $v : B \rightarrow B'$ follows from the fact that the diagram

$$\begin{array}{ccccc}
 A & \xrightarrow{f \circ u} & B & \xrightarrow{\Phi} & [A', B] \xrightarrow{[u, B]} [A, B] \\
 & \searrow & \downarrow \ddagger & \downarrow \ddagger & \downarrow [A', v] \\
 & A' & & & \\
 & \swarrow & \downarrow \ddagger & \downarrow \ddagger & \downarrow [A, v] \\
 A & \xrightarrow{f' \circ u} & B' & \xrightarrow{\Phi'} & [A', B'] \xrightarrow{[u, B']} [A, B']
 \end{array} \quad (3.35)$$

commutes in all its parts, using bifunctionality of $[-, -]$ and the assumption that $\ddagger$ and $\ddagger$ are commutative (since v is an arrow in $\mathbf{iMRS}^{A'}$). $\square$

Remark 3.24. On the other hand, due to the covariant-contravariant dependence on the parameter B in the type of an incoherent (M,R)-system, the functor $[-]_B$ is *neither* a fibration nor an opfibration; it only exhibits $\mathbf{iMRS}$ as a *displayed* category over C —which amounts to the fact that the dependence on B of the fibres $\mathbf{iMRS}_B$ is *profunctorial*, and not functorial; moreover, it is only functorial on the groupoid core of C .

Proof. There is nothing new to prove, since the (inverse) Grothendieck-Bénabou construction applied to the functor $[-]_B$ yields a lax normal functor $C \rightarrow \mathbf{Prof}$ sending B to $\mathbf{iMRS}_B$ and an arrow $v : B \rightarrow B'$ to a reindexing profunctor $v_* : \mathbf{iMRS}_B \rightarrow \mathbf{iMRS}_{B'}$; but we define the profunctor in detail for completeness. Let $v : B \rightarrow B'$ be an arrow in the base category C ; then, there is a profunctor $v_* : \mathbf{iMRS}_B^{\text{op}} \times \mathbf{iMRS}_{B'} \rightarrow \mathbf{Set}$ sending

- a pair of objects $(\langle f, \Phi \rangle, \langle f', \Phi' \rangle)$ to the set

$$v_*(\langle f, \Phi \rangle, \langle f', \Phi' \rangle) := \left\{ u : A \rightarrow A' \mid \begin{array}{c} v \circ f = f' \circ u \\ [id, v] \circ \Phi = [u, id] \circ \Phi' \circ v \end{array} \right\} \quad (3.36)$$

- a pair of morphisms $(s : \langle g, \Psi \rangle \rightarrow \langle f, \Phi \rangle, t : \langle f', \Phi' \rangle \rightarrow \langle g', \Psi' \rangle)$ in $\mathbf{iMRS}_B^{\text{op}} \times \mathbf{iMRS}_{B'}$ to the function

$$v_*(\langle f, \Phi \rangle, \langle f', \Phi' \rangle) \longrightarrow v_*(\langle g, \Psi \rangle, \langle g', \Psi' \rangle) \quad (3.37)$$

defined by sending the element $u : A \rightarrow A'$ in the domain to $t \circ u \circ s$; clearly $v \circ g = g' \circ t \circ u \circ s$, because

$$\begin{aligned}
 v \circ g &= (v \circ f) \circ s \\
 &= f' \circ (u \circ s) \\
 &= g' \circ (t \circ (u \circ s))
 \end{aligned}$$

and equally clearly $[id, v] \circ \Psi = [t \circ u \circ s, id] \circ \Psi' \circ v$, because

$$\begin{aligned}
 [id, v] \circ \Psi &= ([id, v] \circ [s, id]) \circ \Phi \\
 &= [s, id] \circ ([id, v] \circ \Phi) \\
 &= ([s, id] \circ [u, id]) \circ (\Phi' \circ v) \\
 &= [u \circ s, id] \circ (\Phi' \circ v) \\
 &= ([u \circ s, id] \circ [t, id]) \circ (\Psi' \circ v) \\
 &= [t \circ u \circ s, id] \circ (\Psi' \circ v)
 \end{aligned}$$

The identity and functoriality of this assignment are evident once unfolded, and this concludes the proof. $\square$

The discussion so far has established the following theorem:

Theorem 3.25. The total category of incoherent (M,R)-systems forms, with the functors of Remark 3.20, a *discrete span* (by which we mean a span $\mathcal{A} \xleftarrow{p} \mathcal{E} \xrightarrow{q} \mathcal{B}$ such that the induced functor $\langle p, q \rangle$ has discrete fibres), whose left leg is a Grothendieck fibration.

Remark 3.26 (On the iterated composites of **iMRS**). Considering the diagram of Remark 3.20 as an object of the monoidal category of spans ($\mathbf{Span}(C, C)$, $\circ$), one can construct the iterated composites

$$\mathbf{iMRS}^{(0)} := C \quad \mathbf{iMRS}^{(n+1)} := \mathbf{iMRS}^{(n)} \circ \mathbf{iMRS}; \quad (3.38)$$

we refer to $\mathbf{iMRS}^{(n)}$ as n -ary incoherent (M, R) -systems.

In particular $\mathbf{iMRS}^{(2)}$ is obtained as the pullback of $[-]_B$ and $[-]_A$ yielding the category having

- as objects the pairs $(\langle f, \Phi \rangle, \langle g, \Psi \rangle)$ of incoherent (M, R) -systems

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\ & & \parallel & & \\ & & X & \xrightarrow{g} & Y \xrightarrow{\Psi} [X, Y] \end{array} \quad (3.39)$$

such that $B = [f, \Phi]_B = [g, \Psi]_A = X$;

- as morphisms the pairs of morphisms $(\ell, r) : \langle f, \Phi \rangle \rightarrow \langle f', \Phi' \rangle$ and $(\ell', r') : \langle g, \Psi \rangle \rightarrow \langle g', \Psi' \rangle$ of incoherent (M, R) -systems such that $r = \ell'$ as arrows $B \rightarrow B'$ in the appropriate commutative diagram;

compositions and identities are inherited from those of the product category $\mathbf{iMRS} \times \mathbf{iMRS}$, of which $\mathbf{iMRS}^{(2)}$ is evidently a regular subobject.

Construction 3.27 (Sequential composition of incoherent (M, R) -systems, $\circ$). There exists a functor

$$- \circ - : \mathbf{iMRS}^{(2)} \longrightarrow \mathbf{iMRS} \quad (3.40)$$

defined as follows:

- on objects, the pair $(\langle f, \Phi \rangle, \langle g, \Psi \rangle)$ goes to the incoherent (M, R) -system based at (A, Y) and obtained as the composition

$$\langle f, \Phi \rangle \circ \langle g, \Psi \rangle := \langle g \circ f, f^* \Psi \rangle : A \xrightarrow{f} B \xrightarrow{g} Y \xrightarrow{\Psi} [B, Y] \xrightarrow{[f, Y]} [A, Y] \quad (3.41)$$

- on morphisms, one acts accordingly; a morphism of incoherent (M, R) -systems is such that the diagram marked in red here

$$\begin{array}{ccccc} & & X' & \xrightarrow{\Psi'} & [B', X'] \\ & \nearrow w & & \searrow [f', X'] & \\ X & & & & [A', X'] \\ \downarrow \Psi & & \swarrow [v, X'] & & \\ [B, X] & \xrightarrow{[B, w]} & [B, X'] & \xrightarrow{[f, X']} & [A, X'] \\ & \searrow [f, X] & & \nearrow [u, X'] & \\ & & [A, X] & \xrightarrow{[A, w]} & [A, X'] \end{array} \quad (3.42)$$

commutes, using only the bifunctionality of $[-, -]$.

Two structural properties of $- \circ -$ are worth recording explicitly. First, $- \circ -$ is strictly associative: for composable incoherent (M, R) -systems $\langle f, \Phi \rangle : A \rightarrow B$, $\langle g, \Psi \rangle : B \rightarrow Y$, $\langle h, \Xi \rangle : Y \rightarrow Z$, both

$$(\langle f, \Phi \rangle \circ \langle g, \Psi \rangle) \circ \langle h, \Xi \rangle \quad \text{and} \quad \langle f, \Phi \rangle \circ (\langle g, \Psi \rangle \circ \langle h, \Xi \rangle) \quad (3.43)$$

compute to $\langle h \circ g \circ f, [f, Z] \circ [g, Z] \circ \Xi \rangle$, using associativity of composition in C together with the identity $[g \circ f, Z] = [f, Z] \circ [g, Z]$ furnished by bifunctionality of $[-, -]$.

Second, and this line of reasoning will be expanded further in Proposition 3.51, $- \circ -$ is only *one-sidedly* unital. For any object A and *any* choice of repair map η , the system $\langle \text{id}_A, \eta \rangle$ based at (A, A) is a left unit: $\langle \text{id}_A, \eta \rangle \circ \langle g, \Psi \rangle = \langle g \circ \text{id}_A, [\text{id}_A, Y] \circ \Psi \rangle = \langle g, \Psi \rangle$, since the formula for $- \circ -$ never references the repair map of its left argument. By contrast, $\langle f, \Phi \rangle \circ \langle \text{id}_B, \eta \rangle = \langle f, [f, B] \circ \eta \rangle$, which recovers $\langle f, \Phi \rangle$ only for those systems whose repair map happens to factor as $\Phi = [f, B] \circ \eta$; no single η achieves this simultaneously for every $\langle f, \Phi \rangle$ based at B . Consequently,

$(C_0, \mathbf{iMRS}(-, -), - \circ -)$ falls short of being an ordinary category: it is associative but merely left-unital, a semigroupoid rather than a category. This is the same phenomenon, seen from a different angle, as the failure of $[-]_B$ to be a fibration recorded in Remark 3.24: the dependence of an incoherent (M, R) -system's repair map on B is profunctorial rather than functorial, and it is exactly this profunctoriality that $- \circ -$ cannot repair when composing on the right.

Remark 3.28. Continuing the induction, one can see that a ternary incoherent (M, R) -system consists of three incoherent (M, R) -systems $\langle f, \Phi \rangle, \langle g, \Psi \rangle, \langle h, \Xi \rangle$ arranged in a diagram of the following type:

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\
 & & \downarrow g & & \\
 & & Y & \xrightarrow{h} & Z \xrightarrow{\Xi} [Y, Z] \\
 & & \downarrow \Psi & & \\
 & & [B, Y] & &
 \end{array} \tag{3.44}$$

It is evident that from this diagram one can extract a 3-simplex (f, g, h) in the nerve of C , and that there is a canonical way to complete the composition $h \circ g \circ f$ to an incoherent (M, R) -system with repair function

$$Z \xrightarrow{\Xi} [Y, Z] \xrightarrow{[g \circ f, Z]} [A, Z]. \tag{3.45}$$

This sets up a functor $\mathbf{iMRS}^{(3)} \rightarrow \mathbf{iMRS}$ and the induction proceeds.

We now collect a number of equivalent presentations for incoherent (M, R) -systems; from time to time we will assume that the base category admits some limits and colimits. Note that some of these abstract characterizations will shed light on the results found in the above examples, in particular Remark 3.13 and Example 3.17.

Proposition 3.29 $((M, R)$ -systems at A as algebras, $\mathcal{U}^A$). For this example, assume that C admits binary coproducts; then (since each tensoring functor $B \otimes -$ commutes with all colimits), one can uncurry Φ and pair the arrows $(f, \hat{\Phi})$ in an incoherent (M, R) -system into a single map $[f, \hat{\Phi}] : A + B \otimes A \rightarrow B$, which in turn corresponds to

$$(I + B) \otimes A \xrightarrow{\sim} I \otimes A + B \otimes A \xrightarrow[\text{unit}^A + B \otimes A]{\sim} A + B \otimes A \xrightarrow{[f, \hat{\Phi}]} B. \tag{3.46}$$

This is the correspondence on objects of an equivalence of categories between $\mathbf{iMRS}^A$ and the category of algebras for the endofunctor

$$K_A = \lambda X. ((I + X) \otimes A) : C \longrightarrow C \tag{3.47}$$

Proof. The discussion above outlines the action on objects of a candidate functor

$$(-)^a : \mathbf{iMRS}^A \longrightarrow \mathbf{Alg}(K_A). \tag{3.48}$$

Clearly this is a bijection on objects whose inverse is defined by precomposition with the coproduct injections, and currying. It is now straightforward to check that a morphism $v : \langle f, \Phi \rangle \rightarrow \langle g, \Psi \rangle$ of incoherent (M, R) -systems based at A corresponds, under this assignment, to a K_A -algebra homomorphism $v^a : \langle f, \Phi \rangle^a \rightarrow \langle g, \Psi \rangle^a$, and bijectively so. The correspondence is built so that the two conditions in (3.6) coincide with the fact that the map $[f, \hat{\Phi}] : K_A B = A + B \otimes A \rightarrow B$ determined in (3.47) makes the square

$$\begin{array}{ccc}
 K_A B & \xrightarrow{(I+v) \otimes A} & K_A B' \\
 [f, \hat{\Phi}] \downarrow & & \downarrow [g, \hat{\Psi}] \\
 B & \xrightarrow{v} & B'
 \end{array} \tag{3.49}$$

commute: the line of reasoning goes in a way similar to this derivation, using the axioms, functoriality of hom , and the adjunction identities: a key lemma, which is very classical in the theory of

adjunctions, is that the left square in the diagram below commutes if and only if the right square commutes:

$$\begin{array}{ccc}
 B & \xrightarrow{\Phi} & [A, B] \\
 \downarrow v & & \downarrow [A, v] \\
 B' & \xrightarrow{\Psi} & [A, B']
 \end{array}
 \quad
 \begin{array}{ccc}
 B \otimes A & \xrightarrow{\hat{\Phi}} & B \\
 \downarrow v \otimes A & & \downarrow v \\
 B' \otimes A & \xrightarrow{\hat{\Psi}} & B'
 \end{array}
 \quad (3.50)$$

Thus, this comparison functor $(-)^a$ is an equivalence. $\square$

Remark 3.30 (Limits and colimits of A -based (M, R) -systems). The previous result comes in handy because the theory of endofunctor algebras provides powerful tools for establishing completeness and accessibility properties of $\mathbf{iMRS}^A$: assuming C is a locally presentable category [AR94, GU71] (thus accessible, complete, and cocomplete) we have that

- the category $\mathbf{iMRS}^A$ is complete, and limits are created in C by the forgetful functor $U = [-]_B|_{\mathbf{iMRS}^A}$;
- the functor U has a left adjoint yielding the free K_A -algebra on an object X , obtained by sending an object X of C to the initial algebra of the endofunctor $Y \mapsto X + K_A Y$, i.e. to the colimit of the free-algebra sequence of [Ad74],

$$X + K_A \emptyset \longrightarrow X + K_A(X + K_A \emptyset) \longrightarrow \dots \quad (3.51)$$

which is $A^* \otimes (1 + X)$;

- the category $\mathbf{iMRS}^A$ is accessible, because K_A is an accessible functor and accessible categories are closed under bilimits [MP89, §5.1]; thus, $\mathbf{iMRS}^A$ is locally presentable, hence also cocomplete; the initial K_A -algebra is then the colimit of the sequence

$$\emptyset \longrightarrow K_A \emptyset \longrightarrow K_A K_A \emptyset \longrightarrow K_A K_A K_A \emptyset \longrightarrow \dots \quad (3.52)$$

which is easily seen to be the free monoid on A , with structure maps

$$A \xrightarrow{(-)} A^* \quad A^* \xrightarrow{a \mapsto \lambda x.(a : x)} [A, A^*] \quad (3.53)$$

where we use the notation $\underline{a} \mapsto \lambda x.(\underline{a} : x)$, thinking of a λ -term, to denote the arrow $\sum_{n \geq 0} A^{\otimes(n+1)} \rightarrow \sum_{m \geq 0} A^{\otimes m}$ induced by $\iota_{n+1} : A^{\otimes(n+1)} \rightarrow A^*$.

Proposition 3.31 ((M, R)-systems at B as a slice category). In a very similar vein, we can characterize the fibres $\mathbf{iMRS}_B$ via a universal property. Assume that C is symmetric monoidal closed and that it admits binary products; then one can uncurry, swap the domain objects, and re-curry the repair arrow Φ of an incoherent (M, R) -system into a single map

$$A \longrightarrow B \times [B, B]. \quad (3.54)$$

If in addition C admits coproducts with the monoidal unit I , we can rewrite this as a map

$$A \longrightarrow [I, B] \times [B, B] \stackrel{\sim}{=} [I + B, B]; \quad (3.55)$$

sending $\langle f, \Phi \rangle$ to this map constitutes the action on objects of an equivalence between $\mathbf{iMRS}_B$ and the slice category $C/[I + B, B]$.

Proof. The argument is a straightforward check that the correspondence just defined on objects extends to morphisms, sending a diagram (3.7) to a homomorphism in the slice category; by way of example, we show part of the argument.

Denote by $\tilde{p}$ the mate of an arrow $p : X \otimes Y \rightarrow Z$ according to the monoidal closed structure on C , i.e. the arrow

$$Y \xrightarrow{\eta_Y} [X, X \otimes Y] \xrightarrow{[X, p]} [X, Z] \quad (3.56)$$

Then the claim of functoriality amounts to the fact that the triangle

$$\begin{array}{ccc}
 A & \xrightarrow{u} & A' \\
 \searrow \langle \widetilde{f \circ \rho_A}, \hat{\Phi} \rangle & & \swarrow \langle \widetilde{g \circ \rho_{A'}}, \tilde{\Psi} \rangle \\
 & [I, B] \times [B, B] &
 \end{array}
 \quad (3.57)$$

is commutative, which can be checked ‘componentwise’ as $\langle g \circ \widetilde{\rho_{A'}}, \tilde{\Psi} \rangle \circ u = \langle g \circ \widetilde{\rho_{A'}} \circ u, \tilde{\Psi} \circ u \rangle$ and

$$\begin{aligned} [I, g \circ \rho_{A'}] \circ \eta_{A'} \circ u &= [I, g \circ \rho_{A'}] \circ [I, u \otimes I] \circ \eta_A \\ &= [I, g \circ \rho_{A'} \circ u \otimes I] \circ \eta_A \\ &= [I, f \circ \rho_A] \circ \eta_A \\ &= \eta_{A'} \circ f \circ \rho_A \\ &= \widetilde{f \circ \rho_A}. \end{aligned}$$

One reasons similarly, if more tediously, for $\tilde{\Psi} \circ u$ and this concludes the proof. $\square$

Remark 3.32 (On the symmetry hypothesis). This is the only result that seems to require symmetry, or at least biclosure of C , because it is the only one that seems to rely on the uncurrying-recurring operation (Definition 2.19).

We nonetheless assume symmetry, in the statement and in the formalization, because in a biclosed setting the two internal homs must be carried, related and reasoned about separately throughout, and the codebase would not have benefitted from this extra generality. The Agda development makes the choice visible: `Incoherent/Slice.agda` takes a `Symmetric` structure as an explicit module parameter, and the braiding occurs there exactly where the argument above predicts.

Remark 3.33. As a consequence of the equivalence just outlined, a set of corollaries similar to the ones in Remark 3.30 applies to $\mathbf{iMRS}_B$ and the existence of co/limits therein, as well as its accessibility.

The existence of co/limits in the fibres $\mathbf{iMRS}^A$ and $\mathbf{iMRS}_B$ is a first step towards establishing completeness and cocompleteness of the total category $\mathbf{iMRS}$. We now embark on providing a modern, explicit argument for the existence of co/limits in the $\mathbf{iMRS}$ of Definition 3.5 using the language of accessible categories.

The treatment here settles a matter the existing literature leaves genuinely obscure. Several authors—among them Rosen, Arbib and Warner [Ros66a, Arb66, War82]—assert the existence of co/limits in subcategories of $\mathbf{iMRS}$, but none proves completeness or cocompleteness directly, and no two of them work with quite the same category.

Băianu states a completeness theorem, in the form of a claim that ‘projective systems’ of general (M, R) -systems admit ‘projective limits’, and that discrete families admit ‘direct sums’ and ‘cokernels’ [Băi73, Theorems 1–2]; the latter clearly echoes the classical results reducing cocompleteness to the existence of arbitrary coproducts and coequalizers. The proof, however, is attributed to an older paper of Căzănescu couched in the language of automata [Căz67], which is effectively unfindable; the Băianu–Marinescu paper cited alongside it is likewise unavailable for verification [BM74].

Louie, for his part, exhibits a neighbouring category of systems [Lou09, §7.10, 13.23], only to observe that it fails to admit all equalizers.

The net effect is a tower of Babel: definitions differing in details seldom made explicit, carrying theorems that variously affirm and deny the existence of the same co/limits, and no dictionary for passing between them. There is great disorder under heaven; the situation is excellent, in that the disagreements turn out to concern not mathematics but which category one is standing in. We therefore begin by rectifying names, in the sense of *Analectis* XIII.3; Proposition 3.36 below then establishes completeness and cocompleteness for the category $\mathbf{iMRS}$ that we defined in Definition 3.5.

3.1. Limits and colimits in $\mathbf{iMRS}$. The cocompleteness theorem makes heavy use of the following construction, that we recall from [ACC⁺24, Example 2.24].

Definition 3.34 (The fibration of $-$ -algebras for the regular action of C). Define a category $\int_A \mathbf{Alg}(- \otimes A)$ of *parametric algebras*, having

- objects the tuples $(X, \xi)^A$ consisting of an object A of C , an object X of C , and a map $\xi : X \otimes A \rightarrow X$ in C ;

- arrows $(X, \xi)^A \rightarrow (Y, \theta)^B$ the pairs (u, v) of arrows $u : A \rightarrow B$ and $v : X \rightarrow Y$ in C such that the diagram

$$\begin{array}{ccc} X \otimes A & \xrightarrow{v \otimes u} & Y \otimes B \\ \xi \downarrow & & \downarrow \theta \\ X & \xrightarrow{v} & Y \end{array} \quad (3.58)$$

commutes.

Evidently, all the category axioms are satisfied, and the assignment $(X, \xi)^A \mapsto A$ defines a functor $p_\otimes : \int_A \mathbf{Alg}(- \otimes A) \rightarrow C$.

In [ACC⁺24] it is argued that this category deserves the name of *total category of semiautomata*, given that a semiautomaton is precisely an algebra for the endofunctor $- \otimes A$ on C , and that the typical object of $\int_A \mathbf{Alg}(- \otimes A)$ is a pair (parameter, algebra for the functor defined at that parameter).

There we also argue, among other things, that the category $\int_A \mathbf{Alg}(- \otimes A)$ can be seen as a ‘semidirect product’ of C acting on itself by the ‘regular’ representation $(X, Y) \mapsto X \otimes Y$. We thus adopt the shorthand $C \ltimes_\otimes C := \int_A \mathbf{Alg}(- \otimes A)$ from now on.

Remark 3.35. The category $C \ltimes_\otimes C$ comes equipped with two functors

$$C \xleftarrow{p_\otimes} C \ltimes_\otimes C \xrightarrow{V} C \quad (3.59)$$

defined respectively by sending the typical object $(X, \xi)^A$ to $X = V((X, \xi)^A)$ and to $A = p_\otimes((X, \xi)^A)$. The functor $p_\otimes$ is moreover a *Grothendieck fibration*, and the fibre over an object A is precisely the category $\mathbf{Alg}(- \otimes A)$ of algebras for the endofunctor $- \otimes A$; this explains in a nutshell the notation and the relation between $\int_A \mathbf{Alg}(- \otimes A)$ and $p_\otimes$: the latter is the Grothendieck fibration associated, under the equivalence of Theorem 2.21, to the pseudofunctor

$$C^{\text{op}} \longrightarrow \mathbf{Cat} : A \longmapsto \mathbf{Alg}(- \otimes A) \quad (3.60)$$

Proposition 3.36. Let C be a closed monoidal category, which in addition is *locally presentable* in the sense of [AR94, Theorem 1.20];⁶ then the category $\mathbf{iMRS}_C$ of Definition 3.7 is itself locally presentable, and thus (complete and) cocomplete.

The proof of this proposition revolves around the following lemma, which we record separately since it is of independent interest: it provides a highly conceptual characterization of the total category $\mathbf{iMRS}$ of incoherent (M,R)-systems.

Lemma 3.37. Let C be a closed monoidal category, and let $C \ltimes_\otimes C$ be the total category of semi-automata as above. Then the category $\mathbf{iMRS}$ of incoherent (M,R)-systems (in uncurried form, see Remark 3.8) fits in a pullback of categories

$$\begin{array}{ccccc} \mathbf{iMRS} & \longrightarrow & \mathcal{Y} & \longrightarrow & C^{[0 \rightarrow 1]} \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \text{dom} \\ \mathcal{X} & \longrightarrow & C \ltimes_\otimes C & \xrightarrow{p_\otimes} & C \\ \downarrow & \lrcorner & \downarrow V & & \\ C^{[0 \rightarrow 1]} & \xrightarrow{\text{cod}} & C & & \end{array} \quad (3.61)$$

Proof. The proof is straightforward: an object of $\mathcal{X}$, the pullback of the pair (cod, V) , is a pair $(f : X \rightarrow B, \Phi : B \otimes A \rightarrow B)$ for certain objects $X, A \in C_0$; at the same time, an object of $\mathcal{Y}$, the pullback of the pair $(\text{dom}, p_\otimes)$, is a pair $(g : A \rightarrow Y, \Psi : Z \otimes A \rightarrow Z)$ for certain objects $Y, Z \in C_0$. It is clear how the functors in the cospan $\mathcal{X} \rightarrow C \ltimes_\otimes C \leftarrow \mathcal{Y}$ act, and that their pullback is exactly

⁶Recall that this means that C is cocomplete, and it is generated under filtered colimits by the subcategory of its κ -compact objects.

an incoherent (M, R) -system based at (A, B) . The same reasoning applies to morphisms, and this concludes the proof.

It is worth observing that the destructor functors $[-]_A, [-]_B$ defined in Remark 3.20 are ‘hidden’ in (3.61), and thus admit a more fundamental (and universal) characterization in terms of this limit diagram. $\square$

Proof of Proposition 3.36. The proof is a patchwork of results that are classical in the theory of accessible and locally presentable categories, but it consists of rather technical arguments that we spell out in detail for the reader’s convenience. We proceed in three steps.

- First, we recall that [MP89, Theorem 5.1.6] proves the *completeness theorem for accessible categories*, that we need to invoke in the following baby example: given a pullback square

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{P} & \mathcal{X} \\ Q \downarrow & \lrcorner & \downarrow G \\ \mathcal{Y} & \xrightarrow{F} & \mathcal{A} \end{array} \quad (3.62)$$

of categories, if F and G are accessible functors between accessible categories, then the pullback $\mathcal{E}$ is itself an accessible category, and the projection functors P, Q are accessible.

- In particular, we would like to deduce that if $\mathcal{A}$ is locally presentable, then so is $\mathcal{E}$, in the same notation and under the same assumptions. This follows from an application of [ML98, V.6, Exercise 3], where it is asserted that in any strict pullback square of categories like (3.62), assuming G preserves and F creates limits, then P creates limits (thus makes $\mathcal{E}$ complete if $\mathcal{X}$ is). A category is then locally presentable (i.e. it is accessible and complete) if and only if it is accessible and *complete*, and we conclude that $\mathcal{E}$ is locally presentable.
- Now, we can apply the above result once we have characterized **iMRS** via a(n iterated) pullback of categories; this is exactly what the presentation of **iMRS** outlined in (3.61) attains. The functors involved in the pullback are easily seen to be accessible, provided C is, and the base category C is cocomplete (thus locally presentable) by assumption; thus, repeatedly applying [MP89]’s completeness theorem, we conclude that **iMRS** is locally presentable.

We conclude that **iMRS** is accessible and complete, thus locally presentable. $\square$

A similar line of reasoning works for the category of twisted elements of incoherent (M, R) -systems, that we define below.

Now for something different. The categorical theory of lenses, and more generally of *optics*, has by now an extensive literature; the reader can consult [FGM⁺07] for the original definition of a lens, and its motivating application to the view-update problem for bidirectional transformations, [JRW12] for the identification of lenses with certain fibrational data, [Ril18] for the unifying account of optics (lenses among them) as morphisms in a category built from a monoidal action, and [CEG⁺24] for the profunctor representation of optics that we tacitly use below; for applications of lenses across computer science, from modular data accessors [PGW17, BG18] to open games [GHWZ18], supervised learning [FST19] and dependent lenses [Spi19], the literature is by now vast.

Proposition 3.38 ((M,R)-systems as lenses). The type of an incoherent (M, R) -system is that of a lens $[\begin{smallmatrix} A \\ B \end{smallmatrix}] \rightarrow [\begin{smallmatrix} B \\ B \end{smallmatrix}]$.

Proof. Recall from [FGM⁺07, JRW12] that a (bimorphic) lens $[\begin{smallmatrix} S \\ S' \end{smallmatrix}] \rightarrow [\begin{smallmatrix} V \\ V' \end{smallmatrix}]$ in a Cartesian category consists of a pair ($\text{get} : S \rightarrow V$, $\text{put} : S \times V' \rightarrow S'$). In the Cartesian setting the order of the two factors of put carries no information, since $S \times V' \cong V' \times S$ naturally; in the monoidal, not necessarily symmetric setting of the present paper it does, and we fix once and for all the convention

$$\text{get} : S \rightarrow V, \quad \text{put} : V' \otimes S \rightarrow S', \quad (3.63)$$

placing the residual to the *right* of the update. With this convention the correspondence is immediate and appeals to no braiding: an incoherent (M, R) -system based at (A, B) is the lens

$\begin{bmatrix} A \\ B \end{bmatrix} \rightarrow \begin{bmatrix} B \\ B \end{bmatrix}$ whose **get** is the process map $f : A \rightarrow B$, and whose **put** is the uncurried repair map $\hat{\Phi} : B \otimes A \rightarrow B$ of Notation 3.1. This identification is a bijection on the nose, not merely up to isomorphism.

The choice of side is forced, rather than conventional. Since **get** = f pins the source object to $S = A$, the opposite convention **put** : $S \otimes V' \rightarrow S'$ would demand an isomorphism $A \otimes V' \cong B \otimes A$, which is unavailable for *any* choice of target pair $\begin{bmatrix} V \\ V' \end{bmatrix}$ unless $A \cong B$; in particular, retargeting the lens to $\begin{bmatrix} B \\ A \end{bmatrix}$ does not repair the mismatch. $\square$

Remark 3.39 (Lenses versus optics). The word ‘lens’ is used above for the bare (**get**, **put**) pair, which is its original meaning in [FGM⁺07] and the one that the (M,R)-system diagram instantiates. It should not be conflated with the *optic* for the $\otimes$ -action, i.e. the coend

$$\int^{M \in C} C(S, M \otimes V) \times C(M \otimes V', S') \quad (3.64)$$

of [Ril18, CEG⁺24]. The two agree when C is Cartesian, where the co-Yoneda lemma collapses the coend at $M = S$ and returns exactly a (**get**, **put**) pair; but in a general monoidal closed category the coend instead computes the *linear lens* $C(S, [V', S'] \otimes V)$, which is a different notion. It is the pair, not the optic, that Proposition 3.38 identifies with an incoherent (M,R)-system.

Remark 3.40. At first sight this proposition does not appear to be of any use beyond restating the definition in different notation: the general theory of (polymorphic) lenses composes morphisms $\begin{bmatrix} S \\ S' \end{bmatrix} \rightarrow \begin{bmatrix} V \\ V' \end{bmatrix}$ and $\begin{bmatrix} V \\ V' \end{bmatrix} \rightarrow \begin{bmatrix} U \\ U' \end{bmatrix}$ only when the target pair $\begin{bmatrix} V \\ V' \end{bmatrix}$ of the first matches the source pair of the second *exactly* in both coordinates. An incoherent (M,R)-system based at (A, B) , read as a lens $\begin{bmatrix} A \\ B \end{bmatrix} \rightarrow \begin{bmatrix} B \\ B \end{bmatrix}$, has target pair $\begin{bmatrix} B \\ B \end{bmatrix}$; a subsequent incoherent (M,R)-system based at (B, Y) is a lens $\begin{bmatrix} B \\ Y \end{bmatrix} \rightarrow \begin{bmatrix} Y \\ Y \end{bmatrix}$, with *source* pair $\begin{bmatrix} B \\ Y \end{bmatrix} \neq \begin{bmatrix} B \\ B \end{bmatrix}$ unless $Y = B$. Consequently the general lens-composition formula (which threads the first **put** through the second, $\text{put}(s, u') := \text{put}_1(s, \text{put}_2(\text{get}_1(s), u'))$), which duplicates s and therefore presupposes a diagonal, so that it is in any case unavailable in a non-Cartesian C) is not even type-correct here, and cannot be invoked to compose incoherent (M,R)-systems. This explains, from the optics perspective, why the sequential composition $- \circ -$ of Construction 3.27 had to be defined by hand rather than inherited from the general theory of lenses, and why it turned out to be associative but only left-unital rather than assembling **iMRS** into a genuine category of lenses.

In addition to the functors of Remark 3.20, it would be desirable to have functors $[-]_f$ and $[-]_\Phi$, selecting the process and the repair map of an (M,R)-system, respectively, i.e. some correspondence of type $[\mathcal{M}]_f = f$ and $[\mathcal{M}]_\Phi = \Phi$ if $\mathcal{M} = \langle f, \Phi \rangle$; however, due to the variance of the diagrams in (3.7), the best one can do starting from the category **iMRS** is to define a span

$$C^{[0 \rightarrow 1]} \xleftarrow{[-]_f} \mathbf{iMRS} \quad \mathbf{iMRS}^A \xrightarrow{[-]_\Phi} C^{[0 \rightarrow 1]} \quad (3.65)$$

In particular, one can destructure an (M,R)-system into its repair map, functorially, only if the domain of the (M,R)-system is fixed.

This is undesirable, and a possible adjustment is to resort to a modified version of the category **iMRS**.

Definition 3.41 (The category of incoherent twisted elements, $\mathcal{E}(\mathbf{iMRS})$). The category of *twisted elements* for incoherent (M,R)-systems, denoted $\mathcal{E}(\mathbf{iMRS})$, is defined having

- objects the (M,R)-systems as in Definition 3.5;
- arrows the *twisted* morphisms of (M,R)-systems: $\langle f, \Phi \rangle \rightarrow \langle g, \Psi \rangle$ consists of a pair $\ell : A' \rightarrow A, r : B \rightarrow B'$ such that (equationally)

$$r \circ f \circ \ell = g \quad \Psi \circ r = [\ell, r] \circ \Phi \quad (3.66)$$

or diagrammatically, the following square and twisted square both commute.

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\
 \uparrow \ell & (i) & \downarrow r & (ii) & \downarrow [\ell, r] \\
 A' & \xrightarrow{g} & B' & \xrightarrow{\Psi} & [A', B']
 \end{array} \tag{3.67}$$

It is immediate to verify that

Proposition 3.42 ($\mathcal{U}/\mathcal{U}$). $\mathcal{E}(\mathbf{iMRS})$ is a category.

Remark 3.43. The category $\mathcal{E}(\mathbf{iMRS})$ fits in a pullback diagram similar to (3.61),

$$\begin{array}{ccccc}
 \mathcal{E}(\mathbf{iMRS}) & \longrightarrow & \mathcal{Y} & \longrightarrow & C^{[0 \rightarrow 1]} \\
 \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \text{dom} \\
 X & \longrightarrow & C \ltimes_{\otimes} C & \xrightarrow{p_{\otimes}} & C \\
 \downarrow & \lrcorner & \downarrow V & & \\
 \mathbf{Tw}(C) & \xrightarrow{\text{cod}} & C & &
 \end{array} \tag{3.68}$$

(where cod is a small abuse of notation for the composite functor $\mathbf{Tw}(C) \rightarrow C^{\text{op}} \times C \xrightarrow{\pi'} C$) but we have to be content with a weaker statement: the twisted arrow category of C might not inherit colimits from C (a blatant example is $C = \mathbf{Set}$, where $\mathbf{Tw}(\mathbf{Set})$ lacks an initial object).

Proposition 3.44 ($\mathcal{U}/\mathcal{U}$). There is a span of “projection” functors

$$\mathbf{Tw}(C) \xleftarrow{\lambda - \mathfrak{J}_f} \mathcal{E}(\mathbf{iMRS}) \xrightarrow{\lambda - \mathfrak{J}_{\Phi}} C^{[0 \rightarrow 1]} \tag{3.69}$$

defined on objects respectively as $\lambda f, \Phi \mathfrak{J}_f = f$ and $\lambda f, \Phi \mathfrak{J}_{\Phi} = \Phi$, and mapping morphisms to the squares (i) and (ii) in (3.67).

The theory of coherent (M, R) -systems will yield another ‘ $[-]_{\Phi}$ accessor’, defined only on a category where the typical object is not just an arrow $u : X \rightarrow Y$, but a pair $(A, \Phi : id \Rightarrow [A, -])$ (the resulting category **Rep** is the *fibration of repair maps*). In Definition 4.28 we will construct a coreflective subcategory

$$J : \mathbf{Rep}^{\hookrightarrow} \mathbf{iMRS} \tag{3.70}$$

of the fibration of repairs into the total category of incoherent (M, R) -systems, whose right adjoint is a functor in the opposite direction, defined on objects by sending $\langle\langle f, \Phi \rangle\rangle$ to Φ (the definition of **Rep** is engineered to make K functorial).

Remark 3.45 (Functorial abstraction of incoherent (M, R) -systems). We can devise the following very general abstraction of (M, R) -system; let now C be any category, and $F : C \rightarrow C$ an endofunctor which admits a right adjoint R ; clearly, this abstracts the situation where C is monoidal closed and $(F \dashv R)$ is the parametric adjunction $- \otimes A \dashv [A, -]$. An F -(M, R)-system now consists of a pair

$$A \xrightarrow{f} B \quad FB \xrightarrow{\Phi} B \tag{3.71}$$

where f is any arrow of C , and (B, Φ) (in uncurried form) is an F -algebra (equivalently, we can take a Φ' in curried form, an arrow of C of type $B \rightarrow RB$, making B into an R -coalgebra).

There is a total category $\mathbf{iMRS}_{[F]}$ having the tuples $(A, B; \langle\langle f, \Phi \rangle\rangle_F)$ of the above type as objects, and where a morphism $(A, B; f, \Phi) \rightarrow (A', B'; g, \Psi)$ consists of the obvious choice of arrows $u : A' \rightarrow A, v : B \rightarrow B'$ such that $v \circ f \circ u = g$ and in addition, v is an F -algebra homomorphism.

3.2. Cartesian incoherent (M,R)-systems. The aim of this section is to specialize the previous discussion to the case where C is Cartesian, and to explore the consequences of this assumption. In particular, we will see that in this case, the category of incoherent (M,R)-systems fits into additional functors and adjunctions, which are not available in the general monoidal closed setting. Ultimately, what one gains in this specialization is a ‘trivial’ (M,R)-system, in the sense clarified by the following lemma.

Definition 3.46 (Cartesian incoherent (M,R)-system). When C is a Cartesian monoidal category, i.e. $\otimes = \times$ is the categorical product and $[A, B]$ is the actual exponential object B^A , we refer to an incoherent (M,R)-system of Definition 3.5 simply as a *Cartesian incoherent (M,R)-system*; unlike the general monoidal case, C is automatically symmetric and equipped with diagonal maps $\Delta_X : X \rightarrow X \times X$ and projections to the terminal object 1 , both of which we will use below to construct the ‘trivial’ (M,R)-systems.

Definition 3.47 (Trivial and identity (M,R)-systems). Let C be a Cartesian category, and let A be an object of C . The *terminal (M,R)-system* based at A , denoted $\mathcal{M}_A^\top$ (or just $\mathcal{M}^\top$ for short), is the incoherent (M,R)-system $\langle !_A, \Phi_A \rangle$ where $!_A : A \rightarrow 1$ is the unique arrow to the terminal object, and $\Phi_A : 1 \rightarrow [A, 1] \cong 1$ is the unique arrow to the internal hom.

By a similar token, the *identity (M,R)-system* based at A , denoted $\mathcal{M}_A^0$, is the incoherent (M,R)-system $\langle \text{id}, \Phi^0 \rangle$ where $\Phi^0 : A \rightarrow [A, A]$ is the curried projection (which means $\Phi^0 = \lambda xy.x$ in lambda notation).

Lemma 3.48. For every object $A \in C$, the terminal (M,R)-system $\mathcal{M}_A^\top$ based at A is the terminal object of the fibre $\mathbf{iMRS}^A$ of Definition 3.21.

Proof. Let $\langle g, \Psi \rangle$ be any incoherent (M,R)-system based at (A, B) . The unique arrow $B \rightarrow 1$ to the terminal object of C is automatically the unique candidate arrow $v : (B; \langle g, \Psi \rangle) \rightarrow (1; \langle !_A, \Phi_A \rangle)$ in $\mathbf{iMRS}^A$: the first commutativity condition $v \circ g = !_A$ holds because 1 is terminal in C , so any two arrows $A \rightarrow 1$ coincide; the second condition $[A, v] \circ \Psi = \Phi_A \circ v$ holds for the same reason, since $[A, 1]$ is terminal too (being isomorphic to 1), so any two arrows into it agree. Hence $\mathcal{M}^\top$ receives a unique arrow from every object of $\mathbf{iMRS}^A$, i.e. it is terminal. $\square$

This is the precise sense in which $\mathcal{M}^\top$ is ‘trivial’: it is the canonical, terminal choice of (M,R)-system attachable to any process map $f : A \rightarrow B$, and this terminality is exactly what makes the assignment $f \mapsto \langle f, \mathcal{M}^\top \rangle$ functorial in the theorem below. By contrast, the identity (M,R)-system $\mathcal{M}_A^0$ (or just $\mathcal{M}^0$ for short) is trivial in a complementary, more elementary sense: its repair map $\Phi^0 = \lambda xy.x$ simply discards the environment argument and returns the resource unchanged, so that the corresponding right action $A \times A \rightarrow A$ is literally the first projection: the simplest repair rule available in a Cartesian category is ‘no repair at all’...

The presence of the trivial (M,R)-systems allows one to prove the existence of functors that we do not have in the purely monoidal case, and that are adjoint to the ‘destructors’ defined in (3.30), where we have established the existence of a span

$$C \xleftarrow{[-]_A} \mathbf{iMRS} \xrightarrow{[-]_B} C$$

Theorem 3.49 ($\mathcal{C}^{\mathcal{U}}$). There exists a functor $L : C^{[0 \rightarrow 1]} \rightarrow \mathbf{iMRS}$ defined as follows:

- on objects, $L \left[\begin{smallmatrix} A \\ \downarrow \\ B \end{smallmatrix} \right] f$ is the (M,R)-system $\langle f, \Phi_{A,B}^0 \rangle$, where $\Phi_{A,B}^0 : B \rightarrow [A, B]$ is the *constant repair map*, i.e. the transpose of the projection $B \times A \rightarrow B$; in lambda notation $\Phi_{A,B}^0 = \lambda xy.x$. This is the evident extension to a variable codomain of the identity (M,R)-system’s repair map Φ^0 of Definition 3.47, which is the case $B = A$;
- on arrows, L sends a morphism $f \rightarrow g$ in $C^{[0 \rightarrow 1]}$,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ h \downarrow & & \downarrow k \\ X & \xrightarrow{g} & Y \end{array} \quad (3.72)$$

to the same square, carrying the constant repair maps along: the compatibility square for Φ^0 commutes because both composites discard their A -argument.

Theorem 3.50 ($\mathcal{U}$). There exists a functor $L' : \mathbf{Tw}(C) \rightarrow \mathcal{E}(\mathbf{iMRS})$ defined as follows:

- on objects, $L' \left[\begin{smallmatrix} A \\ \downarrow \\ B \end{smallmatrix} \right] f$ is the (M, R) -system $\langle\langle f, \Phi_{A,B}^0 \rangle\rangle$ of Theorem 3.49;
- on arrows, L' sends a twisted morphism $f \rightarrow g$ in $\mathbf{Tw}(C)$,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ h \uparrow & & \downarrow k \\ X & \xrightarrow{g} & Y \end{array} \quad (3.73)$$

to the same twisted square, again carrying the constant repair maps along.

Observe that then we are in the situation depicted by the diagram

$$\begin{array}{ccc} C^{[0 \rightarrow 1]} & \xrightarrow{L} & \mathbf{iMRS} \\ \xleftarrow{[-]_f} & & \end{array} \quad \begin{array}{ccc} \mathbf{Tw}(C) & \xrightarrow{L'} & \mathcal{E}(\mathbf{iMRS}) \\ \xleftarrow{[-]_f} & & \end{array} \quad (3.74)$$

It is now natural to conjecture that these two pairs form adjunctions (in fact, coreflections, since L, L' are fully faithful and their definition suggests they are left adjoints). However, one can check that L and L' fall just short of satisfying the adjunction property.⁷

Proposition 3.51 (Semisimplicial structure on $\mathbf{iMRS}^{(n)}$, $\mathcal{U}$). The assignment that sends a finite nonempty ordinal $[n]$ to the category $\mathbf{iMRS}^{(n)}$ of Remark 3.26 has a canonical semisimplicial structure, at least in the low degrees

$$C \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathbf{iMRS} \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathbf{iMRS}^{(2)} \quad \dots \quad (3.75)$$

More precisely, let Δ_{inj} be the category of finite nonempty total orders, and *injective* monotone maps; the categories $\mathbf{iMRS}^{(n)}$ for $n = 0, 1, 2$ (under the convention that $\mathbf{iMRS}^{(0)} = C$), together with suitably defined face and degeneracy functors, form a *2-truncated semisimplicial object* $\Delta_{\text{inj}}^{\text{op}} \rightarrow \mathbf{Cat}$.

The natural choice of degeneracies does not, however, make this into a full simplicial object in the category of all incoherent (M, R) -systems: there seems to be no way to extend the above diagram by filling in the dotted arrow with functors satisfying all simplicial identities.

Proof. Represent an incoherent (M, R) -system in the usual way as

$$A \xrightarrow{f} B \xrightarrow{\Phi} [A, B] \quad (3.76)$$

A 2-simplex $A \xrightarrow{\xi} B \xrightarrow{\eta} C$ in $\mathbf{iMRS}^{(2)}$ consists of a composable pair that we can depict as in (3.44).

The two outer faces select ξ and η , while the middle face is given by

$$\xi \circ \eta = (g \circ f, [f, \text{id}_C]_1 \circ \Psi). \quad (3.77)$$

The three face–face identities assert only that this composite has source A and target C , and that the two original edges have the common endpoint B . These identities hold naturally. Composition is also associative up to the canonical equality supplied by functoriality of the internal hom, since

$$[f, \text{id}]_1 \circ [g, \text{id}]_1 \circ \Omega = [g \circ f, \text{id}]_1 \circ \Omega. \quad (3.78)$$

⁷The obstruction is easy to locate: the counit at an (M, R) -system $\langle\langle f, \Phi \rangle\rangle$ would be a morphism $\langle\langle f, \Phi_{A,B}^0 \rangle\rangle \rightarrow \langle\langle f, \Phi \rangle\rangle$ over the identities, and this forces $\Phi = \Phi_{A,B}^0$. Passing to *coherent* (M, R) -systems does repair it, but only at a price: the same computation then requires $\text{Nat}(\text{id}, [A, -])$ to be a singleton for every A , i.e. that the Bernstein centre of C be a terminal object (Proposition 4.19); in the Cartesian case this is exactly the condition that $\mathfrak{z}C \cong 1$, a condition guaranteed by well-pointedness, though strictly weaker than it, cf. Remark 4.6. That is a severe restriction, and it is exactly the hypothesis that Proposition 4.23 shows one must abandon in order to have any nontrivial coherent (M, R) -system at all. The adjunction is therefore available only in the setting where the objects it relates are all trivial, and we do not pursue it.

The current Agda formalization verifies this through degree 2.

In the Cartesian case, let $\kappa_{A,B}$ be the transpose of the projection, i.e.

$$\kappa_{A,B}: B \longrightarrow [A, B], \quad \kappa_{A,B}(b)(a) = b, \quad (3.79)$$

and let $\mathbf{1}_A = (\text{id}_A, \kappa_{A,A})$ be the evident degenerate edge. One unit identity holds: $\mathbf{1}_A \circ \xi \cong \xi$, as is immediate to check.

The other identity does not. Indeed,

$$\xi \circ \mathbf{1}_B = (f, [f, \text{id}_B]_1 \circ \kappa_{B,B}) = (f, \kappa_{A,B}), \quad (3.80)$$

which is independent of the original repair map Φ . The simplicial identity $d_1^2 s_0^1 \cong \text{id}_{\mathbf{iMRS}}$ would require every incoherent repair map to satisfy

$$\Phi(b)(a) = b. \quad (3.81)$$

This is false in general; we will see in Lemma 4.2 that for coherent (M, R) -systems in a Cartesian category this is, however, the only possible choice.⁸

To see a concrete counterexample, take $A = \{*\}$, $B = \{0, 1\}$ in the category **Set** of sets and functions; let $f(*) = 0$, and define $\Phi(b)(*) = 0$. The normalized repair satisfies $\kappa_{A,B}(1)(*) = 1$, so the original system and its right-unit composite are not equal. $\square$

Remark 3.52 (Possible corrections). One may simply omit the degeneracies and use the semi-simplicial object described above. There are several ways to remove the obstruction, depending on which information one wishes to retain.

- SE1) If one studies (M, R) -systems with trivial repair function, where $\Phi = \kappa_{A,B}$, one essentially obtains the ordinary nerve of the arrow category $C^{[0 \rightarrow 1]}$. We will see in Lemma 4.2 that for Cartesian coherent (M, R) -systems in the category of sets this is the only possible choice.
- SE2) One may enlarge the notion of simplex. For example, a 2-simplex may consist of a complete triangle

$$(\xi_{01}, \xi_{12}, \xi_{02}), \quad U\xi_{02} = U\xi_{12} \circ U\xi_{01}, \quad (3.82)$$

where U forgets the repair map and the repair carried by ξ_{02} is stored independently. The three faces are then the three projections, and the degeneracies are $s_0(\xi) = (\xi, \mathbf{1}, \xi)$, $s_1(\xi) = (\mathbf{1}, \xi, \xi)$. In higher degrees one stores an independently decorated edge ξ_{ij} for every $i < j$, and faces and degeneracies act by reindexing. This produces a natural candidate for a genuine simplicial object retaining arbitrary repairs, at the cost of no longer requiring the diagonal repair to be computed from the two adjacent repairs.

Proposition 3.53 ((M, R) -systems as dynamical systems in a coKleisli category). By yet another similar token, if C is a Cartesian category, we can present an incoherent (M, R) -system $A \rightarrow B \rightarrow [A, B]$ in yet another way; first, uncurry Φ to obtain a pair $A \rightarrow B, B \times A \rightarrow B$; then, Cartesianity implies that each $S_A = _ \times A$ is a comonad on C , and the maps (f, Φ) are coKleisli maps $S_A 1 \rightarrow B, S_A B \rightarrow B$, i.e. arrows

$$1 \xrightarrow{f} B \quad B \xrightarrow{\Phi} B \quad (3.83)$$

in $\mathbf{coKl}(S_A)$; thus, Φ is a dynamical system, one orbit of which is the sequence $\{f, \Phi \cdot f, \Phi \cdot (\Phi \cdot f), \dots\}$ (here $\cdot$ denotes coKleisli composition); we can think of (M, R) -systems as *parametric algebras* of some sort, in the sense that follows: first, observe that one can lift the “Maybe” endofunctor $1 + _$ to $\mathbf{coKl}(S_A)$, because for every A there exists a natural transformation $S_A(1 + _) \rightarrow 1 + S_A _$ whose components are

$$A \times (1 + X) \cong A + A \times X \xrightarrow{!+A \times X} 1 + A \times X, \quad (3.84)$$

⁸It is worth recording how narrowly the construction misses: of the thirteen identities of a truncated simplicial object, twelve hold, and $d_1^2 s_0^1 \cong \text{id}$ is the only casualty. This is not an impression but a checked statement — the Agda development builds the record carrying all twelve, $\mathcal{C}$, alongside the semisimplicial object of Proposition 3.51 that omits the degeneracies altogether.

and which is compatible with the comonad structure. Thus, we get a lifting $1 +_{[A]} -$ filling the square⁹

$$\begin{array}{ccc} \mathbf{coKl}(S_A) & \xrightarrow{1+_{[A]}-} & \mathbf{coKl}(S_A) \\ \uparrow & & \uparrow \\ C & \xrightarrow{1+_{-}} & C \end{array} \quad (3.85)$$

into a commutative one (the vertical arrows being the cofree functors), together with an identification $\mathbf{iMRS}^A \cong \mathbf{Alg}(1 +_{[A]} -)$.

This concludes our survey of incoherent (M,R)-systems and their equivalent presentations, specialized to the Cartesian setting. The remaining task, taken up in §4, is to upgrade the repair map Φ to a genuinely natural transformation, yielding the *coherent* (M,R)-systems for which the total category assembles into an honest two-sided fibration.

Before doing that, however, the Cartesian theory of (M,R)-systems raises the question of whether there is something to gain in implementing an (M,R)-system as a true datatype like

```
data MRS a b = MRS
{ f    :: a -> b
, phi  :: b -> (a -> b)
}
```

From this angle, the repair map can be seen as a certain kind of ‘unfold’ where the seed is a function, rather than a value, and it can be modelled as a machine of sorts. Building on the work of Rosen [Ros64a] and Arbib [Arb66] we will observe in section 5 that there exists a standard procedure to obtain a special kind of *Mealy automaton* from an MR system. For the moment, however, let’s keep the discussion at a more elementary level and employ the pseudocode above as intuition, working in a Cartesian closed category **Hask** of Haskell types (see [Has, Ell17] even though strictly speaking such a category does not exist [Bau16], though reasoning in it is justified for the total fragment [DHJG06]).

Definition 3.54. We read an *MRS record*

$$\mathbf{MRS} A B = (f : A \rightarrow B, \phi : B \rightarrow (A \rightarrow B)) \quad (3.86)$$

as above as describing *two-stage, feedback-sensitive* computation: the field f produces a provisional output $x = f(a)$, and ϕ is a *post-processor* that, given that provisional output, returns a new higher-order term $\phi(x) : A \rightarrow B$.

Acting together, they yield a revised output function $k = \lambda a. \phi(f(a))(a) : A \rightarrow B$, of which we can then study the iterates

$$k^0 := f \quad k^{(n+1)} = \lambda a. \phi(k^{(n)}(a))(a) \quad (3.87)$$

In such a computation, the output is observed and then immediately *revises* the future $A \rightarrow B$ behaviour.

In **Hask** the uncurrying-recurrying operation of Definition 2.19 is certainly available, whence we get an A -parametric family of endomorphisms

$$\phi_a : B \rightarrow B, \quad \phi_a(b) = \phi(b, a), \quad (3.88)$$

(and obviously for every fixed b_0 we get a function $a \mapsto \phi(b_0, a)$).

This viewpoint can be coupled with the assumption that A or B carry more structure, such as that of a monoid or semigroup in **Hask**, to obtain more specific patterns of behaviour. In general, ϕ equips B with the structure of a right A -set, i.e. there is a right action of the free monoid A^* on B . If such an action has a nontrivial kernel, or if A is not free, and we assume that ϕ uncurries to an action $B \times A \rightarrow B$, the situation becomes more complex and interesting, as the dynamics of the system is constrained by algebraic relations.

⁹Note that the object 1 remains terminal in $\mathbf{coKl}(S_A)$; this motivates the intuition of the lifting in question as a lifted Maybe monad.

Example 3.55 (Assuming B is a semigroup / monoid). When B has a semigroup (resp. monoid) structure, certain relatively canonical ways to obtain a post-processor ϕ arise: let $*$ be the monoid operation and u_B the unit of B . We can define

- A ‘semigroup augment’ pattern: one uses the monoid operation of B and sets

$$\text{MRS}_h = \begin{cases} f : A \rightarrow B \\ \phi : B \times A \rightarrow B : (x, a) \mapsto x * h(a). \end{cases} \quad (3.89)$$

for some function $h : A \rightarrow B$ (possibly constant at an element of B).

```
semigroupAugment :: Semigroup b =>
  (a -> b) -> (a -> b) -> MRS a b
semigroupAugment base extra =
  MRS
  { f    = base
  , phi = \x a -> x <> extra a
  }
```

Closure requires a single seed x_0 with

$$f(a) = x_0 * \text{extra}(a) \quad \text{for all } a. \quad (3.90)$$

- Similarly, one can define a ‘monoid gate/override’: let $p : A \rightarrow \mathbf{Bool}$ be a predicate on A , and set

$$\text{MRS}_p = \begin{cases} f : A \rightarrow B \\ \phi : B \times A \rightarrow B : (x, a) \mapsto \begin{cases} x & \text{if } p(a), \\ e_B & \text{otherwise.} \end{cases} \end{cases} \quad (3.91)$$

Closure forces f to be constant on the p -true locus and equal to e_B on its complement.

Note that in nondegenerate cases (f nonconstant) the image of the seed map $\lambda a. \phi(b_0, a)$ is small inside the set of functions $A \rightarrow B$, so closedness is a ‘non-generic’ property.

Example 3.56 (Assuming B is of coproduct / partial type $\mathbf{1} + T$). The category of sets and partial functions has products and coproducts, so upon uncurrying ϕ into a map $B \times A \rightarrow B$, we can study MR systems whose components f and ϕ are partially defined.

In particular, if f is a partial function, and it is not defined on its input, $\phi : 1 \rightarrow T$ pipes a fallback value instead. This can be used to ‘patch’ a partial function into a total one, by providing a default value for the cases where it is not defined.

```
repair :: (a -> Maybe t) -> t -> MRS a (Maybe t)
repair base fallback =
  MRS
  { f    = base
  , phi = \mx a -> case mx of
    Just _ -> mx
    Nothing -> Just fallback
  }
```

- Repair:

$$\phi(\text{Nothing}, a) = \text{Just}(\text{fallback}(a)). \quad (3.92)$$

- Validate:

$$\phi(mx, a) = \begin{cases} mx & \text{if } \text{ok}(a), \\ \text{Nothing} & \text{otherwise.} \end{cases} \quad (3.93)$$

```
validate :: (a -> Maybe t) -> (a -> Bool) -> MRS a (Maybe t)
validate base test =
  MRS
  { f    = base
  , phi = \mx a -> if test a then mx else Nothing
  }
```

- Enrich:

$$\phi(\text{Lefte}, a) = \text{Left}(e * \text{ctx}(a)). \quad (3.94)$$

```

enrichLeft :: Semigroup e =>
  (a -> Either e t) -> (a -> e) -> MRS a (Either e t)
enrichLeft base ctx =
  MRS
  { f    = base
  , phi = \ex a -> case ex of
    Left e  -> Left (e <> ctx a)
    Right y -> Right y
  }

```

4. COHERENT (M,R)-SYSTEMS

Having introduced and studied incoherent (M,R)-systems at length in section 3, we now turn to the question left open there: under what notion of naturality can the repair map Φ be upgraded from an arbitrary morphism into a genuinely coherent (natural) one, without either trivializing the theory or losing the fibrational structure we are after?

There are at least two reasons to seek a more coherent presentation for (M,R)-systems.

- The first motivation is purely mathematical. In Remark 3.20 we constructed a span

$$C \xleftarrow{[-]_A} \mathbf{iMRS} \xrightarrow{[-]_B} C \quad (4.1)$$

whose left leg is a fibration, and whose fibres over (A, B) are the discrete sets $\mathbf{iMRS}(A, B)$ of Remark 3.22. However, we then observed that $[-]_B$ is not an opfibration: the assignment $B \mapsto \mathbf{iMRS}_B$ is not a covariant functor. It would be desirable to obtain, instead, a *two-sided discrete fibration* of (M,R)-systems—or equivalently (given the equivalence in Theorem 2.21), a profunctor

$$\mathbf{MRS}(_, _) : C^{\text{op}} \times C \longrightarrow \mathbf{Set} : (A, B) \longmapsto \mathbf{MRS}(A, B) \quad (4.2)$$

which depends functorially not only on A , but also on B . (Note that we lose the ‘i’ of the $\mathbf{iMRS}$ in Definition 3.5). If we denote as $(A, B) \xrightleftharpoons[u]{u} (A', B')$ a morphism of the category $C^{\text{op}} \times C$, by which we mean a pair $u : A' \rightarrow A, v : B \rightarrow B'$, the fact that $\mathbf{MRS}$ is a profunctor means that there exists an induced *function*

$$\mathbf{MRS}(A, B) \xrightarrow{\mathbf{MRS}(u,v)} \mathbf{MRS}(A', B') \quad (4.3)$$

between sets. This would significantly enhance the expressive power of the theory, because a plethora of techniques from fibred category theory [Gra66, Jac99, Bén85] could then be brought to bear on producing meaningful invariants of (M,R)-systems. For instance, certain functors that exist for *coherent* (M,R)-systems fail to exist for incoherent ones, and functors that exist—like the ones introduced in Theorem 3.49 and 3.50—fail to be adjoints for incoherent systems.

- The second motivation comes from the intended semantics of an (M,R)-system. If $\langle f, \Phi \rangle$ is to be understood as a process–repair pair, then it is somewhat natural to consider an entire *family* of repair functions of a fixed polymorphic shape

$$\forall X. \Phi_X : X \rightarrow [A, X], \quad (4.4)$$

that are allowed to vary with the parameter X , rather than just one component at $B = \text{cod } f$. Being able to do this would allow us to interpret Φ as a *blueprint for repair*, rather than as a single map; given an input of any type X , Φ produces an output of type $[A, X]$ coherently in X .

In engineering this definition, it would be desirable for a coherent (M, R) -system to yield an incoherent one—for instance, by evaluating Φ at a component that depends on B —if possible through a functor

$$\mathbf{MRS} \xrightarrow{(-)^b} \mathbf{iMRS} \quad (4.5)$$

that ‘flattens’ a coherent (M, R) -system into an incoherent one.

In a few words, we are thus seeking to strengthen the definition given in Definition 3.5, in such a way that the coherent one specializes to the incoherent one when considering the component of Φ at $B = \text{cod}f$.

This section aims to analyze what possibilities one has to achieve this goal, outlining the pros and cons of every design choice. In short, imposing a naturality condition can, in several cases, turn out to be rather restrictive (as in Lemma 4.2), or some intended semantics for a promotion of incoherent systems to coherent ones may fail to give rise to a two-sided fibration (as in §4.3.3); this is rooted in the notion of ‘naturality as polymorphism’ and ultimately follows from the core argument of the Yoneda lemma [Yon60, Str74].

We summarize the situation as follows.

- Each notion of coherent (M, R) -system has pros and cons; loosely speaking, making Φ natural comes at the price of considerably cutting down the number of possible choices for what such a family of maps can do.
- The issue is twofold: on one side, we want to define a notion of (M, R) -system where repair is natural in its input; on the other, we have to ensure that the notion is not so strict as to admit only trivial examples; last, and equally important, one would like to bundle any new given notion of coherent (M, R) -system into a *fibred category*, giving rise to a profunctor

$$\mathbf{MRS} : C^{\text{op}} \times C \longrightarrow \mathbf{Set} \quad (4.6)$$

- There is a hierarchy of notions of increasing abstraction (that we describe in subsections 4.1—4.3.5 below) that we can employ to define total categories of (M, R) -systems that are, or are not, two-sided discrete fibrations with typical fibre $\mathbf{MRS}(A, B)$. Again, there is a tension between two opposing needs: with a little foreshadowing of notation, (M, R) -systems that are A/C -natural (§4.3.2) ‘make sense semantically’ but do not yield a profunctor; (M, R) -systems that are cod -natural (§4.3.1) do give a profunctor of coherent (M, R) -systems, but turn out to be potentially very few in Cartesian categories. Other choices are possible—for instance, (M, R) -systems that are $(A/C)^{\text{op}} \times (C/B)$ -natural, or $(C^{[0 \rightarrow 1]})^{\text{op}} \times (C^{[0 \rightarrow 1]})$ -natural—but these tend to be unwieldy settings for computation. At the top of the abstraction ladder, we can define ‘fully abstract’ (M, R) -systems of profunctorial type (§4.3.5), at the cost of losing completely most of the intuition.

The attitude we take here is ‘as agnostic as possible’: this project being ultimately foundational, our goal is to offer several frameworks—each determined by different assumptions about the external world the model describes—in which to carry out deductions and computations. The ‘correct’ design choice can only be motivated by what the theory aims to describe and achieve.

The purpose of this section is therefore twofold.

We begin by analyzing in detail various choices for defining coherent (M, R) -systems (we might adopt the shorthand of calling these simply ‘ (M, R) -systems’); most of the discussion that we develop uses one of the following two enhancements of incoherent (M, R) -systems.

Definition 4.1 (Coherent and **cod**-coherent (M, R) -systems). Let A, B be objects of C .

- An *id-coherent* (M, R) -system (or simply a *coherent* one) is a pair $\langle f, \Phi \rangle$ where $f : A \rightarrow B$ and $\Phi : \text{id} \Rightarrow [A, -]$ is a natural transformation, $\mathcal{U}$; thus, one can recover the incoherent (M, R) -system $\langle f, \Phi_B \rangle$ by evaluating Φ at the component B ;
- a **cod-coherent** (M, R) -system is a pair $\langle f, \Phi \rangle$ where $f : A \rightarrow B$ and $\Phi : \text{cod} \Rightarrow [A, \text{cod}]$ is a natural transformation, $\mathcal{U}$; thus, one can recover the incoherent (M, R) -system $\langle f, \Phi_B \rangle$ by evaluating Φ at the component f (which is, obviously, an object of $C^{[0 \rightarrow 1]}$).

We prove in Remark 4.34 that these two notions are equivalent, due to a special property of the codomain functor $\text{cod} : C^{[0 \rightarrow 1]} \rightarrow C$ (the property of being a *lax epimorphism*, see [LNS22]). We can choose to work with either without losing any information.

There are alternatives to this choice, which we outline in §§4.3.1–4.3.5 to a certain degree of detail; to a large extent, the results that hold for the profunctor of *id*-natural (M, R) -systems also hold for other profunctors of (M, R) -systems (so, for *cod*-natural systems, profunctorial systems, and—although for a trivial reason—fully polymorphic systems; see again §§4.3.1–4.3.5 for this terminology). The subsequent theory is general enough to provide invariants (such as the tabulator of Definition 2.24) attached to ‘any’ profunctor of (M, R) -systems. This culminates in Definition 4.10, at the end of the first half of the section, establishing that there exists a *fibration of (M, R) -systems* giving the title to this paper.

This ultimately allows us to reinterpret Rosen’s ideas using the machinery of fibred categories and to obtain invariants—for example, comparison functors from the Rosen fibration to other categories—that are not generally available for the incoherent (M, R) -systems of Definition 3.5.

Last, in §4.5 we draw further consequences and turn to a question persistently pursued by Rosen himself and by several other authors [Ros12, LSAGA⁺06, Lou17]: extending the notion of (M, R) -system to a sequence of the form

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \longrightarrow \dots \quad (4.7)$$

This extension was studied especially in connection with *closure under efficient causation*, a major theme in Rosen’s work, to which we intend to apply the language developed in this first foundational paper. We define a system that ‘exhibits metabolic closure’ in Definition 4.42 and show that having a ‘universal closure point’ is a sufficient condition for it (Definition 4.44; the converse does not hold in general, since a closure point witnesses the condition at the single component Φ_B , while a universal one witnesses it uniformly in the input object); in Proposition 4.49 we show that the reindexing of an (M, R) -system having a universal closure point still has one, so that the reindexing operation of a profunctor of coherent (M, R) -systems is compatible with closure; in Remark 4.55 we show how to extend this notion of closure to longer (M, R) -systems.

Notably, the extension (4.7) can also be carried out for incoherent (M, R) -systems, and is thus of independent value even for those who want to work exclusively with incoherent (M, R) -systems. We focus on the construction of a chain

$$(i)\mathbf{MRS} =: (i)\mathbf{MRS}_2 \longleftarrow (i)\mathbf{MRS}_3 \longleftarrow (i)\mathbf{MRS}_4 \longleftarrow (i)\mathbf{MRS}_5 \longleftarrow \dots \quad (4.8)$$

of categories of coherent (or incoherent) (M, R) -systems in §4.5, see in particular Construction 4.50 and Definition 4.51; both the incoherent and coherent constructions are fully formalized in the Agda library.

4.1. *id*-natural (M, R) -systems. The simplest possible way to obtain natural repair maps is to regard $\Phi : B \rightarrow [A, B]$ as a single component, at $B \in C_0$, of a natural transformation of type

$$\Phi : id \Rightarrow [A, _] : C \rightarrow C \quad \text{or in cell form,} \quad C \begin{array}{c} \xrightarrow{id} \\ \Downarrow \Phi \\ \xrightarrow{[A, _]} \end{array} C. \quad (4.9)$$

An *id*-natural (M, R) -system is then a pair $\langle f, \Phi \rangle$ as above. We denote by $\text{id-MRS}(A, B)$, or simply $\mathbf{MRS}(A, B)$, the class of *id*-natural (M, R) -systems.

The naturality requirement makes this quite a rigid object; in short, a natural transformation of this type amounts to specifying an $(_ \otimes A)$ -algebra structure on each object $B \in C_0$. As a consequence of the following observations, *there are no nontrivial examples of id-natural (M, R) -systems in the category of sets.*

Lemma 4.2 ($\hookrightarrow$). Let $C = \mathbf{Set}$, with its usual Cartesian closed structure, and let A be a set; then the unique natural transformation $id \Rightarrow (_)^A$ is the one having components the constant maps $X \rightarrow X^A : x \mapsto \lambda a. x$.

Proof. This result is, in a sense, a consequence of the Yoneda lemma, but we offer an elementary argument instead of a one-liner, proving that a natural transformation $\alpha : id \Rightarrow (_)^A$ is determined

by its component at any singleton set $1 = \{\bullet\}$. Indeed, an element $x \in X$ can be thought of as an arrow $x : 1 \rightarrow X$, and thus we can invoke ‘naturality at x ’ for α , obtaining a commutative square

$$\begin{array}{ccc} 1 & \xrightarrow{\alpha_1} & [A, 1] \\ x \downarrow & & \downarrow x \circ - \\ X & \xrightarrow{\alpha_X} & [A, X] \end{array} \quad (4.10)$$

that is, $\alpha_X(x)a = x \circ \alpha_1(\bullet)$ as desired. $\square$

In a generic closed monoidal category $(C, \otimes, I)$ the situation is more interesting, even though the naturality condition is still quite restrictive. First, observe that the identity functor of C is isomorphic to the functor $[I, -]$ (I the monoidal unit of C); it is then possible to define a map

$$p : [C, C](id, [A, -]) \rightarrow C(A, I) \quad (\text{⌘})$$

of ‘projection’ so that the component of $\alpha : id \Rightarrow [A, -]$ at the monoidal unit I is in turn uncurried and composed with a unitor:

$$A \xrightarrow{\lambda_A^{-1}} I \otimes A \xrightarrow{\alpha_I} [A, I] \otimes A \xrightarrow{\epsilon} I. \quad (4.11)$$

It is also possible to define a map ι in the opposite direction ‘extending’ an arrow $a : A \rightarrow I$ into a natural transformation as follows: on the component at $X \in C_0$, $\alpha_X^a : X \rightarrow [A, X]$ is the composition

$$X \xrightarrow{\eta} [I, I \otimes X] \xrightarrow{[I, \lambda_X^{-1}]} [I, X] \xrightarrow{[a, X]} [A, X]. \quad (\text{⌘})$$

Lemma 4.3 (⌘). The composite $p \circ \iota$ is the identity of $C(A, I)$.

Proof. The composite $p \circ \iota$ sends an arrow $a : A \rightarrow I$ of C to the composite arrow marked in red in the diagram

$$\begin{array}{ccccccc} & & & & I \otimes I & & \\ & & & & \uparrow \lambda_I & & \\ & & & & \epsilon_{I \otimes I} & & \\ & & & & \uparrow \eta_I \otimes I & & \\ I & \xrightarrow{\lambda_I^{-1}} & I \otimes I & \xrightarrow{\eta_I \otimes I} & [I, I \otimes I] \otimes I & \xrightarrow{\epsilon_{I \otimes I}} & [I, I] \otimes I & \xrightarrow{\lambda_I} & I \\ \uparrow a & & \uparrow I \otimes a & & \uparrow id \otimes a & & \uparrow id \otimes a & & \uparrow \epsilon_I^{(A)} \\ A & \xrightarrow{\lambda_A^{-1}} & I \otimes A & \xrightarrow{\eta_I^{(I)} \otimes A} & [I, I \otimes I] \otimes A & \xrightarrow{[I, \lambda_I] \otimes A} & [I, I] \otimes A & \xrightarrow{[a, I] \otimes A} & [A, I] \otimes A \end{array} \quad (4.12)$$

Each square commutes, because of naturality, bifunctionality, or the cowedge condition of ϵ ; thus, $(p \circ \iota)(a) = \lambda_I \circ \lambda_I^{-1} \circ a = a$, as desired. $\square$

Remark 4.4 (Representable-coherence). As an immediate consequence of Lemma 4.3, ι is injective (a split monomorphism) and p is surjective (a split epimorphism); we call a coherent (M, R) -system *representable* if it lies in the image of ι .

Remark 4.5 (⌘). On the other hand, the composite $\iota \circ p$ is *not* the identity; we can build a monoidal (even Cartesian) category C such that ι is the inclusion of a singleton in a two-element set. We defer the human-readable proof to Proposition 4.23 below.

An important observation is now in order: despite its size, the literature on (M, R) -systems that we have been able to consult works, without exception, in a category of sets, where an (M, R) -system is made of functions; in that setting Lemma 4.2 shows that no nontrivial coherent (M, R) -system can exist, an obstruction which appears to have gone unremarked.

Remark 4.6 (On (M, R) -systems and classical set theory). The key property that allows us to reproduce the argument in Lemma 4.2 is that the terminal object of **Set** is also a *generator*: parallel maps that agree on all global elements are equal, or in other words the functor $\Gamma = \mathbf{Set}(1, -) : \mathbf{Set} \rightarrow \mathbf{Set}$ is faithful. Fortunately, this condition—called *well-pointedness* in [MLM92, p. 275]—

is a strong property for a Cartesian closed category, one we cannot expect in full generality and that rules out interesting examples of (M, R) -systems.

Indeed, in a (nondegenerate) well-pointed topos $\mathcal{E}$ the global sections functor $\Gamma_{\mathcal{E}} = \mathcal{E}(1, -)$ is faithful, so that element-theoretic reasoning is legitimate; every object $X \neq 0$ admits a global element, since otherwise the two coprojections $X \rightrightarrows 1 + 1$ would agree vacuously on points; and the topos is necessarily two-valued [MLM92, p. 274] and Boolean [MLM92, VI, Proposition 1]. The constraint is severe enough that, classically, **Set** is the only Grothendieck topos that is well-pointed.¹⁰

Examples of non-well-pointed toposes are correspondingly abundant. In the topos $\mathbf{Set}^G$ of G -sets for a nontrivial group G , the global elements of an object are its fixed points, so any free action gives an object with no points at all. In Proposition 4.23 we will indeed use a category of the form $\mathbf{Set}^G$ (with G the cyclic group of order 2) to build the counterexample of (M, R) -system alluded to in Remark 4.5.¹¹ As a rather informal aside, there is a line of research in the foundations of quantum mechanics that relates the absence of global points to the failure of a quantum system to admit a classical presentation, and that finds a natural home for such systems in a universe whose internal logic is intuitionistic.¹²

By contrast, coherent (M, R) -systems are relatively abundant in linear settings: let's consider a toy example in the category of vector spaces.

Corollary 4.7. Fix a field k ; in the category of vector spaces (so, allowing infinite-dimensional ones) over k , the set of coherent (M, R) -systems $\langle f, - \rangle$ such that $[f, -]_f = f$ can be big, and in particular it can be bigger than A in a way that can be estimated; for example, consider the (M, R) -system

$$A \xrightarrow{f} k \quad (-) \xrightarrow{\Phi} [A, -] \quad (4.13)$$

of vector spaces, where $A = k[T]$ is the polynomial ring in one indeterminate; then, $\mathbf{Vect}(A, k) \cong k[[T]]$ is the ring of formal power series over k . There is then a unique (M, R) -system for every formal power series $q(T) = \sum a_n T^n$, determined as the natural transformation $\Phi^q : id \Rightarrow [A, -]$ with components $\Phi_X^q(x) := (p \mapsto (\sum_n a_n p_n)x) \in [A, X]$ for $x \in X$, using the coefficients $p =$

¹⁰A note for the logicians: this is not a purely elementary phenomenon: well-pointedness together with an NNO and choice axiomatizes Lawvere's ETCS, which has many inequivalent models, so a well-pointed topos is better thought of as a *model of a certain classical set theory* than as **Set** itself. It is the non-elementary hypothesis of cocompleteness that collapses the distinction.

¹¹A second note for the logicians: more generally, no presheaf topos $[C^{\text{op}}, \mathbf{Set}]$ over a category with more than one object is well-pointed—witness the Sierpiński topos $\mathbf{Set}^{\rightarrow}$, simplicial sets, or the topos of graphs. Neither is a topos of sheaves $\mathbf{Sh}(X)$ for all but the most degenerate spaces X : already X a two-point discrete space gives $\mathbf{Sh}(X) \simeq \mathbf{Set} \times \mathbf{Set}$, which fails to be two-valued. Realizability toposes such as the effective topos **Eff** (see [V008]) fail for a different reason, being neither Boolean nor two-valued.

¹²More precisely, here's a footnote for the physicists: in topos-theoretic approaches to basic quantum mechanics like [IB98, DI08, HLS09], one replaces the noncommutative algebra $\mathcal{A}$ of observables of a quantum system with the poset $\mathcal{V}(\mathcal{A})$ of its commutative subalgebras, thought of as the 'classical contexts', i.e. those families of observables that admit a joint measurement. With this, it attaches to $\mathcal{A}$ the *spectral presheaf* $\underline{\Sigma} : \mathcal{V}(\mathcal{A})^{\text{op}} \rightarrow \mathbf{Set}$ assigning to each context its *Gel'fand spectrum*. The Kochen–Specker theorem [KS67] then reads, in this language, as the assertion that for $\mathcal{A} = B(\mathcal{H})$ with $\mathcal{H}$ a Hilbert space of dimension at least three, the object $\underline{\Sigma}$ admits no global sections $1 \rightarrow \underline{\Sigma}$. Thus, a quantum system carries no consistent global assignment of values to all its observables, only a coherent family of local, contextual ones. Our Lemma 4.2 records the same tension read from the other side, so that both settings are pushed outside **Set** for what is formally the same reason. A bridge between Rosen and this tradition has in fact been attempted by [Zaf12], who transports the apparatus onto Rosen's *modelling relation* rather than onto (M, R) -systems (no metabolism, repair map, or closure occurs there): *op. cit.* introduces a category $\mathcal{Z}$ of 'complex information structures' having objects partially commutative unital rings of observables, which is shown to be equivalent to a topos of sheaves over a small base $\mathcal{Y}$ of commutative rings. The mathematics underlying that equivalence is the well-known *nerve-realization adjunction*: for any $M : \mathcal{Y} \rightarrow \mathcal{Z}$ with $\mathcal{Y}$ small, the nerve $N_M = \mathcal{Z}(M, -) : \mathcal{Z} \rightarrow [\mathcal{Y}^{\text{op}}, \mathbf{Set}]$ exists unconditionally, and acquires a left adjoint $\text{Lan}_{\mathcal{Z}} M$ precisely when $\mathcal{Z}$ is cocomplete—a hypothesis *op. cit.* imposes on $\mathcal{Z}$ explicitly, citing [MLM92], where the adjunction itself is proved, while the restriction to an equivalence onto sheaves is the density of M for the chosen topology. Given this, *op. cit.* approach seems extendable to other settings: no step of that construction uses the ring structure, let alone its commutativity; the choice is motivated by quantum theory, where observables attached to a system form algebras, rather than from anything in Rosen. In any case, Zafiris' work seems to have no relation with (M, R) -systems, so we take this merely as an interesting aside.

$\sum_n p_n T^n$ of a polynomial $p \in A$; naturality in X is immediate, since for every linear map $g : X \rightarrow Y$, $g(\Phi_X^q(x)(p)) = (\sum_n a_n p_n)g(x) = \Phi_Y^q(g(x))(p)$. Conversely, every id -natural $\Phi : id \Rightarrow [A, -]$ arises this way: evaluating the component Φ_k at $X = k$ recovers a functional $q := \Phi_k(1) \in \mathbf{Vect}(A, k) \cong k[[T]]$, and naturality (applied to the map $k \rightarrow X$, $1 \mapsto x$) forces $\Phi_X(x) = \Phi_k(1) \cdot x$ for every X and $x \in X$, an instance of the general fact that, for F representable, natural transformations $id \Rightarrow F$ correspond bijectively to elements of $F(k)$.

(The remark above, sketching an example in $\mathbf{Chu}(\mathbf{Vect}_k, k)$, is *not* an instance of the present notion: as stated in its own text it produces an *incoherent* (M, R) -system in the different category $\mathbf{Chu}(\mathbf{Vect}_k, k)$, fixed at a single pair rather than natural in a varying target as required here. It is left commented out, and would belong instead in section 3 as a further Chu-space example alongside the Hilbert-space one, should it be developed further.)

Remark 4.8 (*id-coherent* (M, R) -systems and ends). Assume C is symmetric, or at least biclosed.

Under the uncurrying-recurrying operation of Definition 2.19 we can transform the *natural* transformation $\Phi^\natural : id \Rightarrow [A, -]$ into a *dinatural* transformation with components

$$\Phi_X^\natural : A \longrightarrow [X, X]. \quad (4.14)$$

This is a *wedge* in the sense of the dual of (2.21). But then, any id -coherent (M, R) -system can be uncurried-recurried and turned into a single ‘classifying map’

$$\bar{\Phi}^\sharp : A \longrightarrow \int_X [X, X] \quad (4.15)$$

with codomain the *end* of $[-, -] : C^{\text{op}} \times C \rightarrow C$ (of course, provided the limit that defines $\int_X [X, X]$ exists in C). Note that $\int_X [X, X]$ is a $\otimes$ -monoid in C , where the multiplication is the map

$$\left(\int_X [X, X] \right) \otimes \left(\int_Y [Y, Y] \right) \longrightarrow \int_{XY} [X, X] \otimes [Y, Y] \xrightarrow{(\star)} \int_Z [Z, Z] \quad (4.16)$$

where $(\star)$ is induced by $\int_{XY} [X, X] \otimes [Y, Y] \xrightarrow{\pi_{ZZ}} [Z, Z] \otimes [Z, Z] \xrightarrow{\circ} [Z, Z]$ and the unit $I \rightarrow \int_X [X, X]$ is induced by uncurrying and recurrying the unitors; but then, again assuming the colimit that defines $A^* = \sum_n A^{\otimes n}$ exists in C , we get that

Any id -coherent (M, R) -system on a category C that admits the end $\int_X [X, X]$ and countable coproducts induces a monoid homomorphism

$$(\bar{\Phi}^\sharp)^* : A^* \longrightarrow \int_X [X, X]. \quad (4.17)$$

Definition 4.9 (The profunctor of coherent (M, R) -systems, $\mathcal{MRS}$). The assignment $(A, B) \mapsto \mathbf{MRS}(A, B)$ defines a profunctor

$$\mathbf{MRS} : C^{\text{op}} \times C \longrightarrow \mathbf{Set} \quad (4.18)$$

that we call the *profunctor of (id-coherent) (M, R) -systems*.

The above definition contains a statement to prove, namely the fact that the dependence of $\mathbf{MRS}(A, B)$ on the pair (A, B) is indeed a functor of the correct type; given a morphism $(A, B) \xrightarrow[u]{v} (A', B')$ in $C^{\text{op}} \times C$, we must induce a function $\mathbf{MRS}(A, B) \rightarrow \mathbf{MRS}(A', B')$, in a way that preserves identities and composition: the function $\mathbf{MRS}(u, v)$ is defined by sending $\langle f, \Phi \rangle$ to

$$\langle v \circ f \circ u, u^* \Phi \rangle : A' \longrightarrow B'; \{ Y \xrightarrow{\Phi_h} [A, Y] \xrightarrow{[u, Y]} [A', Y] \} \quad (4.19)$$

where $u^* \Phi$ is the natural transformation $[u, \text{cod } -] \circ \Phi$, with components $(u^* \Phi)_h : [u, \text{cod } h] \circ \Phi_h$ at a generic object $h : X \rightarrow Y$ of $C^{[0 \rightarrow 1]}$. Evidently, $u^* \Phi$ defined so is natural, so the correspondence is well-defined.

Now, this assignment is compatible with compositions and it sends the identity arrow $(A, B) \xrightarrow{id_A}{id_B} (A, B)$ of the object (A, B) in $C^{\text{op}} \times C$ to the identity function.

Definition 4.10 (The fibration of (M, R) -systems, $\mathcal{U}$). To the profunctor of *id*-coherent (M, R) -systems one associates the *tabulator* (see Definition 2.24) $\mathcal{O}\mathbf{MRS}$ i.e. the category having

- objects are triples $(A, B; \xi)$ where $A, B \in C_0$ and $\xi = \langle\langle f, \Phi \rangle\rangle \in \mathbf{MRS}(A, B)$ is a coherent (M, R) -system based at (A, B) ;
- arrows $(A, B; \langle\langle f, \Phi \rangle\rangle) \rightarrow (A', B'; \langle\langle g, \Psi \rangle\rangle)$ are pairs (u, v) with $u : A' \rightarrow A$, $v : B \rightarrow B'$ in C (i.e. exactly a morphism $(A, B) \rightarrow (A', B')$ of $C^{\text{op}} \times C$) such that $\mathbf{MRS}(u, v)(\langle\langle f, \Phi \rangle\rangle) = \langle\langle g, \Psi \rangle\rangle$, i.e. $v \circ f \circ u = g$ and $u^* \Phi = \Psi$ (Definition 4.9).

By a similar token, we can define

Definition 4.11 ($\mathcal{U}$). The category of twisted elements of the profunctor in Definition 4.9 is the category $\mathcal{E}(\mathbf{MRS})$ having

- objects are the same triples $(A, B; \langle\langle f, \Phi \rangle\rangle)$ as above;
- arrows $(A, B; \langle\langle f, \Phi \rangle\rangle) \rightarrow (A', B'; \langle\langle g, \Psi \rangle\rangle)$ are pairs (u, v) with $u : A \rightarrow A'$ (reversed relative to the total category above, mirroring Definition 3.41) and $v : B \rightarrow B'$, such that $v \circ f \circ u = g$ and $[u, -] \circ \Psi = \Phi$ as natural transformations (which means, of course, componentwise equal).

Similarly to what happens for incoherent (M, R) -systems, there are equivalent presentations for Definition 4.1, useful to have at hand while working.

Remark 4.12. Given that $f : A \rightarrow B$ is an object of $C^{[0 \rightarrow 1]}$, a coherent (M, R) -system can be seen as a *pointed* natural transformation $\Phi : id_C \Rightarrow [A, id_C]$; thus, one can equivalently present the data of Definition 4.1 above as just a natural transformation Φ of the type above, with a distinguished component $\Phi_f : B \rightarrow [A, B]$ as another arrow of C .

This makes one suspect that there exists a ‘flatten’ functor $(-)^b$, sending a coherent (M, R) -system to the incoherent (M, R) -system $\langle\langle f, \Phi_f \rangle\rangle$; and indeed, when we will have defined the total category $\mathcal{O}\mathbf{MRS}$ of (coherent) (M, R) -systems in Definition 4.10, we will easily see that ‘evaluation at f ’ of Φ is a functor

$$(-)^b : \mathcal{O}\mathbf{MRS} \longrightarrow \mathbf{iMRS}. \quad (4.20)$$

Remark 4.13. With the type given in Definition 4.1, and by the universal property of $\mathbf{coAlg}[A, -]$ as an inserter (cf. Definition 2.7), Φ corresponds in turn to a unique functor

$$C \longrightarrow \mathbf{coAlg}[A, -]. \quad (4.21)$$

Clearly, $b\Phi = \Phi_f$ is a $[A, -]$ -coalgebra. One may therefore regard $\mathbf{MRS}(A, B)$ as the set of ‘basepoint-preserving’ functors

$$(C, f : A \rightarrow B) \longrightarrow (\mathbf{coAlg}[A, -], \Phi_B : B \rightarrow [A, B]). \quad (4.22)$$

Equally clearly, Φ may be regarded as an $(A \otimes -)$ -algebra, and dual considerations apply. We shall freely use either characterisation, according to which is more convenient in the discussion.

As soon as C is symmetric monoidal closed, Remark 4.8 absorbs the entire repair datum of an (M, R) -system into a single arrow with a fixed codomain. That codomain deserves a name.

Notation 4.14 (Bernstein centre of a category). We write

$$\mathfrak{z}C := \int_X [X, X] \quad (4.23)$$

for the end of Remark 4.8, we call it the *Bernstein centre* of C , and we say that C is *Bernstein-complete* when this end exists. The terminology is borrowed from representation theory, where Bernstein [Ber84] studies $\mathbf{End}(id_{\mathbf{Rep}(G)})$ for G a reductive p -adic group (and himself writes ‘centre’ in scare quotes); Remark 4.17 records what carries over and what does not. The notion is unrelated to the *Drinfeld–Joyal–Street* centre $\mathcal{Z}(C)$, which is a braided category rather than an object of C , and to the internal (Huq) centre of an internal group.

Lemma 4.15 (Representability lemma). Assume C is symmetric monoidal closed, complete, and Bernstein-complete. Then the profunctor of (M, R) -systems of Definition 4.9 is representable; more precisely, it is isomorphic to the functor $C(-, - \times \mathfrak{z}C)$.

Proof. Send a coherent (M, R) -system $\langle f, \Phi \rangle$ to the pair $(f, \bar{\Phi}^\#)$, where $\bar{\Phi}^\# : A \rightarrow {}_3C$ is the classifying map of Φ obtained by uncurrying and recurring as in (4.15). By Remark 4.8 the assignment $\Phi \mapsto \bar{\Phi}^\#$ is a bijection between repair families based at A and arrows $A \rightarrow {}_3C$, whence a bijection

$$\mathbf{MRS}(A, B) \cong C(A, B) \times C(A, {}_3C) \cong C(A, B \times {}_3C), \quad (4.24)$$

natural in A and in B . $\square$

Corollary 4.16 (Tabulator as a comma object). Under the same hypotheses, the category $\mathbb{O}\mathbf{MRS}$ of coherent (M, R) -systems is isomorphic to the comma category $(C / - \times {}_3C)$.

We use this to characterise the coherent versions of the (M, R) -systems sketched in Example 3.9—3.19, and to prove that $\mathbb{O}\mathbf{MRS}$ is complete and cocomplete whenever C is; the latter mirrors the incoherent Proposition 3.36.

Remark 4.17 (Elementary properties of the Bernstein centre). The Bernstein centre ‘classifies repair maps’ in the sense of Lemma 4.15. By Remark 4.8 it is moreover an internal monoid under composition of families, with unit $\eta : I \rightarrow {}_3C$ picking out the identity family; the projection ${}_3C \xrightarrow{\pi_I} [I, I] \cong I$ retracts η , so the unit is always a retract of ${}_3C$, and the only question is how much larger the latter is.

Its global points recover the classical centre of a category, $C(I, {}_3C) \cong \mathbf{End}(id_C)$, a commutative monoid by Eckmann–Hilton; this is the point of contact with [Ber84], and the analogy stops there. Commutativity is not inherited by ${}_3C$: for a small groupoid $\mathcal{G}$ one has

$${}_3([\mathcal{G}^{\text{op}}, \mathbf{Set}]) : x \mapsto \text{Aut}_{\mathcal{G}}(x), \quad (4.25)$$

with restriction along $u : x' \rightarrow x$ given by conjugation $\gamma \mapsto u^{-1}\gamma u$, and global points $\prod_{x \in \pi_0 \mathcal{G}} Z(\text{Aut}(x))$. Taking points is thus lossy: for $\mathcal{G} = BG$ the centre is G acting on itself by conjugation, while $\mathbf{End}(id) \cong Z(G)$ is a singleton as soon as G has trivial centre. It is the internal object that governs the theory, a coherent (M, R) -system carrying an arrow $A \rightarrow {}_3C$ and not a point of it. Note finally that $\mathbf{End}(id)$ requires no monoidal structure, whereas the definition of ${}_3C$ requires closedness.

Remark 4.18 (A category without a Bernstein centre). Bernstein-completeness does not follow from completeness of C : the end is indexed by the whole of C , and may fail to exist for size reasons already when C is essentially small and rigid, so that $\mathbf{Nat}(A \otimes -, id)$ is unproblematically a set.

Let k be a field and $C = \mathbf{Vect}_{\mathbf{Z}\text{-gr}}^{\text{fd}}$ the category of finite-dimensional $\mathbf{Z}$ -graded k -vector spaces and degree-preserving maps, with the convolution structure

$$(V \otimes W)_n = \bigoplus_{p+q=n} V_p \otimes W_q, \quad [V, W]_n = \prod_{m \in \mathbf{Z}} \text{Hom}_k(V_m, W_{m+n}), \quad (4.26)$$

and unit $I = k(0)$, where $k(n)$ denotes k placed in degree n . Both constructions stay finite-dimensional, so C is symmetric monoidal closed, and it is rigid, with $(V^\vee)_n = (V_{-n})^*$. Every object is a finite direct sum of copies of the $k(n)$, with $C(k(m), k(n)) = 0$ for $m \neq n$ and $C(k(n), k(n)) = k$; writing $\alpha_{k(n)} = c_n \cdot id$ for $\alpha \in \mathbf{End}(id_C)$, naturality along the inclusions $k(n) \rightarrow V$ forces α_V to act on V_n by c_n , and conversely every family $(c_n)_{n \in \mathbf{Z}}$ is natural, since morphisms preserve degrees. Hence $\mathbf{End}(id_C) \cong k^{\mathbf{Z}}$. If some $E \in C$ satisfied $C(A, E) \cong \mathbf{Nat}(A \otimes -, id)$, taking $A = I$ would give $E_0 \cong k^{\mathbf{Z}}$, contradicting $\dim_k E < \infty$.

Lemma 4.15 is not a triviality result, and the reduction to pairs $(A \rightarrow B, A \rightarrow I)$ that it seems to promise must be read with care. What it shows is that the repair datum is concentrated in the single arrow $\bar{\Phi}^\# : A \rightarrow {}_3C$; the theory therefore degenerates exactly when ${}_3C$ is *terminal*, which by the retraction of Remark 4.17 happens if and only if ${}_3C \cong I$ and I is itself terminal. (If ${}_3C$ is terminal, every $C(X, I)$ is nonempty, π_I being composable with the unique $X \rightarrow {}_3C$, and injects into $C(X, {}_3C)$ along η .) Outside the Cartesian world a trivial centre thus still leaves the datum $A \rightarrow I$, which can be substantial: ${}_3\mathbf{Vect}_k \cong k$, and the repair families based at A are the functionals $A \rightarrow k$ of Corollary 4.7. There remains to determine when ${}_3C$ collapses onto I .

Proposition 4.19 (Trivialisation criteria). Let C be monoidal closed. Each of the following implies that C is Bernstein-complete with ${}_3C \cong I$:

- (i) the unit I is a generator of C ;
- (ii) C is Cartesian closed and its subterminal objects form a generating family (for a topos, this says exactly that it is *localic*).

Whenever ${}_3C \cong I$ one has $\mathbf{Nat}(id, [A, -]) \cong C(A, I)$ for every A , the pair (p, ι) of Remark 4.4 consists of mutually inverse bijections, and every coherent (M, R) -system is representable in the sense of Remark 4.4.

Proof. Both cases rest on the same remark: if a family $u : U \rightarrow X$ is jointly epimorphic, so is $(A \otimes u)_u$, since $f \circ (A \otimes u) = g \circ (A \otimes u)$ for all u transposes to $f^\# \circ u = g^\# \circ u$ for all u , whence $f^\# = g^\#$ and $f = g$.

(i) The generalized elements $x : I \rightarrow X$ are jointly epimorphic, hence so are the $A \otimes x$; naturality of $\theta : A \otimes - \Rightarrow id$ along x gives $\theta_X \circ (A \otimes x) = x \circ \theta_I$ modulo the unitor $A \otimes I \cong A$, so $\theta \mapsto \theta_I$ is injective. Modulo the transposition $\mathbf{Nat}(A \otimes -, id) \cong \mathbf{Nat}(id, [A, -])$ this map is the p of Lemma 4.3, surjective because ι is a section of it; hence $\mathbf{Nat}(A \otimes -, id) \cong C(A, I)$ naturally in A , which is the universal property of ${}_3C$, and ${}_3C \cong I$ by Yoneda.

(ii) Here $I = 1$, and the generating objects need not have points: it suffices that they be too rigid to carry an operation. If U is subterminal then $U \rightarrow 1$ is monic, so any two arrows with codomain U and common domain agree; in particular $\theta_U = \pi_U$ for every natural $\theta : A \times - \Rightarrow id$. Naturality along $u : U \rightarrow X$ gives $\theta_X \circ (A \times u) = u \circ \pi_U = \pi_X \circ (A \times u)$, and the $A \times u$ are jointly epimorphic, so $\theta_X = \pi_X$. Thus $\mathbf{Nat}(A \times -, id)$ is a singleton naturally in A , as is $C(A, 1)$, and $1 = I$ has the universal property of ${}_3C$.

In either case $\mathbf{Nat}(id, [A, -]) \cong C(A, {}_3C) \cong C(A, I)$ by Lemma 4.15, and under this identification p becomes a bijection with inverse ι . $\square$

Remark 4.20 (Neither criterion is necessary). ${}_3C \cong I$ is not a disguised way of saying that I generates. Criterion (ii) refutes the converse of (i): in the Sierpiński topos $\mathbf{Set}^\rightarrow$, the smallest non-well-pointed presheaf topos (Remark 4.6), the object $P = (\emptyset \rightarrow 1)$ has no global points while admitting two distinct maps to $(\emptyset \rightarrow \{0, 1\})$, so 1 does not generate; but the representables $(1 \rightarrow 1)$ and P are subterminal and do, whence ${}_3(\mathbf{Set}^\rightarrow) \cong 1$. Nor is (ii) necessary: in $\mathbf{Cat}$ the only subterminals are 0 and $\mathbf{1}$, while $\mathbf{N}$ -graded sets under the Cauchy product and $\mathbf{Chu}(\mathbf{Set}, K)$ are not Cartesian at all, and the centre is trivial in all three (Remark 4.22). What survives is one implication: a nontrivial centre *requires* the unit to fail to generate. So it should be, since the centre measures the isotropy of C and not its supply of points: $\mathbf{Set}^{C_2}$ is nontrivial because its group acts on every object at once, by a natural automorphism of the identity, whereas $\mathbf{Set}^\rightarrow$ has as few points and no such automorphism.

Remark 4.21 (The enriched end always collapses). All of the above concerns the *ordinary* end, indexed by the underlying category of C . The self-enriched end is always I , by the enriched Yoneda lemma:

$$\int_X^C [X, X] \cong \int_X [[I, X], X] \cong I. \quad (4.27)$$

The construction is non-vacuous only in its unenriched reading, and a reader who enriches it silently will conclude that the whole notion collapses.

Remark 4.22 (Where the Bernstein centre is trivial, and where it is not). The centre trivialises in

- $\mathbf{Set}$, $\mathbf{Vect}_k$ and $R\mathbf{Mod}$ for R commutative, where it is 1 , k and R : criterion (i);
- localic toposes (sheaves $\mathbf{Sh}(X)$ on a space, $\mathbf{Sh}(L)$ on a locale, presheaves on a poset), by criterion (ii);
- $\mathbf{Chu}(\mathbf{Set}, K)$, where the unit does not generate and neither criterion applies: points $I \rightarrow X$ fix the first component of a natural $\theta : A \otimes - \Rightarrow id$ to be the projection, and, every state of X being a copoint $X \rightarrow I^\perp$, naturality along copoints determines the second component from the single component $\theta_{I^\perp}$; the resulting bijection $\mathbf{Nat}(A \otimes -, id) \cong \mathbf{Chu}(A, I)$ gives ${}_3C \cong I = (1, K, \text{ev})$;
- $\mathbf{N}$ -graded sets under the Cauchy product, and graded objects over any cancellative monoid, where the unit, a singleton in degree 0 , is far from generating, and triviality is read off the base as in the last item below;

- **Cat** under the Cartesian and under the funny tensor product, where **1** and **2** generate strongly and the end is **1**; and **Set**^N under the degreewise Cartesian product, where it is terminal.

It does not trivialise in

- **Set**^G, where $\int_X X^X \cong G$ with the conjugation action, the family being $\alpha_X(z, x) = z \cdot x$: our test case, since the leg $A \rightarrow {}_3C$ of Lemma 4.15 is then substantial data, and the internal monoid is noncommutative although its global points $Z(G)$ commute;
- **Set**^{B^N}, sets with an endomorphism, where the end is the constant object **N** with $\alpha_X(n, x) = f_X^n(x)$;
- Joyal species under the Cauchy product, i.e. Day convolution along (**Bij**, +), where the end is $I \amalg I$, the extra element acting on $F[S]$ by the transposition for $|S| = 2$ and trivially elsewhere; k -linear species give $k[\mathbf{Z}/2]$. Under the Cartesian structure one gets instead the groupoid formula of Remark 4.17, $S \mapsto \text{Aut}(S)$: the two tensors have different internal homs, hence different centres;
- more generally $[\mathcal{D}^{\text{op}}, \mathcal{V}]$ with Day convolution, for $\mathcal{D}$ monoidal with nontrivial centre: the criterion is read off the base, since $({}_3[\mathcal{D}^{\text{op}}, \mathcal{V}])(d) \cong \mathbf{Nat}(d \otimes -, id_{\mathcal{D}})$, so the end is I exactly when $\mathcal{D}$ has no nontrivial central endomorphisms, as for $\mathcal{D} = (\mathbf{N}, +)$ discrete.

The canonical nontrivial example thus remains **Set**^G, or **Set**^G for a groupoid $\mathcal{G}$. More generally still, **Set**^D is often Bernstein-complete, with ${}_3(\mathbf{Set}^D) : \mathcal{D} \rightarrow \mathbf{Set}$ sending B to the endomorphisms of $\text{dom} : \mathcal{D}/B \rightarrow \mathcal{D}$ in **Cat**/ $\mathcal{D}$; this is nontrivial as soon as some object of $\mathcal{D}$ has nontrivial endomorphisms, each $k : B \rightarrow B$ inducing $k_* : \mathcal{D}/B \rightarrow \mathcal{D}/B$ by postcomposition.

In a topos the two criteria have familiar names: (i) is well-pointedness (Remark 4.6), and (ii) is locality. A well-pointed topos is generated by 1, which is subterminal, so (i) implies (ii), and the triviality that Lemma 4.2 established under a hypothesis that classically pins down **Set** itself in fact holds over an arbitrary locale. Triviality is therefore not an artefact of set-theoretic foundations but the normal state of affairs in every space-like topos, and a nontrivial coherent (M,R)-system requires isotropy: one must leave the localic world, as the toposes **Set**^M of monoid and group actions do, their only subterminals being 0 and 1 and their global points the fixed points. The smallest such topos already suffices, and settles the debt contracted in Remark 4.5.

Proposition 4.23 ($\mathcal{C}_2$). In the topos **Set**^{C₂} of C_2 -sets, there is an object A (namely $A := I$, the monoidal unit/terminal object) admitting strictly more than one natural transformation $id \Rightarrow [A, -]$; consequently $\iota \circ p \neq id$ on $\text{Nat}(id, [A, -])$, as promised in Remark 4.5.

Proof. Write $C_2 = \{e, \sigma\}$ and $\mathcal{E} = \mathbf{Set}^{C_2}$, the Cartesian closed topos of sets with an involution; its unit I is the terminal object, a fixed point. Take $A := I$. Every morphism of $\mathcal{E}$ is equivariant, so the family $\tau_X := \sigma_X$ is a natural transformation $\tau : id \Rightarrow id$; composing with the isomorphism $i : id \xrightarrow{\cong} [I, -]$ of Definition 2.15 gives $i \circ \tau : id \Rightarrow [I, -]$. It differs from i : at the regular C_2 -set $A_{\text{reg}} := C_2$ and the base point e , the first returns $\sigma \cdot e = \sigma$ and the second e , the translation action being free.

Now $\mathcal{E}(A, I) = \mathcal{E}(I, I)$ is a singleton, so $p(i) = p(i \circ \tau) = id_I$, whence $\#\text{Nat}(id, [A, -]) > \#\mathcal{E}(A, I) = 1$; and $\iota(p(i \circ \tau)) = \iota(id_I) = i \neq i \circ \tau$, so $\iota \circ p \neq id$. $\square$

This is consistent with Proposition 4.19 in the only way it could be: 1 does not generate **Set**^{C₂}, a free action having no global points, ${}_3(\mathbf{Set}^{C_2}) \cong C_2$, and $i \circ \tau$ is the second global point of the centre.

Remark 4.24. Note that the rather heavy-handed machinery of topos theory plays no essential role here; a topos is just a category that is as close as possible to **Set** without having a terminal generator.

It is an interesting observation that in the category **Cat** of categories and functors, which is not a topos, the terminal category is *not* a generator either (the two-element subcategory $\{\bullet, [0 \rightarrow 1]\}$ is, see [AR94]), and yet a *id*-coherent (M,R)-system doesn't recover a truly new notion:

An *id*-coherent (M, R) -system in **Cat** boils down to an incoherent (M, R) -system in funny categories, as described in Example 3.18.

This is not in tension with Proposition 4.19: as recorded in Remark 4.22, **1** and **2** generate **Cat** strongly, and the end is **1**: failure of the terminal object alone to generate is necessary, but not sufficient, for a nontrivial centre.

4.2. Invariants of coherent (M, R) -systems. The categories we defined in Definition 4.10–4.11 come equipped with canonical forgetful functors, that we define at once.

Definition 4.25 (The first edge functor, $\mathcal{U}$). The *first edge functor*

$$[-]_f : \mathcal{O}\mathbf{MRS} \longrightarrow C^{[0 \rightarrow 1]} \quad (4.28)$$

sends a coherent (M, R) -system $\langle f, \Phi \rangle$ based at (A, B) to the first edge of its defining diagram, i.e. to the process map $f : A \rightarrow B$, regarded as an object of the arrow category. On a morphism (l, r) of $\mathcal{O}\mathbf{MRS}$, with $l : A \rightarrow A'$ and $r : B \rightarrow B'$, it returns the commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ l \downarrow & & \downarrow r \\ A' & \xrightarrow{g} & B' \end{array} \quad (4.29)$$

whose commutativity is the first of the two conditions defining a morphism of (M, R) -systems. Note that this is the very functor $[-]_f$ that Remark 3.1 wishes for and shows to be unavailable for incoherent (M, R) -systems: passing to the tabulator resolves the variance obstruction, and the wished-for process selector exists after all.

Definition 4.26 (The last edge functor, $\mathcal{V}$). The *last edge functor*

$$\mathcal{V}[-]_\Phi : \mathcal{E}(\mathbf{MRS}) \longrightarrow C^{[0 \rightarrow 1]} \quad (4.30)$$

sends a coherent (M, R) -system $\langle f, \Phi \rangle$ based at (A, B) , regarded as an object of the category of twisted elements, to the last edge of its defining diagram, i.e. to the component $\Phi_B : B \rightarrow [A, B]$ of its repair family, again as an object of the arrow category. A morphism of $\mathcal{E}(\mathbf{MRS})$ consists of $l : A' \rightarrow A$ and $r : B \rightarrow B'$, twisted as in Definition 2.26, and $\mathcal{V}[-]_\Phi$ returns the square

$$\begin{array}{ccc} B & \xrightarrow{\Phi_B} & [A, B] \\ r \downarrow & & \downarrow [l, r] \\ B' & \xrightarrow{\Psi_{B'}} & [A', B'] \end{array} \quad (4.31)$$

obtained by acting on the internal hom with l contravariantly and r covariantly. It is the coherent counterpart of the right leg of the span of Proposition 3.44, and the notation is deliberately the paper's own: square brackets $[-]_f$ mark a functor out of a total category, bags $\mathcal{V}[-]_\Phi$ one out of twisted elements, and the asymmetry between the two edge functors is exactly that the process map is already functorial on $\mathcal{O}\mathbf{MRS}$ while the repair map is only so after twisting.

Definition 4.27 (The twisted edge functor, $\mathcal{W}$). The *twisted edge functor*

$$\mathcal{W}[-]_f : \mathcal{E}(\mathbf{MRS}) \longrightarrow \mathbf{Tw}(C) \quad (4.32)$$

sends a coherent (M, R) -system $\langle f, \Phi \rangle$ based at (A, B) , regarded as an object of the category of twisted elements, to its process map $f : A \rightarrow B$ — like $[-]_f$ of Definition 4.25, but landing in the *twisted* arrow category rather than in $C^{[0 \rightarrow 1]}$. On a morphism of $\mathcal{E}(\mathbf{MRS})$, given by $l : A' \rightarrow A$ and $r : B \rightarrow B'$, it returns the twisted square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ l \uparrow & & \downarrow r \\ A' & \xrightarrow{g} & B' \end{array} \quad (4.33)$$

which is the shape of a morphism of $\mathbf{Tw}(C)$.

Taken together, the three functors just defined are the coherent counterparts of the incoherent ones, and the notation records which is which: $[-]_f$ is the process selector out of the total category — the one Remark 3.1 wishes for and shows to be unavailable incoherently — while $\wr[-]_f$ and $\wr[-]_\Phi$ are the two legs of the span of Proposition 3.44, out of twisted elements.

Definition 4.28 (The fibration of repairs, $\mathcal{R}$). The *fibration of repairs* **Rep** is the category of elements of the assignment $A \mapsto \text{Nat}(id, [A, -])$; equivalently, it is what remains of an (M, R) -system once the process map is forgotten and only the repair family is kept. Explicitly:

- objects are pairs (A, Φ) with $A \in C_0$ and $\Phi : id \Rightarrow [A, -]$ a natural transformation, i.e. the repair datum of Definition 4.1 on its own;
- an arrow $(A, \Phi) \rightarrow (A', \Phi')$ is a map $u : A \rightarrow A'$ of C such that $[u, -] \circ \Phi' = \Phi$, where $[u, -] : [A', -] \Rightarrow [A, -]$ is the natural transformation induced by u on internal homs;
- identities and composition are those of C .

The condition on arrows is precisely the second of the two conditions defining a morphism of (M, R) -systems, the first — the one constraining process maps — being simply absent. Proposition 4.30 makes that observation exact.

Remark 4.29. It is evident that when a repair map is presented in uncurried form, **Rep** is just another name for the category $C \ltimes_{\otimes} C$ of Definition 3.34, and the functor K of Proposition 4.30 is precisely the comparison that forgets the process map and keeps only the repair datum. [how does it fit in the square of \(3.61\)?](#)

Proposition 4.30 (The repairs coreflect into the total category, $\mathcal{R}$). The fibration of repairs **Rep** of Definition 4.28 is a coreflective subcategory of the total category $\oplus \mathbf{MRS}$ of Definition 4.10: there is an adjunction

$$\mathbf{Rep} \begin{array}{c} \xrightarrow{J} \\ \xleftarrow{K} \end{array} \oplus \mathbf{MRS} \quad (4.34)$$

in which J sends a repair system (A, Φ) to the (M, R) -system $\langle id_A, \Phi \rangle$ based at (A, A) , and K forgets the process map, retaining only the repair datum. The unit of $J \dashv K$ is an identity; equivalently J is fully faithful (which has already been observed), and thus K is a coreflector.

Remark 4.31. The content of the proposition is that a repair blueprint carries no less information than an (M, R) -system whose process map is an identity: choosing Φ and choosing $\langle id_A, \Phi \rangle$ are the same choice. Contrast this with the situation for the functors L, L' of Theorem 3.49 and Theorem 3.50, where the analogous comparison fails for incoherent systems.

We end the subsection stating the result that proves the completeness and cocompleteness of $\oplus \mathbf{MRS}$, which will make use of the ‘representability lemma’ Lemma 4.15 above.

4.3. Related notions of coherence. Here we collect several variations on the notion of *id*-naturality introduced above, each offering a different balance between semantic transparency and the goal of producing a genuine profunctor of (M, R) -systems: **cod**-natural systems (§4.3.1), which we show to be equivalent to *id*-natural ones; *A/C*-natural systems (§4.3.2), a semantically finer variant restricted to processes sharing the domain of f ; fully polymorphic, dinatural systems (§4.3.3); *U*-natural systems (§4.3.4) for a generic functor U ; and, at the top of the abstraction ladder, fully profunctorial systems (§4.3.5).

4.3.1. cod-natural (M, R) -systems. In a ‘cod-natural’ (M, R) -system based at A , there is a process $f : A \rightarrow B$, and the repair function has type

$$\Phi : \text{cod} \Rightarrow [A, \text{cod}]. \quad (4.35)$$

We denote $\text{cod-MRS}(A, B)$ the class of the (M, R) -systems in this sense.

It is evident how any *id*-natural (M, R) -system $\langle f, \Phi \rangle$ gives rise to a **cod**-natural one by whiskering: $\langle f, \Phi * \text{cod} \rangle$. One could then suspect that **cod**-natural (M, R) -systems constitute a more general concept than *id*-natural ones, but this is not true. We prove it in a moment, in Lemma 4.33 below.

Before that, let's spend a word on why, then, we choose **cod**-natural systems as a putative notion of coherent (M,R)-system. The reason in short is that these seem to be ‘semantically meaningful’.

In the intuition on (M,R)-systems that regards them as process-repair pairs, it is natural to think that Φ is a ‘blueprint for reconstruction’ modelled on processes more than on objects: Φ has a component $\Phi_h : Y \rightarrow [A, Y]$ for every *arrow* $h : X \rightarrow Y$, not just for any *object* Y of C . Evidently this recovers an incoherent (M,R)-system as the component at f of Φ :

$$\begin{array}{ccc} A & \xrightarrow{f} & B \longrightarrow [A, B] \\ & \parallel & \parallel \\ & \text{cod } f & \xrightarrow{\Phi_f} [A, \text{cod } f] \end{array} \quad (4.36)$$

Naturality in this case takes the following form:

For every morphism $(p, q) : h \rightarrow h'$ of $C^{[0 \rightarrow 1]}$, i.e. every commutative square

$$\begin{array}{ccc} X & \xrightarrow{h} & Y \\ p \downarrow & & \downarrow q \\ X' & \xrightarrow{h'} & Y' \end{array} \quad (4.37)$$

in C , the diagram

$$\begin{array}{ccc} Y & \xrightarrow{\Phi_h} & [A, Y] \\ q \downarrow & & \downarrow [A, q] \\ Y' & \xrightarrow{\Phi_{h'}} & [A, Y'] \end{array} \quad (4.38)$$

commutes, i.e. $[A, q] \circ \Phi_h = \Phi_{h'} \circ q$.

Now, despite the intuitive idea behind this choice having solid foundation, we are in the following situation:

Definition 4.32 (Lax epimorphism [LNS22]). A functor $f : A \rightarrow B$ is a *lax epimorphism* if, for every category C , the precomposition functor

$$_ - \circ f : \mathbf{Cat}(B, C) \longrightarrow \mathbf{Cat}(A, C) \quad (4.39)$$

is fully faithful.

It is easy to see that the following is true; we provide an elementary proof in the form of a remark, since the result can be established by means of a very elementary inspection.

Lemma 4.33 ($\mathcal{C}^{[0 \rightarrow 1]}$). The codomain functor $\text{cod} : C^{[0 \rightarrow 1]} \rightarrow C$ is a lax epimorphism. As a consequence, there is a bijection between *id*-natural (M,R)-systems and **cod**-natural ones.

Remark 4.34. We can cast the naturality condition of Φ on various ‘degenerate’ morphisms of $C^{[0 \rightarrow 1]}$ to obtain the following equations; the last one will entail the claim that whiskering with **cod** induces a bijection between classes of natural transformations

$$\mathbf{Cat}(C, C)(\text{id}, [A, _]) \cong \mathbf{Cat}(C^{[0 \rightarrow 1]}, C)(\text{cod}, [A, \text{cod}]). \quad (4.40)$$

- On degenerate squares of the form $\begin{array}{ccc} X & \xrightarrow{ku} & Z \\ u \downarrow & \parallel & \\ Y & \xrightarrow{k} & Z \end{array}$, for composable arrows k, u , naturality of Φ

means that the diagram

$$\begin{array}{ccc} Y & \xrightarrow{k} & Z \\ \Phi_u \downarrow & & \downarrow \Phi_{\text{id}_Z} \\ [A, Y] & \xrightarrow{k \circ _ -} & [A, Z] \end{array} \quad (4.41)$$

is commutative; this imposes the equation $k \circ \Phi_u \equiv \Phi_{id_Z} \circ k$. If $k = f$ we have that for all $u : X \rightarrow A$ the equation $f \circ \Phi_u \equiv \Phi_{id_B} \circ f$, whose right hand side does not depend on u , holds.

- So, when $k = id_Y$, and $u : X \rightarrow Y$, naturality of Φ means that the diagram

$$\begin{array}{ccc} Y & \xlongequal{\quad} & Y \\ \Phi_u \downarrow & & \downarrow \Phi_{id_Z} \\ [A, Y] & \xlongequal{\quad} & [A, Y] \end{array} \quad (4.42)$$

is commutative; this imposes the equation $\Phi_u \equiv \Phi_{id_Z}$.

The last equation is a strong requirement, in that it entails that $\Phi_u \equiv \Phi_{id_{\text{cod}_u}}$; hence the components of Φ are constant on the slices C/Y (because as a consequence, if u, u' have the same codomain, $\Phi_u = \Phi_{u'}$); the lax epimorphism condition can then be rephrased as the fact that the assignment $X \mapsto \Phi_{id_X}$ is a natural transformation $\Phi^\natural$ of type $\text{id} \Rightarrow [A, -]$, to which Φ is then equivalent.

As a corollary of Lemma 4.33 one associates to cod -coherent (M, R) -systems the same tabulator of Definition 4.10.

4.3.2. A/C -natural (M, R) -systems. In a ‘coslice-natural’ (M, R) -system, the repair function has type

$$\Phi : \text{cod}^A \Rightarrow [A, \text{cod}^A] \quad (4.43)$$

where $\text{cod}^A : A/C \rightarrow C$ is the codomain functor restricted to the subcategory $A/C \subseteq C^{[0 \rightarrow 1]}$ of arrows with domain A . A pair $\langle f, \Phi \rangle$ of this type, where $f : A \rightarrow B$, is a cod^A -coherent (M, R) -system; we denote by $\text{cod}^A\text{-MRS}(A, B)$ the class of all such systems.

The semantic interpretation for such a structure is even more straightforward: Φ is defined and natural on a narrower class of processes, those sharing the same domain with f ; it is then able to regenerate terms of type f if it can access the shape of the domain A of f .

Naturality for cod^A -coherent systems takes the following form:

For every morphism $q : h \rightarrow h'$ of the coslice category A/C , i.e. for every triangle

$$\begin{array}{ccc} & A & \\ h \swarrow & & \searrow h' \\ X & \xrightarrow{\quad q \quad} & X' \end{array} \quad (4.44)$$

in C , the diagram

$$\begin{array}{ccc} X & \xrightarrow{\Phi_h} & [A, X] \\ q \downarrow & & \downarrow [A, q] \\ X' & \xrightarrow{\Phi_{h'}} & [A, X'] \end{array} \quad (4.45)$$

commutes, i.e. $[A, q] \circ \Phi_h = \Phi_{h'} \circ q$.

Evidently this recovers an incoherent (M, R) -system as the component at f of Φ :

$$A \xrightarrow{f} B \xrightarrow{\Phi_f} [A, B] \quad (4.46)$$

Yet, under this notion of naturality there seems to be no way to turn the assignment $(A, B) \mapsto \text{cod}^A\text{-MRS}(A, B)$ into a profunctor

$$\text{cod}^A\text{-MRS} : C^{\text{op}} \times C \longrightarrow \mathbf{Set}. \quad (4.47)$$

Being able to ‘reindex’ along a morphism $(A, B) \xrightleftharpoons[u]{u} (A', B')$ means that one is able to take an (M, R) -system $\Phi : \text{cod}^A \Rightarrow [A, \text{cod}^A]$ and send it to one of type

$$u^* \Phi : \text{cod}^{A'} \Rightarrow [A', \text{cod}^{A'}], \quad (4.48)$$

which amounts to being able to fill the question marks in this diagram

$$\begin{array}{ccccc}
 A'/C & \xleftarrow{\quad ? \quad} & A/C & \xleftarrow{\quad ? \quad} & A'/C \\
 \downarrow \text{cod}^{A'} & & \downarrow \text{cod}^A & & \downarrow \text{cod}^{A'} \\
 & \swarrow & C & \xrightarrow{\quad \Phi \quad} & C & \searrow \\
 & & \downarrow [u, -] & & \\
 & & [A', -] & &
 \end{array}
 \quad (4.49)$$

The dotted arrow on the left poses no difficulty: it is filled by the functor $u^* : A/C \rightarrow A'/C$ sending an object $h : A \rightarrow X$ to $h \circ u : A' \rightarrow X$, so that the left-hand triangle commutes even strictly, since $\text{cod}^{A'}(h \circ u) = X = \text{cod}^A(h)$ by definition.

The dashed arrow is the problem. Filling it so that the whole diagram pastes correctly would require a comparison 2-cell running in the direction opposite to the one furnished by u^* itself. Indeed, if C admits finite colimits, u^* does have a left adjoint $u_!$, given by pushout along u ; but the canonical arrow this adjunction supplies points the wrong way to be pasted into the diagram above. What would be needed to correct this is a *right* adjoint P_u to u^* , which exists if and only if u is an isomorphism: assuming u^* is a left adjoint, it preserves the initial object of A/C , namely id_A ; but $u^*(\text{id}_A) = u$, regarded as an object of A'/C , and for this to be initial a necessary and sufficient condition is that u is an isomorphism in C . So, except in the degenerate case where u is invertible, no functorial reindexing along u exists, confirming that cod^A -coherent (M, R) -systems do not assemble into a profunctor.

The best we can do is assemble the cod^A -coherent (M, R) -systems into a *displayed* category, much like in Remark 3.24 above.

Definition 4.35 (The displayed category of cod^A -coherent (M, R) -systems). Define a category $\text{MRS}_{/C}$ as follows:

- objects are the ‘ cod^A -coherent’ (M, R) -systems as defined above, i.e. pairs

$$f : A \longrightarrow B \quad \Phi : \text{cod}^A \Longrightarrow [A, \text{cod}^A]; \quad (4.50)$$

- arrows of type $\langle\langle f, \Phi \rangle\rangle_{\text{cod}^A} \rightarrow \langle\langle g, \Psi \rangle\rangle_{\text{cod}^{A'}}$ consist of pairs $\ell : A \rightarrow A', r : B \rightarrow B'$ of arrows in C , such that the following equations hold:

$$\begin{array}{ccc}
 \begin{array}{ccc} A & \xrightarrow{f} & B \\ \ell \downarrow & & \downarrow r \\ A' & \xrightarrow{g} & B' \end{array} & \begin{array}{ccc} A/C & \xleftarrow{\ell^*} & A'/C \\ \downarrow & \swarrow & \downarrow \\ & C & \xrightarrow{\quad \Psi \quad} & C \\ & \downarrow [\ell, -] & \\ & [A', -] & \end{array} & = & \begin{array}{ccc} A/C & \xleftarrow{\ell^*} & A'/C \\ \downarrow & \swarrow & \downarrow \\ & C & \xrightarrow{\quad \Phi \quad} & C \\ & \downarrow [\ell, -] & \\ & [A, -] & \end{array}
 \end{array}
 \quad (4.51)$$

Thinking of Φ as a pointed natural transformation, one can clearly recover an incoherent (M, R) -system by ‘evaluation at f ’.

4.3.3. Fully polymorphic (M, R) -systems. In a ‘fully polymorphic’ (M, R) -system, the repair function has type

$$\Phi : \text{cod} \Rightarrow [\text{dom}, \text{cod}] \quad (4.52)$$

where $\text{dom}, \text{cod} : C^{[0 \rightarrow 1]} \rightarrow C$ are the obvious domain and codomain functors. Semantically, here one is abstracting the shape of Φ so that evaluation at $f : A \rightarrow B$ yields $\Phi_f : \text{cod } f \rightarrow [\text{dom } f, \text{cod } f]$ i.e. $\Phi_f : B \rightarrow [A, B]$.

The first caveat is that the repair cell Φ is now a *dinatural* transformation filling the cell

$$\begin{array}{ccc}
 (C^{[0 \rightarrow 1]})^{\text{op}} \times (C^{[0 \rightarrow 1]}) & \xrightarrow{\text{dom}^{\text{op}} \times \text{cod}} & C^{\text{op}} \times C \\
 \pi^r \downarrow & \Phi \nearrow & \downarrow [-, -] \\
 C^{[0 \rightarrow 1]} & & C \\
 & \searrow \text{cod} & \\
 & & C
 \end{array}
 \quad (4.53)$$

Dinaturality takes the following form:

For every morphism $(p, q) : f \rightarrow f'$ of $C^{[0 \rightarrow 1]}$, i.e. every commutative square $q \circ f = f' \circ p$ with $f : A \rightarrow B$, $f' : A' \rightarrow B'$, the diagram

$$\begin{array}{ccc}
 B & \xrightarrow{\Phi_f} & [A, B] \\
 q \downarrow & & \downarrow [A, q] \\
 B' & & \\
 \Phi_{f'} \downarrow & & \\
 [A', B'] & \xrightarrow{[p, B']} & [A, B']
 \end{array} \tag{4.54}$$

commutes, i.e. $[A, q] \circ \Phi_f = [p, B'] \circ \Phi_{f'} \circ q$; this is the classical hexagon condition for a dinatural transformation, here collapsing to a square because $\text{cod} \circ \pi'$ does not depend on the contravariant argument.

This yields a fibration, but for a very trivial reason: it does not involve the parameters A or B , so it just results in a ‘trivial bundle’ over $C^{\text{op}} \times C$.

Remark 4.36. To a fully polymorphic (M, R) -system we can associate the tabulator of Definition 2.24, and armed with the definition the reader will easily verify that, denoting by $\mathbf{N}(\text{dom}, \text{cod})$ the class of (di)natural transformations $\Phi : \text{cod} \Rightarrow [\text{dom}, \text{cod}]$ as in (4.53) above, the tabulator universal functor is isomorphic to the span

$$C \xleftarrow{\text{dom} \circ \pi^l} C^{[0 \rightarrow 1]} \times \mathbf{N}(\text{dom}, \text{cod}) \xrightarrow{\text{cod} \circ \pi^l} C \tag{4.55}$$

Remark 4.37 (Fully polymorphic (M, R) -systems and ends). Concretely, since Φ does not depend on (A, B) , a ‘fully polymorphic’ (M, R) -system is really just a single dinatural transformation $\Phi : \text{cod} \rightarrow [\text{dom}, \text{cod}]$, i.e. (via the classical correspondence between dinatural transformations and ends) an element of the end $\int_{f \in C^{[0 \rightarrow 1]}} [\text{dom} f, \text{cod} f]$, chosen once and for all, independently of any particular process $f : A \rightarrow B$ under consideration. Every incoherent (M, R) -system $\langle f, \Phi_f \rangle$ recovered by evaluation therefore shares literally the same underlying Φ , so this notion discards exactly the freedom (choosing a repair map tailored to the specific pair (A, B)) that made incoherent (M, R) -systems interesting in the first place; the ‘trivial bundle’ is trivial precisely because it carries no information about (A, B) beyond what is already fixed once, globally, by the single chosen element of the end.

4.3.4. *U-natural (M, R) -systems.* In a U -natural (M, R) -system, the repair function has type

$$\Phi : U \Rightarrow [A, U] \tag{4.56}$$

where $U : \mathcal{E} \rightarrow C$ is a generic functor, or a fibration over C .

This is, in our view, a reasonable notion more general than *id*-natural systems on which one might settle to obtain meaningful nontrivial invariants, with a small caveat that we explain below. One issue to overcome is the excessive freedom in choosing a natural canonical candidate for U ; how many ‘obvious’ functors of the type of U can be attached to a monoidal closed category like C ? One can require U to be a monoidal fibration in the sense of [Shu08], or a functor naturally induced by the monoidal structure; for example, if M is a $\otimes$ -monoid in C , we can consider the forgetful functor $U_M : C^M \rightarrow C$ from the category of objects equipped with an action of M ; or we can consider a monoidal closed category of modules over a von Neumann regular ring R , and leveraging the main result of [Agu97] we can consider a notion of ‘linear’ internal category $\mathbf{Cat}_{\otimes}(C)$ and the associated forgetful functor $U_{\text{Cat}, R} : \mathbf{Cat}_{\otimes}(C) \rightarrow C$. The choice is virtually unlimited: we can also consider C -shaped graphs and their forgetful functor, $U_{\text{Gph}} : C^{[0 \Rightarrow 1]} \rightarrow C$; C -valued (co)simplicial objects $U_{\Delta} : C^{\Delta} \rightarrow C$, with their evaluation at a given dimension, their realization, or totalization (assuming C has enough limits and colimits); we can consider comma categories, or strict pullbacks, obtained from either of the previous choices.

For a fixed U as in (4.56), naturality takes the following form:

For every morphism $h : X \rightarrow Y$ of $\mathcal{E}$, the diagram

$$\begin{array}{ccc} UX & \xrightarrow{\Phi_X} & [A, UX] \\ Uh \downarrow & & \downarrow [A, Uh] \\ UY & \xrightarrow{\Phi_Y} & [A, UY] \end{array} \quad (4.57)$$

commutes, i.e. $[A, Uh] \circ \Phi_X = \Phi_Y \circ Uh$.

This definition clearly yields a profunctor $U\text{-MRS} : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$, where the reindexing of the repair maps is simply given by composition: $\ell^* \Phi : U \Rightarrow [A, U] \Rightarrow [A', U]$ if $(A, B) \xrightleftharpoons[r]{\ell} (A', B')$ is an arrow in $\mathcal{C}^{\text{op}} \times \mathcal{C}$. Then we can attach a tabulator to U -coherent systems, specializing Definition 2.24.

However, the notion is a bit ‘slippery’ as it requires ad hoc constraints to recover incoherent systems.

Remark 4.38. As it is evident, one way in which we can recover incoherent (M,R)-systems is to fix the domain of U to be terminal, i.e. to require that $U : \{*\} \rightarrow \mathcal{C}$ picks the object B ; in greater generality some sort of compatibility between U and f has to be hardcoded to recover an incoherent (M,R)-system: something along the lines of ‘the fibre over f of U must be a singleton’, or at least pointed by a distinguished element $h : X \rightarrow Y$ such that

$$\begin{array}{ccc} UX & \xrightarrow{Uh=f} & UY \\ \parallel & & \parallel \\ A & & B \end{array} \quad (4.58)$$

Such conditions are difficult to ensure in practice given their vagueness; even more problematic is the fact that the compatibility with the pair $(h, Uh = f)$ has to be hardcoded as well in all proofs and constructions. So, this flavour of (M,R)-system is only a convenient one if the user abandons the idea of always being able to recover an incoherent system from a U -coherent one.

4.3.5. Profunctorial (M,R)-systems. This is the most abstract approach we have found so far (and it relies heavily on techniques from bi- and double category theory, for which we refer the reader to [Shu08, GP99a]).

Let $\mathcal{X}$ be a category with a terminal object; we call a profunctor $p : \mathcal{X} \times \mathcal{Y}^{\text{op}} \rightarrow \mathbf{Set}$ *reduced* if the curried functor $p^\dagger : \mathcal{X} \rightarrow [\mathcal{Y}^{\text{op}}, \mathbf{Set}]$ preserves the terminal object.

We can consider (M,R)-systems that are ‘fully profunctorial’: in those, Φ is a natural transformation of type $p \Rightarrow [A, \text{cod}]_*$, where p is reduced and $[A, \text{cod}]_*$ is the *companion* of the functor $[A, \text{cod}]$, defined as the profunctor

$$[A, \text{cod}]_* : \mathcal{C}^{\text{op}} \times A/C \longrightarrow \mathbf{Set} : (X, h : A \rightarrow Y) \longmapsto C(X, [A, Y]) \cong C(X \otimes A, Y). \quad (4.59)$$

In other words, in a language that is closer to double categories [GP99a], we consider a 2-cell

$$\begin{array}{ccc} C/B & \xrightarrow{p} & A/C \\ \text{dom} \downarrow & \not\cong_\Phi & \downarrow \text{cod} \\ C & \xrightarrow{[A, -]_*} & C \end{array} \quad (4.60)$$

in the double category $\mathbb{Cat}$ of categories and profunctors.

Note that since p is reduced, the curried functor $p^\dagger : C/B \rightarrow [(A/C)^{\text{op}}, \mathbf{Set}]$ maps the terminal object of C/B , namely id_B , to the terminal object of $[A/C, \mathbf{Set}]$, namely the constant functor at a terminal object of $\mathbf{Set}$; in other words, $p(\text{id}_B, v)$ is a singleton for every $v \in A/C$. It is then immediate to see that this notion recovers the incoherent (M,R)-systems of Definition 3.5 by evaluating Φ at the component $(\text{id}_B, f : A \rightarrow B)$, since

$$\Phi_{\text{id}_B, f} : p(\text{id}_B, f) = \{*\} \longrightarrow C(\text{dom}(\text{id}_B), [A, \text{cod}(f)]) = C(B, [A, B]). \quad (4.61)$$

Definition 4.39 (Profunctorial (M,R)-system, $\llbracket \mathcal{U} \rrbracket$). Let C be a monoidal closed category; let A, B be objects of C .

A *profunctorial (M,R)-system* based at (A, B) consists of a triple $\llbracket f, p, \Phi \rrbracket$ where

- $f : A \rightarrow B$ is an arrow of C ;
- $p : (C/B) \times (A/C)^{\text{op}} \rightarrow \mathbf{Set}$ is a profunctor;
- $\Phi : p \Rightarrow [A, \text{cod}]_*$ is a cell as in (4.60).

Call $\mathbf{\Pi}\text{-MRS}(A, B)$ the collection of all profunctorial (M,R)-systems based at (A, B) .

Evidently, the profunctor p varies with the fibre because it is (more precisely, its domain and codomain are) indexed over A, B . Resorting to the well-known graphical representation of cells in the double category of profunctors, we can depict a given profunctorial (M,R)-system based at (A, B) as a pair

$$\begin{array}{ccc} A & C/B \xrightarrow{p} A/C & \\ f \downarrow & \text{dom} \downarrow \quad \not\Rightarrow_{\Phi} \quad \downarrow \text{cod} & \\ B & C \xrightarrow{[A, -]_*} C & \end{array} \quad (4.62)$$

We now aim to prove that

Proposition 4.40. The assignment $(A, B) \mapsto \mathbf{\Pi}\text{-MRS}(A, B)$ is a profunctor

$$\mathbf{\Pi}\text{-MRS} : C^{\text{op}} \times C \longrightarrow \mathbf{Set}. \quad (4.63)$$

Remark 4.41. This proposition carries no Agda marker. The file `Variants/Profunctorial.agda` formalizes the profunctorial (M,R)-systems of Definition 4.39 and proves that they form a setoid, but it does not build the infrastructure for the conjoint of v .

To this end, given an arrow $(A, B) \xrightleftharpoons[v]{u} (A', B')$ in $C^{\text{op}} \times C$, we have to exhibit reindexings

$$\mathbf{\Pi}\text{-MRS}(u, v) : \mathbf{\Pi}\text{-MRS}(A, B) \longrightarrow \mathbf{\Pi}\text{-MRS}(A', B'). \quad (4.64)$$

Let's argue graphically; given a system like in (4.62), and $(A, B) \xrightleftharpoons[v]{u} (A', B')$, we define the image of $\llbracket f, p, \Phi \rrbracket$ under $\mathbf{\Pi}\text{-MRS}(u, v)$ as the cell

$$\begin{array}{ccccccc} A' & \xrightarrow{u} & A & & C/B' & \xrightarrow{v^\dagger} & C/B \xrightarrow{p} A/C \xrightarrow{u_\dagger} A'/C \\ & & f \downarrow & & \text{dom} \downarrow & \text{dom} \downarrow & \not\Rightarrow_{\Phi} \quad \downarrow \text{cod} \quad \downarrow \text{cod} \\ B' & \xleftarrow{v} & B & & C & \xrightarrow{[A, -]_*} & C \xrightarrow{[A', -]} C \\ & & & & & \downarrow u^* & \\ & & & & & [A', -] & \end{array} \quad (4.65)$$

where $u_\dagger$ and $v^\dagger$ are the profunctors respectively defined as follows; ‘companion’ and ‘conjoint’ have here the meaning they have in double category theory [Shu08], and the notation for the former is the one already used for $[A, \text{cod}]_*$ above.

- $u_\dagger : A/C \rightarrow A'/C$ is the *companion* of the functor

$$u/C : A/C \longrightarrow A'/C : (a : A \rightarrow X) \longmapsto (a \circ u : A' \rightarrow X) \quad (4.66)$$

induced by $u : A' \rightarrow A$ between the coslices: its value at $a : A \rightarrow X$ and $a' : A' \rightarrow X'$ is the hom-set $(A'/C)((u/C)a, a')$, i.e. the set of arrows $h : X \rightarrow X'$ of C such that $h \circ a \circ u = a'$.

- $v^\dagger : C/B' \rightarrow C/B$ is the *conjoint* of the functor

$$C/v : C/B \longrightarrow C/B' : (x : X \rightarrow B) \longmapsto (v \circ x : X \rightarrow B') \quad (4.67)$$

induced by $v : B \rightarrow B'$ between the slices: its value at $x' : X' \rightarrow B'$ and $x : X \rightarrow B$ is the hom-set $(C/B')(x', (C/v)x)$, i.e. the set of arrows $h : X' \rightarrow X$ of C such that $v \circ x \circ h = x'$.

Both exist for every functor, $\mathbb{Cat}$ being an equipment [Shu08], and the choice of one or the other is forced by the direction of the reindexing: u/C and C/v both point from the slices of (A, B) to those of (A', B') , so p cannot be restricted along them and must be extended instead. This is exactly what pasting a conjoint on the left and a companion on the right achieves: the composite $v^\dagger \odot p \odot u_\dagger$ is the left extension of p along the pair $(C/v, u/C)$, with value

$$(v^\dagger \odot p \odot u_\dagger)(x', a') \cong \int^{x \in C/B} \int^{a \in A/C} (C/B')(x', (C/v)x) \times p(x, a) \times (A'/C)((u/C)a, a'). \quad (4.68)$$

The remaining datum of the reindexed system is the cell $\Phi' : v^\dagger \odot p \odot u_\dagger \Rightarrow [A', \text{cod}]_*$, obtained by pasting Φ with the 2-cell $u^* : [A, -]_* \Rightarrow [A', -]_*$ of the diagram above, itself induced by precomposition with u ; the metabolism of the reindexed system is $v \circ f \circ u : A' \rightarrow B'$.

4.4. Closure under efficient causation. Closure under efficient causation is the claim that a system contains, within its own organization, the causes responsible for its activity. An “efficient cause” here is a function responsible for a transformation, not merely a material component. Rosen’s relational formulation therefore concerns organizational rather than material closure: every such cause is “entailed within the system” [Lou09, pp. 215–220]. For an (M, R) system this is expressed by a metabolism $f : A \rightarrow B$, a repair/replacement map $\Phi : B \rightarrow [A, B]$, and a product $b \in B$ such that $\Phi(b) = f$. The product therefore provides the value at which repair reconstructs the metabolism, so no external efficient cause is required. This is the Set-level formulation of metabolic closure. “Organizational invariance” means that this functional organization is maintained even while the material components participating in it change; Letelier et al. identify it with metabolic closure [LSAGA⁺04, pp. 450–455]; [LSAGA⁺06, pp. 949–958]. Louie also gives its diagrammatic counterpart: each connected component must contain a closed path through all efficient-cause arrows [Lou09, pp. 219–220].

Several authors have tried to make this proposal mathematically precise. In Rosen’s extended diagram, the higher-order map β is the “repair-of-repair” map which assigns to a metabolism f the repair map Φ that realizes it. Letelier et al. emphasize that β must recover repair from f , so closure involves evaluation and uniqueness/invertibility, not mere feedback [LSAGA⁺06, pp. 953–961]. More explicitly, with

$$\text{ev} : [A, B] \times A \longrightarrow B, \quad (g, a) \longmapsto g(a), \quad (4.69)$$

closure asks for a section of the relevant evaluation map in the following pointwise sense: the metabolism f must lie in the image of evaluation at a product b , namely $\Phi(b) = f$; β records the corresponding inverse choice. Cárdenas et al. similarly insist that the internal witness must not be confused with the product object B , and argue that this inverse-evaluation structure is missing from a simple lambda-calculus encoding [CLG⁺10, pp. 80–91]. By contrast, Mossio, Longo, and Stewart translate Rosen’s three equations into untyped λ -calculus: $B = f(A)$, $f = \Phi(B)$, and $\Phi = B(f)$. Substitution reduces these to a single self-application equation, which they solve explicitly with the fixed-point combinator Y : for a suitable term G , take $f = YG$, so that $f = Gf$; B and Φ are then obtained by substitution [MLS09, pp. 489–498]. Since untyped λ -calculus is a canonical formalism for computation, they conclude that this expression of closure is computable. Cárdenas et al. reply that the construction does not represent the inverse-evaluation requirement, so the dispute concerns what a faithful formalization must preserve: self-reference alone, or self-reference together with repair’s internal inverse structure [CLG⁺10, pp. 80–91].

The definition below transports the Set-level witness into a monoidal closed category: $b \in B$ becomes a generalized element $b_0 : I \rightarrow B$, and $\Phi(b) = f$ becomes the displayed evaluation square. A closure point means that seeding repair at b_0 regenerates the whole process f . This generalized-element formulation, and the equivalence of its two closed-category presentations, are extensions proved here rather than results attributed to Rosen or Letelier. Naturality can strengthen closure for one process to a universal point for all processes with input A , as explained below.

We use *metabolic closure* for this specific (M, R) condition. Related proposals should not be identified with it without further hypotheses: Montévil and Mossio’s account concerns mutual dependence among constraints at different scales [MM15, pp. 179–191], while Brown and Vitadello’s account defines network closure through cycles of mutual process enablement [BV25].

Both are relevant neighboring notions, but neither is equivalent to the single-witness condition $\exists b \Phi(b) = f$ by itself.

Definition 4.42 ($\mathcal{C}/\mathcal{C}$). A coherent (M, R) -system *allows metabolic closure* if there exists a generalized element $b_0 : I \rightarrow B$ such that one of these two equivalent conditions holds:

- the square

$$\begin{array}{ccc} I \otimes A & \xrightarrow{b_0 \otimes A} & B \otimes A \\ \downarrow \rho & & \downarrow \hat{\Phi} \\ A & \xrightarrow{f} & B \end{array} \quad (4.70)$$

is commutative ($\hat{\Phi} : B \otimes A \xrightarrow{\Phi \otimes A} [A, B] \otimes A \xrightarrow{\epsilon_B} B$ is the mate of Φ under the monoidal closed adjunction of $\mathcal{C}$);

- the square

$$\begin{array}{ccc} I & \xrightarrow{b_0} & B \\ j_A \downarrow & & \downarrow \Phi \\ [A, A] & \xrightarrow{[A, f]} & [A, B] \end{array} \quad (4.71)$$

is commutative, where $j_A : I \rightarrow [A, A]$ is the closed-category structure map of Definition 2.15, so that $[A, f] \circ j_A$ is the Eilenberg–Kelly name $\bar{f} : I \rightarrow [A, B]$ of f ; this reformulates the first condition purely in terms of the closed structure, without appeal to the monoidal unit ρ .

In such a situation, b_0 is called a *closure point*, and the system is ‘self-sustained’ in the sense that the repair map, once seeded at the closure point b_0 , already reconstructs the metabolism f in its entirety: no information about the process $A \rightarrow B$ needs to be supplied from outside B itself, since $b_0 \in B$ together with Φ suffices to regenerate f . This is the precise categorical rendering of Rosen’s ‘closure under efficient causation’: the efficient cause of the metabolic map f (namely, the repair component Φ evaluated at b_0) is itself already located inside the system, rather than being an external datum.

Remark 4.43. When studying categorical semantics for type theories (in what is perhaps a broad sense of ‘type theory’), the identity arrow is sometimes referred to as the ‘universal term’, in that it represents the type tautology $\{A\} \vdash A$ (the left hand side being a *context* $\Gamma = \{A\}$ from which A can be deduced). Evaluation at the identity arrow is also what makes Yoneda lemma work—and the two perspectives are intimately connected: in a sense, Yoneda lemma is what justifies, semantically, why the deduction of A from the singleton context $\{A\}$ is special among all deductions.

Here, the presence of all components for Φ means in particular that we can access the *universal* one

$$\Phi_{id_A} : A \longrightarrow [A, A], \quad (4.72)$$

whence the natural question: can there be a relation between metabolic closure, as just defined, and the property of finding a generalized element $a_0 : I \rightarrow A$ with the property that the square

$$\begin{array}{ccc} I \otimes A & \xrightarrow{a_0 \otimes A} & A \otimes A \\ \lambda_A^{-1} \uparrow & & \downarrow \hat{\Phi} \\ A & \xlongequal{\quad} & A \end{array} \quad (4.73)$$

is commutative? (In the category of sets, where A has genuine points and functions can be extensionally equivalent, this question becomes: is there $a_0 \in A$ such that $\Phi(a_0) \equiv id_A : A \rightarrow A$ —extensional equality of functions?)

If yes, then (again using λ -notation, trusting this will not cause confusion in the reader) naturality of Φ entails that $f(a_0)$ is a closure point, because

$$\lambda t. \Phi_f(f(a_0))t = \lambda t. (f \circ \Phi_{1_A}(a_0))t = \lambda t. ft; \quad (4.74)$$

we observed that

Assuming Φ is a natural transformation, the burden of ‘allowing metabolic closure’ for the (M, R) -system of which it is part is on the map $\Phi_{1_A} : A \rightarrow [A, A]$, which is specified in terms of the input A of the system alone; if 1_A is in the image of Φ_{1_A} , no matter what process $f : A \rightarrow B$ is considered, the (M, R) -system $\langle\langle f, \Phi \rangle\rangle$ will have metabolic closure.

Definition 4.44 (Universal closure point, $\mathcal{U}$). When there exists a_0 such that $\Phi_A(a_0) \equiv id_A$, we say that a_0 is a *universal closure point* for the (M, R) -system $\langle\langle f, \Phi \rangle\rangle$.

Universal closure points are scarce, and it is worth recording exactly how scarce, since the question of how many can a single system have, and what distinguishes them, has an unusually complete answer.

Unravelling Definition 4.44, a universal closure point is a point $a_0 \in C(I, A)$ with $\Phi_A \circ a_0 = j_A$, where $j_A : I \rightarrow [A, A]$ is the name of the identity (Definition 2.15); so the universal closure points of $\langle\langle f, \Phi \rangle\rangle$ form the fibre of

$$C(I, \Phi_A) : C(I, A) \longrightarrow C(I, [A, A]) \cong C(A, A) \quad (4.75)$$

over id_A , equivalently the set of points of the pullback of j_A along Φ_A . Three consequences are immediate: the notion depends only on the repair map Φ , and not on the process f (which is what makes it *universal*, cf. the quotation in Remark 4.43); a system has at most $\#C(I, A)$ of them, hence possibly none; and it has at most one as soon as Φ_A is a monomorphism.

That last condition is not an ad hoc hypothesis: ‘two repair maps that agree at one value agree everywhere’ is precisely the stringency condition under which Rosen constructs his replication map as an inverse evaluation [Ros66b, pp. 149–151], and which Louie reads as the abstract form of the one-gene-one-enzyme hypothesis [Lou09, §§11.15–11.17]; we return to it in subsection 4.5. Under representability the picture becomes sharper still.

Proposition 4.45 (Universal closure points, classified). Let C be symmetric monoidal closed and Bernstein-complete, and let $\langle\langle f, \Phi \rangle\rangle$ be a coherent (M, R) -system based at (A, B) , with classifying map $\bar{\Phi}^\# : A \rightarrow {}_3C$ as in Lemma 4.15. For a point $a_0 : I \rightarrow A$ write

$$\text{orb}(a_0) : {}_3C \cong I \otimes {}_3C \xrightarrow{a_0 \otimes {}_3C} A \otimes {}_3C \xrightarrow{A \otimes \pi_A} A \otimes [A, A] \xrightarrow{\epsilon_A} A \quad (4.76)$$

for the associated ‘orbit map’. Then a_0 is a universal closure point if and only if

$$\text{orb}(a_0) \circ \bar{\Phi}^\# = id_A. \quad (4.77)$$

In particular, a coherent (M, R) -system admits a universal closure point only if its classifying map is a split monomorphism; A is then a retract of the Bernstein centre of C .

Proof. By construction of $\bar{\Phi}^\#$ (Remark 4.8 and (4.15)), the mate $\hat{\Phi}_X : X \otimes A \rightarrow X$ of the component Φ_X factors as $\epsilon_X \circ (X \otimes \pi_X) \circ (X \otimes \bar{\Phi}^\#)$. The condition $\Phi_A \circ a_0 = j_A$ of Definition 4.44 transposes to $\hat{\Phi}_A \circ (a_0 \otimes A) = \lambda_A$, and substituting the factorisation turns the left-hand side into $\text{orb}(a_0) \circ \bar{\Phi}^\#$ modulo the unitor. The last assertion is the observation that the displayed equation exhibits $\text{orb}(a_0)$ as a retraction of $\bar{\Phi}^\#$. $\square$

Corollary 4.46 (The Cartesian case is degenerate). Let C be Cartesian closed. A coherent (M, R) -system based at (A, B) admits a universal closure point if and only if $A \cong 1$, and in that case the point is unique.

Proof. If $A \cong 1$ then $[A, A] \cong 1$ is terminal, so $\Phi_A \circ a_0 = j_A$ holds for the unique point a_0 of A . Conversely, let a_0 be a universal closure point and put $h := a_0 \circ !_A : A \rightarrow A$, the constant map at

a_0 . Naturality of Φ at h reads $\Phi_A \circ h = [A, h] \circ \Phi_A$; precomposing with a_0 and using $!_A \circ a_0 = id_1$ gives

$$j_A = \Phi_A \circ a_0 = \Phi_A \circ h \circ a_0 = [A, h] \circ \Phi_A \circ a_0 = [A, h] \circ j_A, \quad (4.78)$$

and $[A, h] \circ j_A$ is the name of h . Names of arrows being in bijection with arrows, $h = id_A$; together with $!_A \circ a_0 = id_1$ this exhibits a_0 as an isomorphism $1 \cong A$. $\square$

Example 4.47 (The linear case: a summand of the unit). Outside the Cartesian world the argument above is unavailable, there being no map $A \rightarrow I$ to build h from, and universal closure points become genuine data, though still a rare one. Take $C = R\mathbf{Mod}$ for R commutative, so that ${}_3C \cong R = I$ by Proposition 4.19, and a coherent repair family is (as in Corollary 4.7) of the form $\Phi_X(x)(a) = q(a) \cdot x$ for a unique $q \in \text{Hom}_R(A, R)$. A point $a_0 \in A$ is a universal closure point precisely when $q(a)a_0 = a$ for every $a \in A$, i.e. when $a \mapsto q(a)a_0$ retracts q . Writing $e := q(a_0)$, applying q to $q(a_0)a_0 = a_0$ gives $e^2 = e$, and q identifies A with the summand $eR \subseteq R$, with a_0 corresponding to e . Conversely every idempotent $e \in R$ yields such a system, on $A := eR$ with q the inclusion. The point is moreover unique: applying q to $q(a)a_0 = a$ gives $q(a)e = q(a)$, hence $q(ea) = e q(a) = q(a)$ and, q being injective, $ea = a$ for every $a \in A$; so if a_1 is another universal closure point then $q(a)(a_0 - a_1) = 0$ for every a , and taking $a = a_0$ yields $a_0 - a_1 = e(a_0 - a_1) = q(a_0)(a_0 - a_1) = 0$. Thus in $\mathbf{Vect}_k$ only $A = 0$ and $A \cong k$ admit universal closure points, and each admits exactly one.

Remark 4.48 (What remains open). In every setting we have computed, the universal closure point is unique when it exists, and Proposition 4.45 explains why: it must be a point of A whose orbit map retracts a fixed arrow $\bar{\Phi}^\sharp$, and retractions are hard to come by in the small objects that survive the retract condition. The fibre description above shows, however, that nothing formal forbids several: a system with two distinct universal closure points is exactly a failure of injectivity of Φ_A concentrated over the name of id_A , and we do not know an example. Note also that Corollary 4.46 does not make the notion vacuous in the Cartesian case, only degenerate in its input: $A \cong 1$ still leaves the process $f : 1 \rightarrow B$ and the repair family Φ entirely free, and by Proposition 4.23 the latter can be nontrivial.

Proposition 4.49 (Reindexing is compatible with metabolic closure, $\mathcal{C} \circ \mathcal{U}$). Let $(u, v) : (A, B) \rightarrow (A', B')$ be a morphism in the indexing category, so $u : A' \rightarrow A$ and $v : B \rightarrow B'$. If $\langle f, \Phi \rangle$ has closure point $b_0 : I \rightarrow B$, then its reindexing has closure point

$$b'_0 := v \circ b_0 : I \rightarrow B'. \quad (4.79)$$

Proof. Write Φ' for the reindexed repair family and $f' = v \circ f \circ u$. We must show that evaluation of Φ' at b'_0 is the curry of $f' \circ \rho^{-1}$. By the definition of reindexing, the left-hand side is $[u, 1_{B'}] \circ \Phi'_{f'} \circ v \circ b_0$. The repair family is constant on arrows with the same codomain, so its component at f' may be replaced by its component at $1_{B'}$. Naturality along v then gives

$$\Phi_{1_{B'}} \circ v = [1_B, v] \circ \Phi_{1_B}. \quad (4.80)$$

Using these identities, followed by the closure equation for b_0 , the left-hand side becomes

$$[u, 1_{B'}] \circ [1_B, v] \circ \text{curry}(f \circ \rho^{-1}). \quad (4.81)$$

Currying commutes with postcomposition, and the defining mate identity for the internal hom says that precomposing the curried map with $[u, 1_{B'}]$ is the same as currying after precomposition by $1_B \otimes u$. Consequently this is

$$\text{curry}((v \circ f \circ \rho^{-1}) \circ (1_B \otimes u)). \quad (4.82)$$

Finally, naturality of the left unitor rewrites the displayed composite as $(v \circ f \circ u) \circ \rho^{-1} = f' \circ \rho^{-1}$. This is exactly the closure equation for the reindexed system, so b'_0 is a closure point. $\square$

4.5. Extended (M,R)-systems, or “quis reficiet ipsos refectores?”. The repair component of an (M,R)-system is a piece of machinery like any other, and machinery wears out. In [Cas88, p. 145] Casti puts the question this subsection is about in the bluntest possible form: ‘who repairs the repairers? It would lead to a useless infinite regress to introduce another level of repair mechanisms, but what is the alternative?’.

The same difficulty is stated by Letelier et al. as the observation that ‘the same wear-and-tear argument that was applied to M applies equally well to the R ’, so that invoking a second-level repairer ‘just raises the new question of how to replace the second-level repairers, and is thus no solution at all’ [LSAGA⁺04, p. 451].

Rosen’s own answer is a *replication* map. Already in [Ros59, Ros66b] he observes that a set X embeds into $H(H(X, Y), Y)$ by sending x to the evaluation $\hat{x} : f \mapsto f(x)$ (‘the abstract version of the familiar embedding of a vector space into its second dual space’, as he puts it in [Ros72]), and applies this with $X = B$, $Y = H(A, B)$: an element $b \in B$ then yields $\hat{b} \in H(H(B, H(A, B)), H(A, B))$, and when $\hat{b}$ is injective its inverse is a map

$$\beta = \hat{b}^{-1} : H(A, B) \longrightarrow H(B, H(A, B)) \quad (4.83)$$

sending the single value $\Phi(b) = f$ to the whole repair map Φ . Louie’s account [Lou09, §§11.11–11.17] isolates what matters in this construction. First, that the *form* of β is incidental and only its codomain is forced: a replication map is anything whose values are repair maps, so $\beta : Z \rightarrow H(B, H(A, B))$ for some Z , and the choice $Z = H(A, B)$ is made only ‘for the convenience of iterative combination’. Second, that with this choice the pair (Φ, β) is again an (M, R) -system,

$$B \xrightarrow{\Phi} H(A, B) \xrightarrow{\beta} H(B, H(A, B)), \quad (4.84)$$

whose metabolism is the repair of the old one, and that ‘one may sequentially extend this formalism ad infinitum’. Third, and crucially, that the injectivity hypothesis on $\hat{b}$ (two repair maps agreeing at b agree everywhere, the abstract form of the one-gene-one-enzyme hypothesis) is what allows the sequence to be *truncated*: ‘the apparent infinite sequence of maps that may arise with the iteration of “repair the repair” is truncated by a mathematical argument that turns it into a hierarchical cycle of maps instead’. This is the same hypothesis we met in Proposition 4.45 as the condition making a universal closure point unique, and Rosen already noted that it fails in categories that are too rich in maps, in particular whenever $H(A, B)$ is the full set of functions $A \rightarrow B$ [Ros66b, p. 150].

The literature is divided on exactly how much of this is essential. Letelier et al. read β as an inverse evaluation ev_b^{-1} and regard its invertibility, unusual for an evaluation map, as the substance of Rosen’s central result [LSAGA⁺04, pp. 451–453]; Cárdenas et al. insist that the internal witness must not be confused with the product object B , and that a formalization dropping the inverse-evaluation requirement has not formalized closure [CLG⁺10, pp. 86–91]; Mossio, Longo and Stewart, on the other hand, hold that the regress is ‘perfectly handled by recursion’, the tower being the Böhm tree of a fixed-point combinator [MLS09, pp. 489–498]. We have already declined to arbitrate this dispute in the discussion preceding Definition 4.42, and we decline again here; suffices to say that one reason why little attention has been given to this circle of ideas is that to any trained mathematician the existence of an injective function $(B^A)^B \rightarrow B^A$ sounds extremely fishy.

Yet, it is worth saying what our formalism does with it. Rather than truncating the tower by fiat, we *build* it, as a genuine cofiltered limit of categories (Construction 4.50 and Definition 4.51), and only then ask which towers are governed by a single global datum. That question is not the same as closure at one level: metabolic closure in the sense of Definition 4.42 asks one level to reconstruct its own process from within, and there is no a priori reason for countably many such reconstructions to cohere. Isolating the extra compatibility is the point of *tower-coherence* (Remark 4.55 and Definition 4.56), which Proposition 4.57 shows to be exactly the existence of a single transformation defined on the colimit of the tower. In this reading Rosen’s truncation is not an axiom but a property, and the ‘hierarchical cycle’ he was after is the statement that an infinite tower is entailed by finitely much data. An *extended* (M, R) -system consists of a chain

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow F_3 \longrightarrow \dots \quad (4.85)$$

that follows the ‘Fibonacci style’ inductive rule

$$\begin{cases} F_0 = A \\ F_1 = B \\ F_{k+2} = [F_k, F_{k+1}] \end{cases} \quad (4.86)$$

So, in particular, an (M, R) -system of length 3 looks like

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \quad (4.87)$$

and one of length 4 looks like

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \longrightarrow [[A, B], [B, [A, B]]]. \quad (4.88)$$

One would like to study these longer (M, R) -systems with the intuition that they provide repair-of-repair functions Φ_n for the process Φ_{n-1} . However, one faces a problem analogous to the one that led us to introduce coherent (M, R) -systems: the dependence on B of a two-step system $A \rightarrow B \rightarrow [A, B]$ is not bifunctorial.

Here we face a similar problem, with one additional layer of complexity in the dependence of an extended (M, R) -system on the pair (A, B) ; even assuming a simple description for (4.87) above, what then with sequences of length four, five, and so on?

The inductive nature of this idea—already a half-baked definition, but requiring refinement—suggests a similarly inductive mechanism for constructing a family of total categories $\mathbf{MRS}(n)$ containing sequences like (4.85) up to length n , each equipped with a canonical “first edge” functor, as follows:

Construction 4.50 ((M, R) -systems of arbitrary length, $\mathcal{U}$).

- Start the induction by defining (M, R) -systems of length 2 as the total category of Definition 4.10, equipped with the functor that chooses f :

$$[-]_f : \mathbf{MRS} \longrightarrow \mathcal{C}^{[0 \rightarrow 1]} \quad (4.89)$$

- Note that by Proposition 3.44 above, one has another functor into $\mathcal{C}^{[0 \rightarrow 1]}$ that chooses Φ , but defined from the *category of elements* of the profunctor $\mathbf{MRS}$:

$$\lambda_{-\} \circ \Phi : \mathcal{E}(\mathbf{MRS}) \longrightarrow \mathcal{C}^{[0 \rightarrow 1]} \quad (4.90)$$

The category $\mathbf{MRS}(3)$ is then defined as the pullback

$$\begin{array}{ccc} \mathbf{MRS}(3) & \longrightarrow & \mathbf{MRS} \\ \downarrow & \lrcorner & \downarrow [-]_f \\ \mathcal{E}(\mathbf{MRS}) & \xrightarrow{[-]_\Phi} & \mathcal{C}^{[0 \rightarrow 1]} \end{array} \quad (4.91)$$

Inspecting the universal property of such a pullback, one easily sees that an object of $\mathbf{MRS}(3)$ consists of the following data

- an (M, R) -system of length 2 in $\mathbf{MRS}$, denoted (f, Φ) , so that $\Phi_f = B \rightarrow [A, B]$;
- an (M, R) -system of length 2 in $\mathcal{E}(\mathbf{MRS})$, denoted (g, Ψ) , so that $\Psi_g = Y \rightarrow [X, Y]$;
- subject to the condition that $\Phi_f = g$:

$$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & [A, B] \\ & & \parallel & & \parallel \\ & & X & \longrightarrow & Y \longrightarrow [X, Y] \end{array} \quad (4.92)$$

- To proceed with the induction, observe that there exists a ‘first edge’ functor

$$[-]_f^3 : \mathbf{MRS}(3) \longrightarrow \mathcal{C}^{[0 \rightarrow 1]} \quad (4.93)$$

such that $[A \xrightarrow{f} B \rightarrow [A, B] \rightarrow [B, [A, B]]] = f$. Then, one can define

$$\begin{array}{ccc} \mathbf{MRS}(n+1) & \longrightarrow & \mathbf{MRS}(n) \\ \downarrow & \lrcorner & \downarrow [-]_f^n \\ \mathcal{E}(\mathbf{MRS}) & \xrightarrow{[-]_\Phi} & \mathcal{C}^{[0 \rightarrow 1]} \end{array} \quad (4.94)$$

which yields ‘gluing data’ for a pair of an n -step and a 2-step (M, R) -system to identify the Φ part of one with the f part of the other.

This provides a reasonable, self-contained interpretation of what a sequence like (4.85) might mean formally, without entangling oneself in explicit reasoning about the parametric dependence of the construction. However, we can go further: the canonical projections from the pullbacks assemble into a cofiltered sequence of categories

$$\mathbf{MRS}(2) \longleftarrow \mathbf{MRS}(3) \longleftarrow \mathbf{MRS}(4) \longleftarrow \mathbf{MRS}(5) \longleftarrow \dots \quad (4.95)$$

Definition 4.51 ($\mathcal{U}$). The category of infinitely extended (M,R)-systems is defined as the limit (in $\mathbf{Cat}$) of the cofiltered chain

$$\mathbf{MRS}_\infty := \lim \left(\mathbf{MRS}(2) \longleftarrow \mathbf{MRS}(3) \longleftarrow \mathbf{MRS}(4) \longleftarrow \mathbf{MRS}(5) \longleftarrow \dots \right) \quad (4.96)$$

In fact, the chain above lives entirely in the slice $\mathbf{Cat}/C^{[0 \rightarrow 1]}$:

Remark 4.52. There exists a cofiltered sequence of *sliced* categories

$$\begin{array}{ccccccc} \mathbf{MRS}(2) & \xleftarrow{\Pi_2} & \mathbf{MRS}(3) & \xleftarrow{\Pi_3} & \mathbf{MRS}(4) & \xleftarrow{\Pi_4} & \mathbf{MRS}(5) \xleftarrow{\Pi_5} \dots \\ & & & & \searrow V_4 & & \downarrow V_5 \\ & & & & & & C^{[0 \rightarrow 1]} \\ & & \searrow V_3 & & & & \\ & & & & & & \\ & \searrow V_2 & & & & & \end{array} \quad (4.97)$$

where the functor V_n is defined as follows:

$$\dots \quad (4.98)$$

Definition 4.53 ($\mathcal{U}$). A similar construction can be performed for incoherent systems: we sketch the construction but we refer the reader to the Agda codebase $\mathcal{U}$. Given that metabolic closure (in the sense of Definition 4.42) only makes sense for coherent systems, however, we mainly work in the setting of Definition 4.51 above.

Remark 4.54. Note also that it is improper to describe an (M,R)-system of length n , or an infinitely extended one, as a sequence of arrows like (4.85); instead (for example) a coherent system of length four consists of an $f : A \rightarrow B$, a natural transformation $\Phi_2 : (-) \Rightarrow [A, -]$ where we single out the component at B , a natural transformation $\Phi_3 : (-) \Rightarrow [B, -]$ where we single out the component at $[A, B]$, etc., in such a way as to obtain a canonical sequence of type (4.88), as

$$A \xrightarrow{f} B \xrightarrow{\Phi_{1,B}} [A, B] \xrightarrow{\Phi_{2,[A,B]}} [B, [A, B]] \xrightarrow{\Phi_3} [[A, B], [B, [A, B]]]. \quad (4.99)$$

In a coherent object of $\mathbf{MRS}_4$ we can access all components of all transformations

$$\Phi_1 : (-) \Rightarrow [A, -]; \quad \Phi_2 : (-) \Rightarrow [B, -]; \quad \Phi_3 : (-) \Rightarrow [[A, B], -] \quad (4.100)$$

and all sorts of obstructions can arise ‘preventing closure’: we might not find a nontrivial transformation of the type we need up to the top level of the chain; we might find a natural transformation, but not such that Φ_n at the F_n component is able to reconstruct $\Phi_{n-1} \dots$

Closure for higher-order systems is thus a rather restrictive condition, imposed on a countable family of natural transformations $\{\Phi_n : id_C \Rightarrow [F_{n-1}, -] \mid n = 1, 2, \dots\}$, each of which is specified by several components, to be able to reconstruct f from Φ and more generally, inductively, Φ_{n-1} from Φ_n by ‘feeding’ each their own closure point.

Remark 4.55. Metabolic closure for longer (M,R)-systems could be defined, at the very least, granting each level of the tower of subsection 4.5 an independently chosen closure point: but Definition 4.42 asks a single level to reconstruct its own process from within, while an extended system asks the same of every level *simultaneously*; there is no a priori reason for these countably many reconstructions to cohere with one another.

This is why we introduce the notion of *tower-coherence*, defined below, to isolate the extra compatibility across levels needed for the hierarchy to be governed by a single global datum rather than by an unrelated family of local ones.

To start the discussion, and give the general idea, let's start from the case $n = 4$ as above; we can take (4.99) and apply the functor $[-, X]$ to obtain

$$[A, X] \xleftarrow{[- \circ f]} [B, X] \xleftarrow{[- \circ \Phi_1]} [[A, B], X] \xleftarrow{[- \circ \Phi_2]} [[B, [A, B]], X] \quad (4.101)$$

Tower-coherence is then the requirement that the components of the successive repair families, probed at one and the same object X , assemble into a cone over this diagram: each $\Phi_{n,X} : X \rightarrow [F_{n-1}, X]$ must be the restriction of $\Phi_{n+1,X}$ along the connecting map, so that the whole tower is determined by probing it as deep as one likes.

The condition is made precise as the requirement that a certain family of maps forms a cone for a diagram, that we now define; assuming the base category C has colimits of sequences, it admits a finite-order weakening, that we outline in Proposition 4.57 below; it relates to the section-retraction pair (p, ι) of Remark 4.4.

Definition 4.56 (Tower-coherence).

$$\begin{array}{c} X \\ \begin{array}{c} \searrow \Phi_{1,X} \\ \downarrow \end{array} \\ [A, X] \end{array} \quad \begin{array}{c} \searrow \Phi_{2,X} \\ \downarrow \end{array} [B, X] \quad \begin{array}{c} \searrow \Phi_{3,X} \\ \downarrow \end{array} [[A, B], X] \quad \begin{array}{c} \searrow \Phi_{4,X} \\ \downarrow \end{array} [[B, [A, B]], X] \\ \begin{array}{c} \xleftarrow{[f,X]} \\ \xleftarrow{[\Phi_{1,B},X]} \\ \xleftarrow{[\Phi_{2,[A,B]},X]} \end{array} \end{array} \quad (4.102)$$

Fix an infinitely extended (M, R) -system in the sense of Definition 4.51, so that each level $n \geq 1$ carries an id -natural transformation $\Phi_n : id \Rightarrow [F_{n-1}, -]$, subject only to the ‘on-diagonal’ requirement, already built into Construction 4.50, that its component at F_n recovers the next stage’s process: writing $c_0 := f$ and $c_n := \Phi_{n,F_n} : F_n \rightarrow [F_{n-1}, F_n] = F_{n+1}$ for $n \geq 1$, the datum of the $(n+1)$ -th level is exactly the pair (c_n, Φ_{n+1}) . Nothing in this on-diagonal gluing relates the components of Φ_n and Φ_{n+1} at any object X other than F_n itself. We say the tower is *tower-coherent*, or *self-coherent*, if it is: for every $n \geq 1$ and every $X \in C_0$,

$$\Phi_{n,X} = [c_{n-1}, X] \circ \Phi_{n+1,X}, \quad (4.103)$$

i.e. the triangles pictured above commute for every X . Informally: the repair prescribed at level n , probed at *any* object X , must be recoverable from the repair prescribed at level $n+1$ by simply restricting along the connecting map – each level of the hierarchy must agree with every deeper unfolding of itself, not merely with the one adjacent level that the bare construction of Construction 4.50 compares it to.

Proposition 4.57 (Tower-coherence as a single transformation out of the colimit). Suppose the colimit $F_\infty := \text{colim}_n F_n$ of the chain (4.85) exists in C , with cocone maps $\kappa_n : F_n \rightarrow F_\infty$ (so $\kappa_{n-1} = \kappa_n \circ c_{n-1}$ for every $n \geq 1$). A tower of id -natural transformations $(\Phi_n)_{n \geq 1}$ satisfying the on-diagonal condition above is tower-coherent if and only if there exists an id -natural transformation $\Phi_\infty : id \Rightarrow [F_\infty, -]$ with $\Phi_n = [\kappa_{n-1}, -] \circ \Phi_\infty$ for every n ; when it exists, Φ_∞ is unique.

Proof. ($\Leftarrow$) Given such a Φ_∞ , contravariant bifactoriality of $[-, X]$ together with the cocone identity $\kappa_{n-1} = \kappa_n \circ c_{n-1}$ give, for every X ,

$$[c_{n-1}, X] \circ \Phi_{n+1,X} = [c_{n-1}, X] \circ [\kappa_n, X] \circ \Phi_{\infty,X} = [\kappa_n \circ c_{n-1}, X] \circ \Phi_{\infty,X} = [\kappa_{n-1}, X] \circ \Phi_{\infty,X} = \Phi_{n,X}, \quad (4.104)$$

so (Φ_n) is tower-coherent.

($\Rightarrow$) Since $- \otimes Y$ is a left adjoint for every Y (biclosedness), $- \otimes -$ preserves the colimit F_∞ separately in each variable, and consequently $[-, X]$ sends it to a limit, naturally in X :

$$[F_\infty, X] \cong \lim_n ([F_n, X], [c_{n-1}, X]). \quad (4.105)$$

Tower-coherence says precisely that, for every fixed X , the family $(\Phi_{n,X})_n$ is a cone over this diagram, hence corresponds to a unique element $\Phi_{\infty,X} \in [F_\infty, X]$ restricting to each $\Phi_{n,X}$ along κ_{n-1} . Naturality of Φ_∞ in X follows because the isomorphism above is itself natural in X , being a limit computed pointwise in a functor category; uniqueness of Φ_∞ is immediate from the universal property of the limit. $\square$

Remark 4.58 (Tower-coherence as compatibility of labels). When the Bernstein centre of C is trivial, ${}_3C \cong I$ (for example when the monoidal unit is a generator, or in any of the other situations of Proposition 4.19 and Remark 4.20; cf. also Remark 4.6, of which this is the appropriate non-Cartesian generalisation, well-pointedness being the special case $I = 1$ of generation), Remark 4.4 and Lemma 4.3 upgrade to a genuine equivalence $\iota : C(C, I) \xrightarrow{\cong} \text{Nat}(id, [C, -])$ for every C , so every Φ_n above is $\iota(a_n)$ for a unique *label* $a_n : F_{n-1} \rightarrow I$. The following elementary fact shows that tower-coherence has a purely algebraic reformulation in terms of these labels.

Lemma 4.59. For every $f : A \rightarrow B$ and $a : B \rightarrow I$, $[f, -] \circ \iota(a) = \iota(a \circ f)$ as natural transformations $id \Rightarrow [A, -]$.

Proof. At a component X , $[f, X] \circ \iota(a)_X = [f, X] \circ [a, X] \circ [I, \lambda_X^{-1}] \circ \eta_X = [a \circ f, X] \circ [I, \lambda_X^{-1}] \circ \eta_X = \iota(a \circ f)_X$, using contravariant bifunctionality of $[-, X]$ in the middle step. $\square$

Applying the lemma with $f = c_{n-1}$: the triangle $\Phi_{n,X} = [c_{n-1}, X] \circ \Phi_{n+1,X}$, for all X , becomes $\iota(a_n) = \iota(a_{n+1} \circ c_{n-1})$, i.e., since ι is injective, exactly $a_n = a_{n+1} \circ c_{n-1}$. Tower-coherence is thus equivalent to the labels $(a_n)_{n \geq 1}$ forming a compatible cone for the inverse system $C(F_0, I) \xleftarrow{c_0^*} C(F_1, I) \xleftarrow{c_1^*} C(F_2, I) \leftarrow \dots$ – restating Proposition 4.57 at the level of labels, this is exactly the existence of a single *global label* $a_\infty : F_\infty \rightarrow I$ with $a_n = a_\infty \circ \kappa_{n-1}$ for every n .

Remark 4.60 (Tower-coherence is vacuous in **Set**, but not elsewhere). In **Set**, where $I = 1$ and $C(C, I)$ is a singleton for every C , every Φ_n of a given type is already unique (Lemma 4.2), so Remark 4.58 shows tower-coherence holds automatically, with no content beyond naturality: there is simply nothing left to independently choose at each level that could fail to cohere. This degeneracy is specific to the terminal unit having no room in its hom-sets, and disappears as soon as $C(C, I)$ is genuinely large – for instance in $\mathbf{Vect}_k$, where the vector-space example given earlier in this section already exhibits a $k[[T]]$ -worth of independent choices at a single level: two independently chosen labels a_n, a_{n+1} generically fail to satisfy $a_n = a_{n+1} \circ c_{n-1}$, and tower-coherence is then a genuine, non-vacuous restriction on an extended (M,R)-system, cutting the tower of independently coherent levels down to those governed by a single label a_∞ for the whole hierarchy.

4.6. A comonad structure on the profunctor of coherent (M,R)-systems.

Remark 4.61. Since the literature on comonads is way more scattered than the one on monads, we recall that by a *comonad in profunctors*, or more shortly a *procomonad*, we mean a profunctor $s : C \multimap C$, i.e. a functor of type

$$s : C^{\text{op}} \times C \longrightarrow \mathbf{Set} \quad (4.106)$$

equipped with natural transformations

$$\begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad (4.107)$$

satisfying appropriate axioms of counitality,

$$\begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad = \text{id}_s = \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad (4.108)$$

and coassociativity

$$\begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad = \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow \epsilon & \downarrow \delta \\ & & C \end{array} \quad (4.109)$$

These diagrams have to be interpreted in the (bi)category of profunctors, where composition $s \circ s$ is defined as the coend

$$(s \circ s)(X, Y) := \int^C s(X, C) \times s(C, Y) \quad (4.110)$$

(see Definition 2.13).

We now aim to prove the following result.

Theorem 4.62 (The procomonad of (M,R)-systems). Let $\mathcal{C}$ be a Cartesian closed category. There is a procomonad structure on the profunctor of Definition 4.9.

Proof. In the notation of Remark 4.61, we have to define a counit natural transformation $\epsilon : \mathbf{MRS} \rightarrow \text{id}_{\mathcal{C}}$, the components of which are the destructor maps

$$\mathbf{MRS}(A, B) \longrightarrow \mathcal{C}(A, B) \quad (4.111)$$

sending an (M,R)-system $\langle\langle f, \Phi \rangle\rangle$ to f . Clearly this is just the fibrewise action of the functor $[-]_f$ defined on page 31, which relates the tabulator $\odot \mathbf{MRS}$ of the $\mathbf{MRS}$ profunctor to the tabulator $\mathcal{C}^{[0 \rightarrow 1]} = \odot \mathcal{C}(-, -)$ of the hom functor.

We also have to define the comultiplication $\delta : \mathbf{MRS} \Rightarrow \mathbf{MRS} \circ \mathbf{MRS}$, which intuitively expresses the idea that an (M,R)-system $\langle\langle f, \Phi \rangle\rangle$ based at (A, B) can be decomposed into a ‘left’ and a ‘right’ part, along a certain intermediate object X :

$$\delta_{AB} : \mathbf{MRS}(A, B) \longrightarrow \int^X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B), \quad (4.112)$$

where the coend accounts for the choice of said intermediate object. Concretely, δ_{AB} makes use of the ‘identity’ (M,R)-system of Definition 3.47; there are essentially two ways to decompose $\langle\langle f, \Phi \rangle\rangle$ that get identified in the quotient that defines the coend, a property which guarantees that the assignment is well defined.

This is not immediately clear, since one has to determine a family of maps *into* a coend –thus into a colimit, thus in the opposite direction of usual universal property arguments.

The idea is to use the fact that for any two given objects A, B the square

$$\begin{array}{ccc} \mathbf{MRS}(A, B) & \xrightarrow{\delta' = \mathcal{M}_A^0 \times \text{id}} & \mathbf{MRS}(A, A) \times \mathbf{MRS}(A, B) \\ \delta'' = \text{id} \times \mathcal{M}_B^0 \downarrow & & \downarrow j_A \\ \mathbf{MRS}(A, B) \times \mathbf{MRS}(B, B) & \xrightarrow{j_B} & \int^X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B) \end{array} \quad (4.113)$$

commutes (if $\mathcal{M}^0$ is the ‘identity’ (M,R)-system defined as in Definition 3.47, j_A is the composite

$$\mathbf{MRS}(A, A) \times \mathbf{MRS}(A, B) \hookrightarrow \sum_X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B) \twoheadrightarrow \int^X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B),$$

and similarly for j_B); thus there is a well-defined diagonal yielding the component at A, B of δ ; it is then a routine check to establish naturality and the comonad equations.

Then we prove the comonad axioms:

- left and right counitality are the easiest to prove, as one can choose in each case the presentation for δ that trivializes the counitality equation;
- coassociativity of δ also leverages on the freedom to pick the easiest representative for the equivalence class of $\delta_{AB} \langle\langle f, \Phi \rangle\rangle$; all in all, the axiom that δ is coassociative boils down to prove that

$$\begin{array}{ccc} \langle\langle f, \Phi \rangle\rangle & \xrightarrow{\quad} & (\mathcal{M}^0, \langle\langle f, \Phi \rangle\rangle) \\ \downarrow & & \downarrow \\ & & ((\mathcal{M}^0, \langle\langle f, \Phi \rangle\rangle), \mathcal{M}^0) \\ \downarrow & & \downarrow \\ (\langle\langle f, \Phi \rangle\rangle, \mathcal{M}^0) & \xrightarrow{\quad} & (\mathcal{M}^0, (\langle\langle f, \Phi \rangle\rangle, \mathcal{M}^0)) \end{array} \quad (4.114)$$

become equivalent in the coend to the same class $\mathcal{M}^0 \otimes \langle f, \Phi \rangle \otimes \mathcal{M}^0$. This is a straightforward check and it concludes the proof of the comonad axioms. $\square$

5. (M,R)-SYSTEMS AND MEALY AUTOMATA

A central aspect of Rosen's intuition, as developed in [Ros64a, Ros64b, Ros66a], is the observation that an (M,R)-system bears some resemblance to a Mealy automaton; this idea was later expanded in [Arb66, AM75, PA07] and others. We now relate the total category of incoherent (M,R)-systems of Definition 3.5 to a total category $\tau\mathbf{Mly}$ of Mealy automata and 'twisted' maps.

The correspondence on objects outlined by Arbib in [*op. cit.*] *does not* extend to a functor into the category of Mealy automata that has been considered by several authors in more recent times, whichever of the total categories we choose as its domain; to obtain such a (desirable) result one has to change the codomain category into a category $\tau\mathbf{Mly}$ of 'twisted automata', with the same fibres $\tau\mathbf{Mly}(A, B)$ containing the Mealy automata with input A and output B ; such a tweak is, however, of little, if any, practical importance, since the fibre $\mathbf{iMRS}(A, B)$ is discrete, while $\tau\mathbf{Mly}(A, B)$ is not; if we want to read the correspondence defined by Arbib on objects in [Arb66] as a functor of type

$$\begin{array}{ccc} \mathbf{iMRS} & \xrightarrow{\quad} & \tau\mathbf{Mly} \\ & \searrow \quad \swarrow & \\ & C^{\text{op}} \times C & \end{array} \quad (5.1)$$

i.e., as it should, as a map compatible with the projection functors, $\mathcal{A}$ must induce fibrewise functors

$$\mathcal{A}_{(A,B)} : \mathbf{iMRS}(A, B) \longrightarrow \tau\mathbf{Mly}(A, B); \quad (5.2)$$

yet, such a functor is nothing more than a function on objects of type

$$(\mathcal{A}_{(A,B)})_0 : \mathbf{iMRS}(A, B) \longrightarrow \tau\mathbf{Mly}(A, B)_0; \quad (5.3)$$

it does not seem worthwhile to pursue this matter further.

As a preliminary to our discussion, we shall review the basics of Mealy automata. We adapt the presentation of [Lor25a], where a rather general version of a 'total category of Mealy automata' is built and studied.

Definition 5.1 ($\mathcal{C}$). Let C be an ambient category (no hypotheses at the moment), and $F : C \rightarrow C$ an endofunctor.

Define the category $\mathbf{Mly}(F, B)$ of *Mealy automata with input F and output B* as the following strict 2-pullback in $\mathbf{Cat}$:

$$\begin{array}{ccc} \mathbf{Mly}(F, B) & \longrightarrow & F/B \\ \downarrow & \lrcorner & \downarrow U \\ \mathbf{Alg}(F) & \xrightarrow{V} & C \end{array} \quad (5.4)$$

where $\mathbf{Alg}(F)$ is the category of *endofunctor algebras* of F , F/B the comma category of arrows $(X, FX \rightarrow B)$ (and U, V are the most obvious forgetful functors).

We observe in passing that using the techniques of [Lor25a] one can show that $\mathbf{Mly}(F, B)$ inherits several good properties from C ; in particular, if C is complete, or cocomplete, or accessible, or locally presentable [AR94, GU71], then so is $\mathbf{Mly}(F, B)$.

Remark 5.2. When C is monoidal and $F : C \rightarrow C$ is the tensor product $A \otimes _$ for a fixed object of C , we shorten $\mathbf{Mly}(F, B)$ to $\mathbf{Mly}(A, B)$; in this particular case, the typical object $\left\| \frac{E}{d,s} \right\|$ of $\mathbf{Mly}(A, B)$ is a span

$$\left\| \frac{E}{d,s} \right\| : E \xleftarrow{d} A \otimes E \xrightarrow{s} B. \quad (5.5)$$

Such models of computation, with the final state depending (on the left) or not depending (on the right) on the inputs A , have a long history, cf. [Mea55, Moo56, Net59]. Their categorical axiomatization also has a long tradition, cf. [Gog72, Gog73, AM75].

Remark 5.3. Consider two endofunctors $F : C \rightarrow C$, $G : \mathcal{H} \rightarrow \mathcal{H}$. If $P : C \rightarrow \mathcal{H}$ is a functor intertwining F, G , i.e. equipped with a natural transformation $\pi : GP \Rightarrow PF$ we can define a functor $\pi^* : \mathbf{Alg}(F) \rightarrow \mathbf{Alg}(G)$ by application of P and precomposition with π , a functor $C/B \rightarrow \mathcal{H}/PB$ in the obvious way, and in turn a unique functor

$$\varpi^* : \mathbf{Mly}_C(F, B) \longrightarrow \mathbf{Mly}_{\mathcal{H}}(G, PB). \quad (5.6)$$

Thus the construction in Definition 5.1 is implicitly functorial in the pair F, B , an observation which allows one to study the properties of the pseudofunctor $(F, B) \mapsto \mathbf{Mly}(F, B)$.

Definition 5.4. The *total Mealy 2-category* $\mathbf{Mly}$ is defined as follows:

- the objects are triples $(C; F, B)$ where $F : C \rightarrow C$ is an endofunctor of a category C , and B an object of C ;
- the morphisms $(P, \pi, u) : (C; F, B) \rightarrow (\mathcal{H}; G, B')$ are triples where $P : C \rightarrow \mathcal{H}$ is a functor, $\pi : GP \Rightarrow PF$ is an *intertwiner* natural transformation between F and G and $u : PB \rightarrow B'$ is a morphism;
- 2-cells $\gamma : (P, \pi, u) \Rightarrow (Q, \theta, v)$ consist of natural transformations $\gamma : P \Rightarrow Q$ compatible with the intertwiners π, θ in the obvious sense, and such that $v \circ \gamma_B = u$.

With such a domain, the assignment sending (C, F, B) to $\mathbf{Mly}(F, B)$ is a strict 2-functor $\mathbf{Mly} \rightarrow \mathbf{Cat}$ ($\mathbf{Cat}$ is the 2-category of categories, strict functors, strict natural transformations).

It is, however, rarely needed to vary the domain C of the automata under study. A simpler (=lower-dimensional) approach is convenient if we are content with keeping C fixed.

Definition 5.5 (The total categories of automata). Definition 5.1 entails at once that the correspondence $(F, B) \mapsto \mathbf{Mly}(F, B)$ is a (pseudo)functor of type $\mathbf{Mly} : \mathbf{Cat}(C, C)^{\text{op}} \times C \rightarrow \mathbf{Cat}$, i.e. a pseudo-profunctor $\mathbf{Cat}(C, C) \rightrightarrows C$ from which we can extract a two-sided fibration, i.e. a span

$$\mathbf{Cat}(C, C) \xleftarrow{p} \mathbf{Mly} \xrightarrow{q} C \quad (5.7)$$

such that p is a fibration, q is an opfibration, p -Cartesian lifts are q -vertical and q -opCartesian lifts are p -vertical. We call the tip $\mathbf{Mly}$ of the span the *total Mealy category* constructed from C .

Similar considerations allow us to construct the total Moore category $\mathbf{Mre}$ from the pseudo-profunctor $(F, B) \mapsto \mathbf{Mre}(F, B)$, and to obtain a two-sided fibration $\mathbf{Cat}(C, C) \leftarrow \mathbf{Mre} \rightarrow C$.

Remark 5.6. Unwinding the definition, it is easy to establish how reindexings of the total Mealy and Moore fibration act. In the particular case where $\alpha : F \Rightarrow G$ is a natural transformation between left adjoints $F \dashv R$ and $G \dashv Q$ and $f : B \rightarrow B'$ a morphism, the reindexing functor $\mathbf{Mly}(\alpha, f) : \mathbf{Mly}(G, B) \rightarrow \mathbf{Mly}(F, B')$ preserves all colimits—and thus, assuming C is a locally presentable category, is a left adjoint; however, it fails to preserve limits (it already fails to preserve terminal objects).

The tensor functor $_ \otimes _ : C \times C \rightarrow C$ of a monoidal category (see Remark 5.2 above) now carries to the ‘left regular representation’ $\lambda : C \rightarrow \mathbf{Cat}(C, C) : A \mapsto A \otimes -$ of C on itself, and as a consequence, we can pull back the total Mealy fibration to obtain the left leg of the diagram

$$\begin{array}{ccc} \mathbf{Mly}^{\otimes} & \xrightarrow{\quad} & \mathbf{Mly} \\ \downarrow & \lrcorner & \downarrow \\ C^{\text{op}} \times C & \xrightarrow{\lambda^{\text{op}} \times C} & \mathbf{Cat}(C, C)^{\text{op}} \times C \end{array} \quad (5.8)$$

which gives rise to the *monoidal Mealy* (two-sided) *fibration*

$$C \xleftarrow{p^{\otimes}} \mathbf{Mly}^{\otimes} \xrightarrow{q^{\otimes}} C \quad (5.9)$$

Remark 5.7. The span (5.9) is associated to the tabulator of a profunctor $\mathbf{Mly}^{\otimes} : C^{\text{op}} \times C \rightarrow \mathbf{Set}$. Its objects are the same Mealy automata of (5.5), and its morphisms $\| \frac{E}{d, s} \| \rightarrow \| \frac{E'}{d', s'} \|$ are triples

$(\ell : A \rightarrow A', r : B \rightarrow B', u : E \rightarrow E')$ such that the diagram

$$\begin{array}{ccccc} E & \xleftarrow{d} & A \otimes E & \xrightarrow{s} & B \\ u \downarrow & & \ell \otimes u \downarrow & & \downarrow r \\ E' & \xleftarrow{d'} & A' \otimes E' & \xrightarrow{s'} & B' \end{array} \quad (5.10)$$

commutes, or in equational form, $s' \circ (\ell \otimes u) = r \circ s$ and $d' \circ (\ell \otimes u) = u \circ d$.

To the same profunctor we can also associate the category of elements, where the direction of arrows is appropriately ‘twisted’: it is this latter total category that provides the correct codomain for promoting Arbib’s construction to a functor.

Definition 5.8 (The twisted category of Mealy automata, $\mathcal{M}^{\ell}$). The *twisted category of Mealy automata*, denoted $\tau\mathbf{Mly}$, is defined as follows

- the objects are the same Mealy automata of (5.5), i.e. tuples $\left\| \frac{X}{d,s} \right\|_{AB} : X \leftarrow A \otimes X \rightarrow B$;
- the morphisms $\left\| \frac{X}{d,s} \right\|_{AB} \rightarrow \left\| \frac{Y}{d',s'} \right\|_{A'B'}$ consist of triples $\ell : A' \rightarrow A, r : B \rightarrow B', u : X \rightarrow Y$ such that

$$u \circ d \circ (id \otimes \ell) = d' \circ (u \otimes id) \quad r \circ s \circ (id \otimes \ell) = s' \circ (u \otimes id) \quad (5.11)$$

Note the difference with (5.10), where the direction of the arrows ℓ is reversed; hence the ‘twisting’ in the name.

Proposition 5.9 ([Arb66] reloaded, $\mathcal{M}^{\ell}$). There exists a family of functions

$$(\mathcal{A}_{(A,B)})_0 : \mathbf{iMRS}(A, B) \longrightarrow \tau\mathbf{Mly}(A, B)_0 \quad (5.12)$$

which, in fact, is the fibrewise decomposition of a functor $\mathcal{A}$ such that $\langle p^{\otimes}, q^{\otimes} \rangle \circ \mathcal{A} = [-]_{(A,B)}$:

$$\begin{array}{ccc} \mathbf{iMRS} & \xrightarrow{\mathcal{A}} & \tau\mathbf{Mly} \\ & \searrow [-]_{(A,B)} & \swarrow \langle p^{\otimes}, q^{\otimes} \rangle \\ & C^{\text{op}} \times C & \end{array} \quad (5.13)$$

sending an incoherent (M, R) -system $A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]$ to the Mealy automaton

$$[A, B] \xleftarrow{\lambda(a,u).\Phi(ua)} A \otimes [A, B] \xrightarrow{\lambda(a,u).ua} B \quad (5.14)$$

where we have adopted the lambda notation for the arrows

$$\epsilon_B : A \otimes [A, B] \longrightarrow B \quad \Phi \circ \epsilon_B : A \otimes [A, B] \longrightarrow B \longrightarrow [A, B] \quad (5.15)$$

(The fact that the process $f : A \rightarrow B$ is seemingly absent from this description should not deceive the reader; indeed, [Arb66] considers automata *pointed* by an initial state $f : [A, B]$.)

Arbib then goes on to prove two fundamental results which, however, can only be stated in the Cartesian case. In the next two statements, then, C is a category with finite products, and we consider *Cartesian* Mealy automata.

Theorem 5.10. A Mealy automaton $\langle d, s \rangle : X \leftarrow A \times X \rightarrow B$ lies in the essential image of $\mathcal{A}$ in Proposition 5.9 if and only if it is of the form

$$X \xleftarrow{w \circ s} A \times X \xrightarrow{s} B \quad (5.16)$$

for some $w : B \rightarrow X$. These Mealy automata are called *simple*.

A convenient diagrammatic representation for a simple Mealy automaton ‘closes’ the span d, s to a commutative triangle, via $w : B \rightarrow X$:

$$\begin{array}{ccc} & A \times X & \\ d \swarrow & & \searrow s \\ X & \xleftarrow{w} & B \end{array} \quad (5.17)$$

More interestingly, Arbib proves that simple Mealy automata satisfy some weak form of lifting property:

Theorem 5.11. If $\langle d, s \rangle : X \leftarrow A \times X \rightarrow B$ is a simple Mealy automaton representing an (M, R) -system, and

$$\begin{array}{ccccc} X & \xleftarrow{\quad} & A \times X & \xrightarrow{\quad} & B \\ f \downarrow & & f \times g \downarrow & & \parallel \\ Y & \xleftarrow{d'} & C \times Y & \xrightarrow{s'} & B \end{array} \quad (5.18)$$

is a morphism in the total fibration of Mealy automata, such that f, g are epi (and h is the identity, in the notation of Remark 5.7 above), then $\langle d', s' \rangle$ is itself simple.

Technically this looks like a ‘fibration where Cartesian lifts exist only for epimorphisms’, but the morphisms in the category **MRS** are never defined in [Arb66] except indirectly, and indeed only the correspondence on objects is defined.

Remark 5.12. These two results are however of little practical interest for us, since

- the proof of Theorem 5.10 in [Arb66] makes use of the fact that the representation $d' : A^* \rightarrow [X, X]$ is injective (in representation-theoretic terms, d' is *faithful*); this does not seem to be a blanket assumption in modern treatments of automata theory;
- the notion of simple (M, R) -system only applies to Cartesian, incoherent (M, R) -systems.

Proposition 5.13. It is interesting to speculate what the property of metabolic closure of an (M, R) -system $\langle f, \Phi \rangle$, defined in Definition 4.42, translates into when one considers the simple Mealy automaton $\mathcal{A}\langle f, \Phi \rangle$: in short, metabolic closure is the condition that the initial state f of $\mathcal{A}\langle f, \Phi \rangle$ has *nontrivial isotropy* [Ste16] when $[A, B]$ is considered as the carrier of a monoid representation; again in representation-theoretic terms, the presence of a universal closure point a_0 (see Definition 4.44) translates into the fact that the *stabilizer* [Ste16] of f contains a_0 .

6. FUTURE WORK

An *autograph*, in the sense of Guitart [Gui14a, Gui14b, Gui17], is a set X equipped with two endomaps $s, t : X \rightarrow X$; equivalently, an action of the free monoid on two generators, so that the category **aGph** of autographs is a presheaf topos. Its defining feature is that vertices and edges are not distinguished: every edge x has a source sx and a target tx , themselves edges, whose own sources and targets are edges again. Autographs are therefore irremediably non-well-founded, the fixpoints of the action, i.e. the joint equalizer of s, t and the identity, being loops $x : x \rightarrow x$; it is exactly this feature that makes them a candidate home for the self-referential diagrams of relational biology; virtually every picture of graph-theoretic dynamics in [Lou17, Ch. 7–8] is an autograph. Guitart pushes the analogy with graphs to a square of adjunctions relating **Gph** $\rightleftarrows$ **Cat** to **aGph** $\rightleftarrows$ **aCat**, the category **aCat** of *autocategories* being monadic over **aGph** as **Cat** is over **Gph**.

The link with the present paper is that the *shape* of an (M, R) -system is an autograph. The 2-step shape $Q(\mathbf{MR}_2)$ has A and B as fixpoints and two further edges $f : A \rightarrow B$ and $\Phi_1 : B \rightarrow f$; the n -step shape of Construction 4.50 adds $\Phi_{k+1} : \Phi_{k-1} \rightarrow \Phi_k$ for $k \geq 1$, with $\Phi_0 := f$, and its two incidence matrices s_n, t_n are obtained from s_{n-1}, t_{n-1} by an evident inductive rule. Every finite (M, R) -system shape thus becomes a representation of the free monoid on two letters, and the natural question is what its representation theory looks like. That question cannot be answered by importing the theory of quiver representations [ASS06, Sch14, Kir16] wholesale, since an autograph has no vertices to index idempotents by, and a representation theory of autographs has instead to be built from scratch, of which [Gui17] is the only precedent we know.

We intend to pursue this programme elsewhere. Future work will be focused on finding an antecedent or an interpretation, in Rosen’s theory, for the following basic facts one can argue about the representation theory of the autographs $\mathbf{MR}_n$.

Theorem. The (class of morphisms of the) free autocategory on $Q(\mathbf{MR}_2)$ is presented by the regular language $[AB]^*(f[AB]^*)? \Phi?$ over the alphabet $X(\mathbf{MR}_2) := \{A, B, f, \Phi\}$, with composition the

concatenation of words, a *partial* operation: precomposition with a word ending in Φ is defined on identities only.

Theorem. For each n , the free autcategory on $Q(\mathbf{MR}_n)$ is presented by a regular language in the same way; the presentation nevertheless fails at infinity, the language $\{\Phi_1\Phi_2\cdots\Phi_k \mid k \geq 0\}$ attached to $\mathbf{MRS}_\infty$ of Definition 4.51 being non-regular by a Myhill–Nerode argument.

Theorem. The path algebra $\mathbb{C}\langle\mathbf{MR}_2\rangle$ (shorthand for $\mathbb{C}\langle Q(\mathbf{MR}_2)\rangle$) is an iterated triangular extension of $\mathbb{C}[a]$ and $\mathbb{C}[b]$ by free bimodules: it is isomorphic to the algebra of lower triangular matrices with diagonal $\mathbb{C}[a], \mathbb{C}[b], \mathbb{C}$, off-diagonal entries the bimodules $M = \langle a^n f b^m \rangle$ and $P = \langle b^m p \rangle \cong \mathbb{C}[b]$, and corner $N \cong M \otimes_{\mathbb{C}[b]} P$. In particular the square built out of a, b, f is not required to commute.

Theorem. The Jacobson radical of $\mathbb{C}\langle\mathbf{MR}_2\rangle$ is the two-sided ideal $\langle f, p \rangle$ generated by the arrows lying on no oriented cycle, and the quotient $\mathbb{C}\langle\mathbf{MR}_2\rangle / \text{rad} \cong \mathbb{C}[a] \oplus \mathbb{C}[b] \oplus \mathbb{C}$ is semiprimitive, though of course not semisimple.

Theorem. Consequently $\mathbf{Rep}(Q(\mathbf{MR}_2))$ is an iterated *recollement* [BBD82, CPS89, AHKL11, PV18]: it is obtained by gluing $\mathbf{Mod}_{\mathbb{C}[a]}$ and $\mathbf{Mod}_{\mathbb{C}[b]}$ along M , and then gluing the result with $\mathbf{Mod}_{\mathbb{C}}$ along (N, P) . The simple, projective and injective representations are computed explicitly in each of the four positions A, B, f, Φ .

7. ACKNOWLEDGMENTS

F. Loregian was supported by the ESF funded Estonian IT Academy research measure (project 2014–2020.4.05.19–0001).

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